



# Participant Handbook

Sector  
**Green Jobs**

Sub-Sector  
**Sustainability**

Occupation  
**Executive**



Reference ID: **SGJ/N5001, Version 1.0**  
**NSQF Level 6**



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## GHG Accounting & Sustainability Reporting

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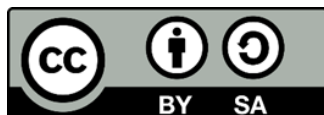
3rd Floor, CBIP Building, Malcha Marg, Chanakyapuri, New Delhi – 110021

Email: [info@sscgj.in](mailto:info@sscgj.in)

Website: <https://sscgj.in/>

Phone: 011-41792866

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## About this Book

This Participant Handbook is designed to enable training for the specific National Occupational (NOS) SGJ/N5001 at workplace which is covered across Unit/s.

Government of India is aiming towards developing a capacity and solution to manage the Greenhouse gases emission in a cost effective and innovative way.

The contents of this book are in simple language, without going into too much theoretical details and calculations. It is envisaged that this training manual will provide the participants with the knowledge and skills required for the optimization of resources complying with all applicable codes, standards, and safety requirements.

## Symbols Used



**Key Learning  
Outcomes**



**Steps**



**Exercise**



**Activity**



**Notes**



**Unit  
Objectives**

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# 1.Introduction to the Climate Challenge and Policy Action

UNIT 1.1 The fundamental concepts of climate change science.

UNIT 1.2 The concepts of the greenhouse effect, global warming potential, greenhouse gases and causes of anthropogenic climate change.

UNIT 1.3 Observed and expected climate trends along with various climate scenarios.

UNIT 1.4 Climate mitigation and adaptation and their significance

UNIT 1.5 International and national Policy Framework and actions on Climate Change

UNIT 1.6 The role of UNFCCC for driving global climate actions

UNIT 1.7 Climate risk Assessment

UNIT 1.8 Principal challenges and opportunities for climate change action and describe the significance of low-carbon development

UNIT 1.9 International mechanisms that support low-carbon development

UNIT 1.10 The role and the landscape of climate finance.

UNIT 1.11 Sector and Industries for adopting low-carbon actions and outline various measures

UNIT 1.12 Incorporating climate change into national planning procedures.

UNIT 1.13 The functions of local, regional, and national institutions in climate change planning and actions.



## Key Learning Outcomes

At the end of this module, you will be able to:

1. Discuss fundamental climate change concepts, including the greenhouse effect, global warming potential (GWP), and greenhouse gases (GHGs).
2. Analyze historical and projected climate trends, identifying key climate scenarios and their implications.
3. Assess the social and economic impacts of climate change, including its effects on human societies, industries, and global economies.
4. Differentiate between climate mitigation and adaptation strategies, exploring their roles in sustainability and resilience building.
5. Understand international and national climate policy frameworks, including agreements like the UNFCCC, Kyoto Protocol, and Paris Agreement.
6. Explain the role of climate finance in low-carbon development, covering financial mechanisms such as the Green Climate Fund (GCF) and carbon pricing systems.
7. Discuss the integration of climate action into national planning, highlighting the significance of climate risk assessments and sustainability reporting.
8. Recognize the importance of a comprehensive approach to climate change, addressing its impacts on ecosystems, societies, and economies.

## Unit 1.1. Fundamental concepts of climate change science

### Unit objectives



At the end of this unit, you will be able to:

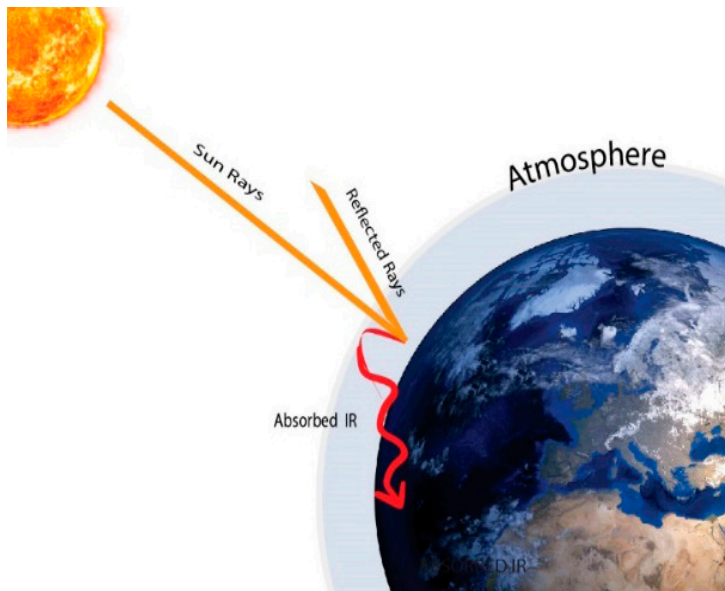
1. What is climate change?
2. What is the Greenhouse effect and Global warming?
3. Introductions to mitigation and adaptation

### 1.1.1 What is Climate change science?

The fundamentals of climate change revolve around the observed and projected changes occurring because of human activities and the scientific understanding of the Earth's climate system.

### 1.1.2 Greenhouse Effects

The earth's atmosphere contains trace gases like carbon dioxide, methane, and nitrogen. These gases can trap the heat reflected by the earth's surface<sup>1</sup>. This natural process helps keep the earth warmer to sustain living organisms.



The figure (fig 1.1.2) shows the simple process of the greenhouse effect.

The excess of these GHGs has negative effects on climate patterns. These topics will be covered extensively in this module.

<sup>1</sup> <https://science.nasa.gov/climate-change/faq/what-is-the-greenhouse-effect>

### 1.1.3 Enhanced Greenhouse Effect

Human activities, primarily the burning of fossil fuels (coal, oil, and natural gas), deforestation, and industrial processes, have significantly increased the concentration of greenhouse gases in the atmosphere<sup>2</sup>. This enhanced greenhouse effect is leading to an imbalance, causing the Earth's temperature to rise.

### 1.1.4 Global Warming

The unbalanced human activity has increased the GHG that has affected the average surface temperature changing the earth's radiative balance – “the give and take “between heat that comes in the day and what the earth emits during the night.

The increase in the amount of heat trapped in the earth's atmosphere (i.e. GHGs absorb the heat and reradiate back to the surface) causes the increase in temperature and alters the precipitation patterns.

### 1.1.5 Effects of global warming

The adverse effects of global warming are given below

- Rise in Global Temperature
- The rise in average global temperature has resulted in frequent and intense heat waves, causing adverse effects on the health of humans and animals.
- Melting ice and Rising sea levels
- The increase in temperature is causing the melting of Polar ice caps and glaciers, causing the rise of sea level, posing a threat to the coastal cities and island nations.

#### Ocean Acidifications

- Oceans act as natural sinks, they absorb carbon dioxide from the atmosphere. The excess absorptions of CO<sub>2</sub> make the ocean acidic, affecting marine life, especially with calcium carbonate shells.

#### Extreme weather events

- Due to global warming there is an increase in frequent and extreme weather events including hurricanes, drought, floods, and wildfires causing adverse impacts on ecosystems and economies.

#### Disruptions of Ecosystem

- Due to the temperature change some species of flora and fauna struggle to adapt, leading to shifts in biodiversity and potentially causing the extinction of vulnerable species.

#### Impact on Agriculture

- Changes in temperature, precipitation, shifts in seasons, pests, and diseases can affect crop yield and agricultural land. This can lead to future food crises.

<sup>2</sup> <https://anjumsspace4.quora.com/Global-warming-refers-to-the-long-term-increase-in-Earths-average-surface-temperature-due-to-the-buildup-of-greenhouse>

### 1.1.6 Mitigation and Adaptation

- Mitigation and adaptation are two key strategies in addressing the challenges posed by global warming and climate change
- Mitigation involves efforts to reduce or prevent the emission of greenhouse gases (GHGs) into the atmosphere, thereby mitigating the extent of climate change.
- Adaptation involves adjusting and preparing for the impacts of climate change to reduce vulnerability and enhance resilience.

(These topics will be covered in detail under the topic “Climate Mitigation and Adaptations and their Significance.”)



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*On October 17, 2009, Maldives President Mohamed Nasheed and 13 government officials held the world's first underwater cabinet meeting off the island of Girifushi. The meeting was intended to draw global attention to the Maldives' vulnerability to climate change and the threat of rising sea levels.\**

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*\*source: Hindustan times*

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## Exercise



1. What are the primary human activities contributing to the enhanced greenhouse effect, and how do they affect the concentration of greenhouse gases in the atmosphere?
2. Explain the process by which greenhouse gases trap heat in the Earth's atmosphere and how this contributes to global warming.
3. What are some of the observed effects of global warming on the environment, particularly regarding temperature changes, sea levels, and ocean acidification?
4. How do extreme weather events and disruptions in ecosystems illustrate the broader impacts of global warming on both natural and human systems?
5. What are the differences between mitigation and adaptation strategies in addressing climate change, and why are both approaches necessary?

## Unit 1.2. Concepts of Greenhouse Effects, Global Warming Potential, GHGs, and Anthropogenic Climate Change

### Unit objectives



After reading this unit, you will comprehend the following points:

1. Greenhouse Gases (GHG).
2. Effects of greenhouse effect.
3. What are the major Greenhouse Gases?
4. What is Global Warming Potential and its application?
5. What is anthropogenic climate change?

### 1.2.1 Greenhouse Effect

The greenhouse effect is the process by which certain gases in the Earth's atmosphere trap and retain heat from the sun, preventing it from escaping into space. To maintain a suitable temperature on Earth for supporting life, this natural phenomenon is crucial. The primary greenhouse gases responsible for the greenhouse effect are water vapor, carbon dioxide (CO<sub>2</sub>), methane (CH<sub>4</sub>), nitrous oxide (N<sub>2</sub>O), ozone (O<sub>3</sub>), and various human-made gases such as chlorofluorocarbons (CFCs)<sup>3</sup>.

The stages involved in the greenhouse effect

- In addition to ultraviolet and visible light, the Sun also generates different types of electromagnetic radiation.
- A portion of the sunlight that reaches Earth is reflected into space by the planet's surface, clouds, and air particles. On the other hand, the Earth's surface absorbs most of the energy.
- The Earth's surface absorbs solar energy and then re-emits it in the form of infrared radiation (heat). This infrared radiation is crucial for maintaining the planet's temperature.

The primary reasons for the enhanced greenhouse effect

- **Burning of Fossil Fuels:** When fossil fuels like coal, oil, and natural gas are used to produce energy, a significant amount of carbon dioxide (CO<sub>2</sub>) is released into the atmosphere. This has a significant role in the rising concentration of greenhouse gases.
- **Deforestation:** Trees function as a natural carbon sink by absorbing CO<sub>2</sub> as part of photosynthesis. Deforestation, or the cutting of forests for development, agriculture, or other reasons, lowers the amount of trees in the atmosphere that can absorb carbon dioxide, which raises greenhouse gas concentrations.
- **Industrial Processes:** Certain industrial processes directly contribute to the atmospheric release of greenhouse gasses. For instance, industrial processes can release nitrous oxide (N<sub>2</sub>O) and methane (CH<sub>4</sub>), as well as CO<sub>2</sub> during the manufacturing of cement.
- **Agricultural Practices:** Agriculture contributes to the greenhouse effect through practices such as rice cultivation, which produces methane, and the use of nitrogen-based fertilizers, which can release nitrous oxide.
- **Waste Management:** Methane emissions may arise from improper organic waste dis-

<sup>3</sup> <https://www.bgs.ac.uk/discovering-geology/climate-change/how-does-the-greenhouse-effect-work>

posal and breakdown in landfills. In addition, CO<sub>2</sub> and other pollutants are released during the burning of waste items.

- **Land Use Changes:** The equilibrium of greenhouse gases in the atmosphere can be impacted by changes in land use, such as urbanization and shifts in land cover.
- **Industrial and Residential Energy Use:** Fossil fuels are frequently burned to provide energy for lighting, cooling, heating, and other industries. This process adds to the production of greenhouse gases.

## 1.2.2 Major Greenhouse Gases

### Carbon Dioxide (CO<sub>2</sub>)

- Combustion of fossil fuels (coal, oil, and natural gas), deforestation, and certain industrial processes. CO<sub>2</sub> is the most prevalent greenhouse gas and is responsible for the majority of human-induced global warming.

### Methane (CH<sub>4</sub>)

- Agricultural activities (livestock digestion and rice cultivation), fossil fuel extraction and processing, and the decay of organic waste in landfills. Although less abundant than CO<sub>2</sub>, methane is more effective at trapping heat in the short term, making it a potent greenhouse gas.

### Nitrous Oxide (N<sub>2</sub>O)

- Agricultural and industrial activities, including the use of nitrogen-based fertilizers, certain industrial processes, and the burning of fossil fuels. Nitrous oxide has a higher warming potential than CO<sub>2</sub> over a longer time frame, contributing to climate change.

### Water Vapor (H<sub>2</sub>O)

- The source of water vapor is from evaporation of oceans, lakes, and rivers. While water vapor is the most abundant greenhouse gas, its concentration in the atmosphere is largely controlled by natural processes. Human activities do not directly emit significant amounts of water vapor.

### Ozone (O<sub>3</sub>)

- Industrial processes, certain chemical reactions, and emissions from vehicles are the Primary sources<sup>4</sup>. Ozone in the stratosphere (upper atmosphere) absorbs and re-emits some of the infrared radiation, contributing to the greenhouse effect. However, ground-level ozone is considered an air pollutant and has detrimental effects on human health<sup>5</sup>.

## 1.2.3 Global Warming Potential

The global warming potential (GWP) is defined as an “index measuring the radiative forcing\* following emission of a unit mass of a given substance, accumulated over a chosen time horizon, relative to that of the reference substance, carbon dioxide (CO<sub>2</sub>).<sup>6</sup>

4 <https://climatechange.chicago.gov/ghgemissions/sources-greenhouse-gas-emissions>

5 <https://www.epa.gov/ground-level-ozone-pollution/ground-level-ozone-basics>

6 [https://en.wikipedia.org/wiki/Global\\_warming\\_potential](https://en.wikipedia.org/wiki/Global_warming_potential)

*\*Radiative forcing (or climate forcing) is a concept used in climate science to quantify the change in energy balance in the Earth's atmosphere caused by various factors, such as concentrations of greenhouse gases, aerosols, and changes in solar radiation*

The global warming potential (GWP) depends on both the efficiency of the molecule as a greenhouse gas and its atmospheric lifetime. GWP is measured relative to the same mass of CO<sub>2</sub> and evaluated for a specific timescale<sup>7</sup>.

When calculating the GWP of a greenhouse gas, the value depends on the following factors:

1. The absorption of infrared radiation by the given gas
2. The time horizon of interest (integration period)
3. The atmospheric lifetime of the gas

GWP is measured relative to the same mass of CO<sub>2</sub> and evaluated for a specific timescale

#### Global Warming Potential Values of some GHG

| Gas                                  | Lifetime | GWP Time Horizon |           |           |
|--------------------------------------|----------|------------------|-----------|-----------|
|                                      |          | 20 years         | 100 Years | 500 Years |
| Methane                              | 12       | 62               | 23        | 7         |
| Nitrous Oxide                        | 114      | 275              | 296       | 156       |
| HFC-134a<br>(Hydrofluorocar-<br>bon) | 13.8     | 3300             | 1300      | 400       |
| HFC-23<br>(Hydrofluorocar-<br>bon)   | 260      | 9400             | 12000     | 10000     |
| Sulfur Hexaflu-<br>oride             | 3200     | 15100            | 22200     | 32400     |

Table 1 GWP

<sup>7</sup> [https://agritech.tnau.ac.in/environment/envi\\_index%20-%20climate%20change-globalwarming%20potential.html](https://agritech.tnau.ac.in/environment/envi_index%20-%20climate%20change-globalwarming%20potential.html)

## Application of Global Warming Potential

Here are some key applications of Global Warming Potential

**Policy and Regulations:** Governments and international bodies use GWP values to inform policies and regulations aimed at reducing greenhouse gas emissions. Regulations often target gases with higher GWPs to achieve more significant climate mitigation.

**Climate Assessments:** GWP values are essential in climate assessments and projections. Scientists use these values to estimate the future impact of various greenhouse gases on global warming and climate change.

**Greenhouse Gas Inventories:** GWP is employed in greenhouse gas inventories, where countries and organizations quantify and report their emissions. This helps track progress in meeting emission reduction targets outlined in international agreements like the Paris Agreement.

**Carbon Offsetting:** In carbon-offset programs, organizations may invest in projects that reduce or remove greenhouse gas emissions equivalent to a specific amount of CO<sub>2</sub>, based on GWP values. This allows for a standardized approach to measuring the climate impact of different activities.

**Life Cycle Assessments:** GWP is utilized in life cycle assessments of products, processes, or services to evaluate their environmental impact. This includes assessing emissions from raw material extraction, production, use, and disposal.

**Technology and Product Development:** Industries and innovators use GWP values when developing new technologies and products to ensure they have a lower environmental impact. This consideration is increasingly important, as sustainability becomes a key focus for businesses.

**Renewable Energy Projects:** GWP is applied in the evaluation of renewable energy projects, helping to compare the overall climate benefits of different sources. This informs decisions about the most effective strategies for reducing greenhouse gas emissions in the energy sector.

**International Collaboration:** GWP facilitates international collaboration and negotiations on climate change. It provides a common metric for assessing the effectiveness of emission reduction efforts across different countries and sectors.

**Corporate Sustainability Reporting:** Many companies include GWP values in their sustainability reports to transparently communicate their greenhouse gas emissions and reduction efforts. This helps stakeholders, including investors and consumers, make informed decisions based on a company's environmental performance.

**Climate Action Planning:** GWP values play a critical role in the development of climate action plans at various levels, guiding decision-makers in identifying and prioritizing strategies to achieve emission reduction targets.

### 1.2.4 Anthropogenic Climate Change



Changes in the Earth's climate system that are predominantly ascribed to human activity are referred to as anthropogenic climate change. The Earth's climate is changing due to several human-caused variables, which cause global warming and other climate-related effects.

Major causes of Anthropogenic Climate Change are

- **Burning of Fossil Fuels:** The combustion of fossil fuels (coal, oil, and natural gas) for energy production, transportation, and industrial processes releases large amounts of carbon dioxide ( $\text{CO}_2$ ), a major greenhouse gas.
- **Methane Emissions:** Methane ( $\text{CH}_4$ ) is a powerful greenhouse gas that is produced by agricultural processes including the decomposition of livestock waste and manure management. When fossil fuels are extracted, processed, and transported, methane is released.
- **Nitrous Oxide Emissions:** The use of nitrogen-based fertilizers and certain agricultural practices can lead to the release of nitrous oxide ( $\text{N}_2\text{O}$ ), another potent greenhouse gas.
- **Industrial Processes:** Greenhouse gases are released during several industrial processes, such as the manufacture of cement and chemicals. Methane is produced in landfills when organic waste is properly disposed of and decomposes.
- **Land Use Changes:** The conversion of natural landscapes into urban areas contributes to changes in surface albedo and disrupts local climate patterns. Converting natural habitats into agricultural land can release stored carbon and alter local and regional climates.
- **Transportation:** Carbon dioxide and other pollutants are released when gasoline and diesel fuels burn in automobiles.
- **Land Use Practices:** Clearing forests for agriculture, logging, and other purposes reduces the capacity of forests to absorb and store carbon. Poor land management practices contribute to the release of stored carbon in soils.

## Exercise



1. What are the primary greenhouse gases responsible for the greenhouse effect, and how do they contribute to maintaining Earth's temperature?
2. Describe the main human activities that lead to the enhanced greenhouse effect and their impact on greenhouse gas concentrations in the atmosphere.
3. How is the Global Warming Potential (GWP) of a greenhouse gas determined, and what factors influence its value?
4. Explain the significance of Global Warming Potential (GWP) in climate policy, regulations, and greenhouse gas inventories.
5. What are the key causes of anthropogenic climate change, and how do they differ from natural climate variations?

## Unit 1.3. Observed and expected climate trends along with various climate scenarios.

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. Shift in climate pattern
2. Different climate scenarios
3. Effect of climate change

### 1.3.1 Observed and Expected Trends

Long-term shifts in temperature, precipitation, and other atmospheric parameters are referred to as climate trends. It is essential to recognize and comprehend these patterns to evaluate the effects of climate change and create adaptation plans. Predicted climatic patterns are based on scientific models that forecast various future climate scenarios.

#### Observed trends

- **Rise in Temperature:** Over the previous century, there has been an increase in global temperatures, with an acceleration of warming during the last several decades. Effects include ecosystem alterations, melting ice caps, and rising sea levels.
- **Severe Meteorological Events:** The frequency and severity of extreme weather events, such as heatwaves, hurricanes, droughts, and floods, have increased. Resulting in Changes affecting agriculture, human settlements, and ecosystems.
- **Rise in Sea Level:** Sea levels are continuously rising as a result of glacier melting<sup>8</sup> and seawater's thermal expansion. Consequences include increased flooding, coastal erosion, and dangers to low-lying places.
- **Retreat of Glaciers:** Glaciers are receding worldwide. Effects include modifications to river flows, alterations to water availability, and effects on ecosystems.
- **Ocean acidification:** Ocean acidity has increased as a result of carbon dioxide absorption, according to observations. Endangering marine life, particularly those living organisms with calcium carbonate shells.

<sup>8</sup> [https://www.reddit.com/r/explainlikeimfive/comments/13nka8r/eli5\\_if\\_a\\_block\\_of\\_ice\\_melts\\_what\\_change\\_would](https://www.reddit.com/r/explainlikeimfive/comments/13nka8r/eli5_if_a_block_of_ice_melts_what_change_would)

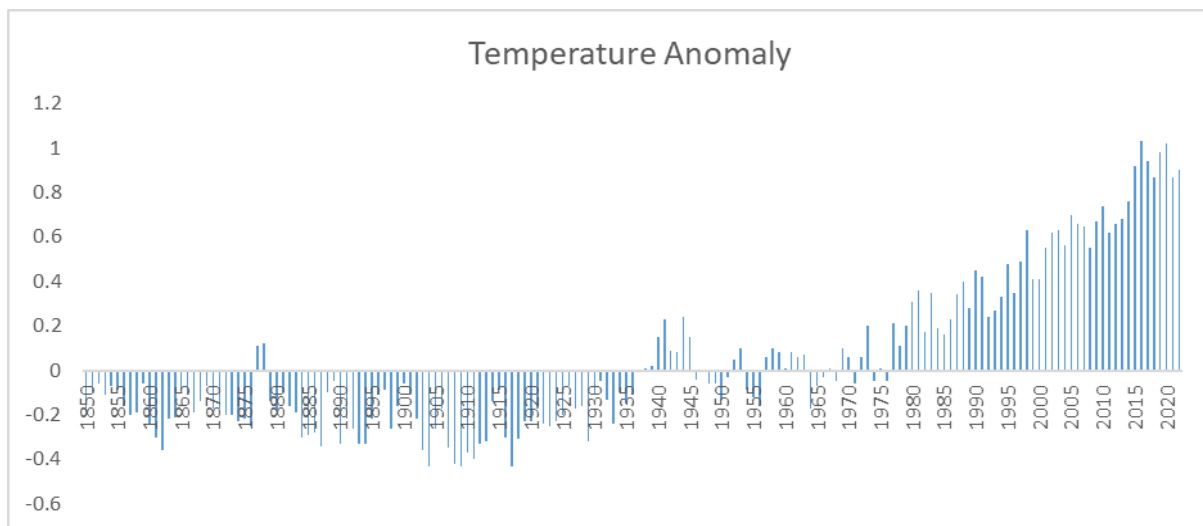


Fig 1.3.2 Trends in climate change

Graph source: NOAA

### Expected Climate Trends

- **Estimates of Temperature:** Persistent global warming, with estimates fluctuating according to scenarios of greenhouse gas emissions. RCP (Representative Concentration Pathway\*) scenarios represent various emission trajectories, with low (RCP2.6) and high (RCP8.5) values indicated.
- **Variations in Rainfall:** Changes in precipitation patterns are anticipated, with some areas expecting greater rainfall and others dealing with longer-lasting droughts. Under various climatic scenarios, models predict variances in precipitation by region.
- **Rise in Sea Level:** Sea levels will continue to rise as a result of glaciers and ice sheets melting. Estimates differ, but higher emissions are thought to cause a more noticeable rise in sea level.
- **Severe Meteorological Events:** A rise in the frequency and power of extreme weather phenomena like heat waves, storms, and hurricanes. Specific details vary depending on emission levels and regional circumstances.
- **Changes in Biodiversity:** As species migrate or adapt to unfamiliar settings, ecosystems and biodiversity will undergo shifts. The rate and extent of climate change determine the impact on ecosystems<sup>9</sup>.
- **Adaptation and Mitigation Situations:** The execution of tactics aimed at diminishing greenhouse gas emissions (mitigation) and adjusting to evolving circumstances (adaptation). The accomplishment of the global climate goals is contingent upon technical

<sup>9</sup> <https://www.noaa.gov/education/resource-collections/climate/climate-change-impacts>

developments, legislative decisions, and international collaboration.

Comprehending and tackling these observed and anticipated climate patterns is imperative for well-informed decision-making, policy development, and worldwide endeavors to alleviate and adjust to climate change



\*Scan to know more about RCP

## Exercise



1. What are some of the observed trends in global climate, and how have they impacted ecosystems and human activities?
2. How do scientific models predict future climate scenarios, and what role do Representative Concentration Pathway (RCP) scenarios play in these predictions?
3. Describe the expected changes in precipitation patterns and sea levels according to various climate scenarios, and how these changes might affect different regions.
4. What are the potential impacts of ocean acidification on marine life, and why is this trend concerning for global biodiversity?
5. How do adaptation and mitigation strategies differ in addressing climate change, and what factors influence their success in achieving global climate goals?

## Unit 1.4 Social and Economic impact of climate change

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. Recognize and analyze the social consequences of climate change
2. Evaluate the economic ramifications of climate change
3. Understand the interconnected nature of social and economic impacts

### 1.4.1 Consequences of climate change

Climate change has considerable social and economic consequences and is primarily caused by human activity, such as the burning of fossil fuels and deforestation. These effects are already noticeable and are anticipated to get worse over the next few decades. Below is a detailed note on the social and economic consequences of climate change

#### Social Impact

- **Food Security:** Altered growing seasons, a rise in the frequency of extreme weather events, and changes in precipitation patterns are some of the ways that climate change impacts agricultural productivity. Developing countries are particularly impacted by food shortages, rising food prices, and malnutrition caused by crop failures and lower yields.
- **Displacement and Migration:** Displacement of communities has been accelerated by catastrophic weather events, shifting precipitation patterns, and rising sea levels. Vulnerable locations run the risk of becoming uninhabitable, which could lead to migration brought on by climate change, particularly in low-lying coastal areas and small island states.
- **Water Scarcity:** Altered precipitation patterns and melting glaciers impact water availability. Increased frequency of droughts and floods jeopardizes access to clean water, particularly in arid and semi-arid regions, leading to water scarcity and sanitation issues.
- **Health Risks:** The health implications of climate change include illnesses connected to food and water, injuries and deaths, diseases of the heart and lungs, diseases caused by pests like West Nile Virus and Lyme disease, and more.
- **Social Inequality:** Research indicates that impoverished populations are frequently forced to relocate to flood-prone locations due to inequality, which increases their vulnerability to climate-related flooding.

## Economic Impact

- **Trade and Agricultural Productivity:** Food production and supply systems worldwide are impacted by decreased crop yields and shifting agricultural circumstances. Economic downturns are experienced by economies that rely heavily on agriculture because of rising resource rivalry and declining output.
- **Impact on Industry:** Changes affect industries that include forestry, fishing, and agriculture that depend on resources that are susceptible to climate change. Regulations intended to reduce greenhouse gas emissions and shifting energy patterns may result in higher prices for energy-intensive companies.
- **Rising Insurance cost:** The recent spike in insurance prices can be partially attributed to climate change and future expenses are expected to rise more. Storms, excessive heat, floods, and other calamities become more frequent because of climate change. Consumers are being charged more by insurers to cover rising claim expenses.
- **Damage to Property and Loss of Infrastructure:** Hurricanes, floods, and wildfires are examples of more frequent and powerful extreme weather events that cause substantial property damage and infrastructure destruction.
- Both national economies and insurance systems are strained by the costs of reconstruction and recovery.
- **Job Displacement:** Sectors like tourism and agriculture that are vulnerable to climate change may see employment losses and economic downturns.

There may be employment displacement in some industries due to the necessity of adaption measures and the shift to a low-carbon economy.

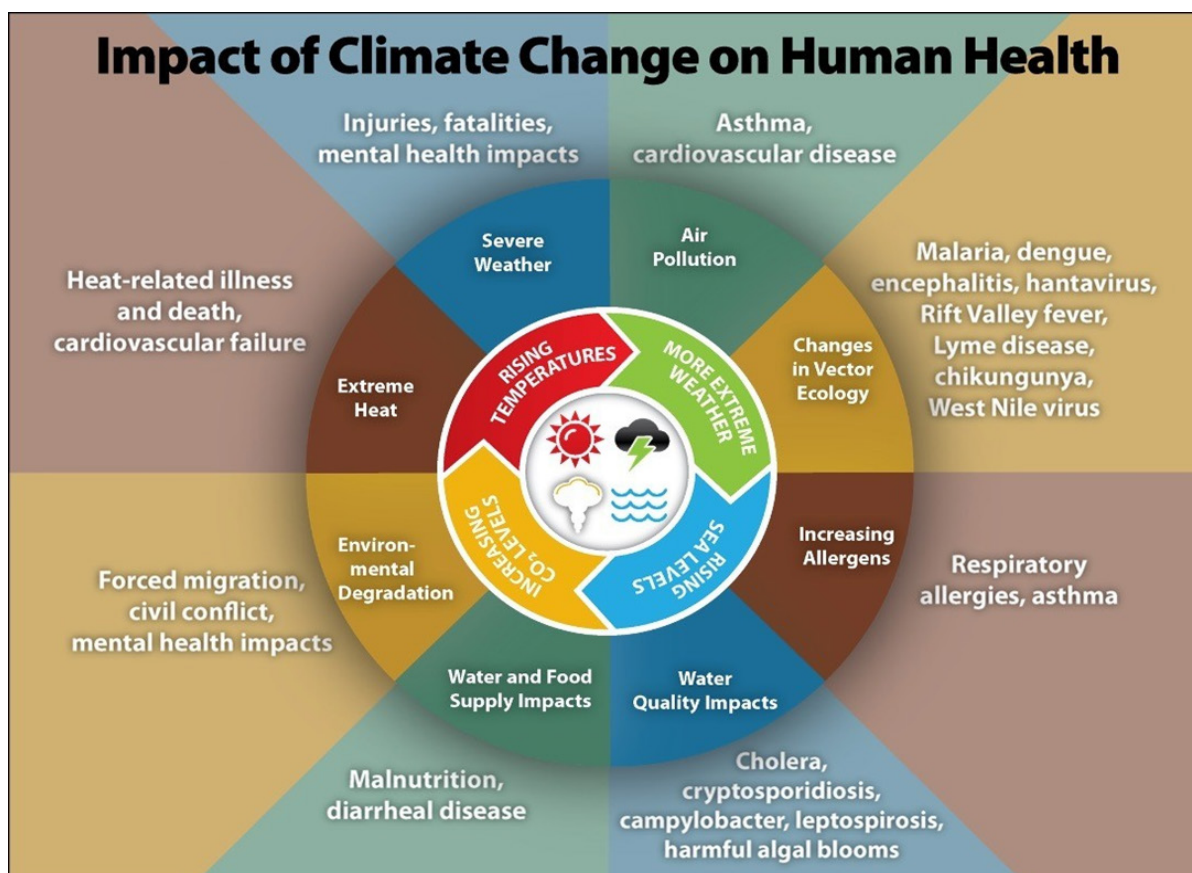


Fig 1.4.1

Image: Centres for Disease Control and Prevention

## Exercise



1. How does climate change affect agricultural productivity and food security, especially in developing countries?
2. What are the key factors contributing to displacement and migration due to climate change, and which regions are most vulnerable?
3. In what ways does climate change impact water availability and lead to water scarcity and sanitation issues?
4. Discuss the various health risks associated with climate change, including those related to food, water, and vector-borne diseases.
5. How does social inequality exacerbate the vulnerability of certain populations to climate-related impacts, such as flooding?
6. What are the economic consequences of climate change on industries dependent on natural resources, and how might this affect global trade and agricultural productivity?
7. How does climate change contribute to rising insurance costs, and what are the implications for consumers and insurers?
8. What are the economic impacts of extreme weather events on property and infrastructure, and how do these events strain national economies and insurance systems?
9. How might job displacement occur in sectors like tourism and agriculture due to climate change, and what measures can be taken to address this issue?
10. What are the challenges and economic implications of transitioning to a low-carbon economy in response to climate change?

## Unit 1.5. Climate Mitigation and Adaptation and Their Significance

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. Differentiating Between Mitigation and Adaptation
2. Exploring Mitigation Strategies
3. Examining Adaptation Measures
4. Assessing the Significance of Mitigation
5. Recognizing the Importance of Adaptation

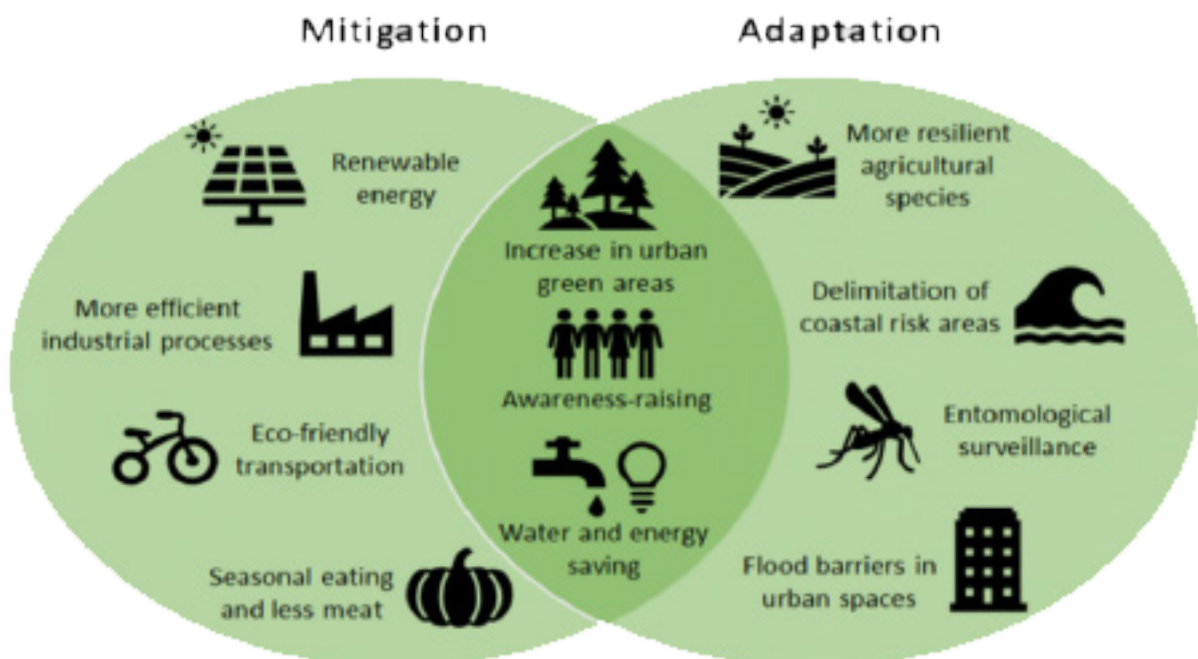


Image source: <https://www.algarveadapta.pt>

Adaptions and Mitigation

Two key strategies for tackling the challenges caused by climate change are adaptation and mitigation. Both serve as essential parts of a comprehensive plan to control climate change's effects and ensure the planet's sustainability in the future. An outline of each idea and its importance is provided below:

### 1.5.1 Climate Mitigation

The Intergovernmental Panel on Climate Change (IPCC) defines mitigation as “a human intervention to reduce emissions or enhance the sinks of greenhouse gases”

#### Significance

- **Stabilization of Global Temperature:** To avoid the worst effects of climate change, such as more extreme heatwaves, rising sea levels, and extreme weather events, mitigation measures seek to stabilize or reduce global temperatures.
- **Protecting Ecosystems:** By reducing the stresses brought on by climate change, such as habitat loss and altered precipitation and temperature patterns, mitigation contributes to the preservation of ecosystems and biodiversity.
- **Sustainable Development:** Several mitigation measures call for switching to greener energy sources and implementing sustainable behaviors to encourage the development of a more robust and ecologically friendly economy.

### 1.5.2 Climate Adaptation

The IPCC defines climate change adaptation as “In human systems, as the process of adjustment to actual or expected climate and its effects to moderate harm or take advantage of beneficial opportunities.”

“In natural systems, adaptation is the process of adjustment to actual climate and its effects; human intervention may facilitate this.”

#### Significance of climate Adaptation

- **Risk Reduction:** By assisting ecosystems and communities in adapting to the present and projected effects of climate change, adaptation strategies lower the likelihood of loss and damage.
- **Infrastructure Resilience:** Adaptation requires planning and modifying infrastructure to endure the effects of climate change, guaranteeing the continuous operation of essential infrastructure such as energy, water supply, and transportation.
- **Food and Water Security:** In the face of climate change, ensuring food and water security requires adapting methods of agriculture to shifting climatic circumstances and enhancing water management.

Both adaptation and mitigation are fundamental components of a comprehensive climate action plan and are interdependent. Mitigation seeks to address the underlying causes of climate change, whereas adaptation concentrates on controlling the effects that are currently transpiring or anticipated to transpire.

Integrated strategies can improve overall efficacy and advance sustainable development by combining adaptation and mitigation activities.

**Exercise**

1. **What is climate mitigation, and why is it significant in stabilizing global temperatures and protecting ecosystems?**
2. **How does climate mitigation contribute to sustainable development and the transition to greener energy sources?**
3. **What are the main goals of climate adaptation, and how does it help in reducing risks associated with climate change?**
4. **In what ways does climate adaptation enhance the resilience of infrastructure to ensure the continuous operation of essential services?**
5. **How can integrating both adaptation and mitigation strategies improve the overall efficacy of climate action plans and promote sustainable development?**

## Unit 1.6 International and National Policy Framework and actions on Climate Change

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. Different international frameworks and their functioning
2. Functions of national frameworks

### Introduction

The worldwide issue of rising temperatures and environmental degradation is the focus of national and international policy frameworks and efforts on climate change. Global accords such as the Paris Agreement establish goals for cutting greenhouse gas emissions and advancing sustainable development. To lessen the effects of climate change and prepare for them, national governments enact policies that encourage sustainable practices, strengthen resilience, and shift to renewable energy. These frameworks are essential for encouraging cooperation, establishing guidelines, and directing group efforts in the direction of a future that is more robust and sustainable.

#### 1.6.1 International Frameworks

The Paris Agreement is a significant international agreement that was adopted in 2015 to keep global warming well below 2<sup>0</sup> Celsius above pre-industrial levels. Nationally determined contributions (NDCs) were made by nations stating their precise goals for adaptation and mitigation of climate change.

The United Nations platform Convention on Climate Change (UNFCCC) serves as the main international platform for climate change cooperation. Meetings known as Conferences of the Parties (COP) bring nations together to debate and agree on climate change mitigation measures.

#### 1.6.2 National Frameworks

- **Nationally Determined Contributions (NDCs):** As part of the Paris Agreement, nations create NDCs that detail their unique plans for mitigating climate change and adapting to it.
- **Regulations and Policies:** To address carbon reductions, support renewable energy, and improve resilience to climate impacts, numerous nations have passed laws and policies on climate change.
- **Investments in Climate-Resilient Infrastructure, Renewable Energy, and Reforestation:** Governments are making more and more investments in climate-resilient infrastructure and sustainable practices and technology.

## Exercise



1. What is the Paris Agreement, and what are its main objectives regarding global warming and sustainable development?
2. How do Nationally Determined Contributions (NDCs) play a role in the Paris Agreement, and what do they entail for individual nations?
3. What is the role of the United Nations Framework Convention on Climate Change (UNFCCC) and its Conferences of the Parties (COP) in international climate change cooperation?
4. What types of regulations and policies have nations implemented to address carbon reductions, promote renewable energy, and enhance resilience to climate impacts?
5. How are governments investing in climate-resilient infrastructure, renewable energy, and reforestation to mitigate and adapt to climate change?

## Unit 1.7. The role of UNFCCC for driving global climate action

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. What is UNFCCC
2. Structure of UNFCCC
3. Role of UNFCCC in combatting climate change

### 1.7.1 . Introduction to UNFCCC

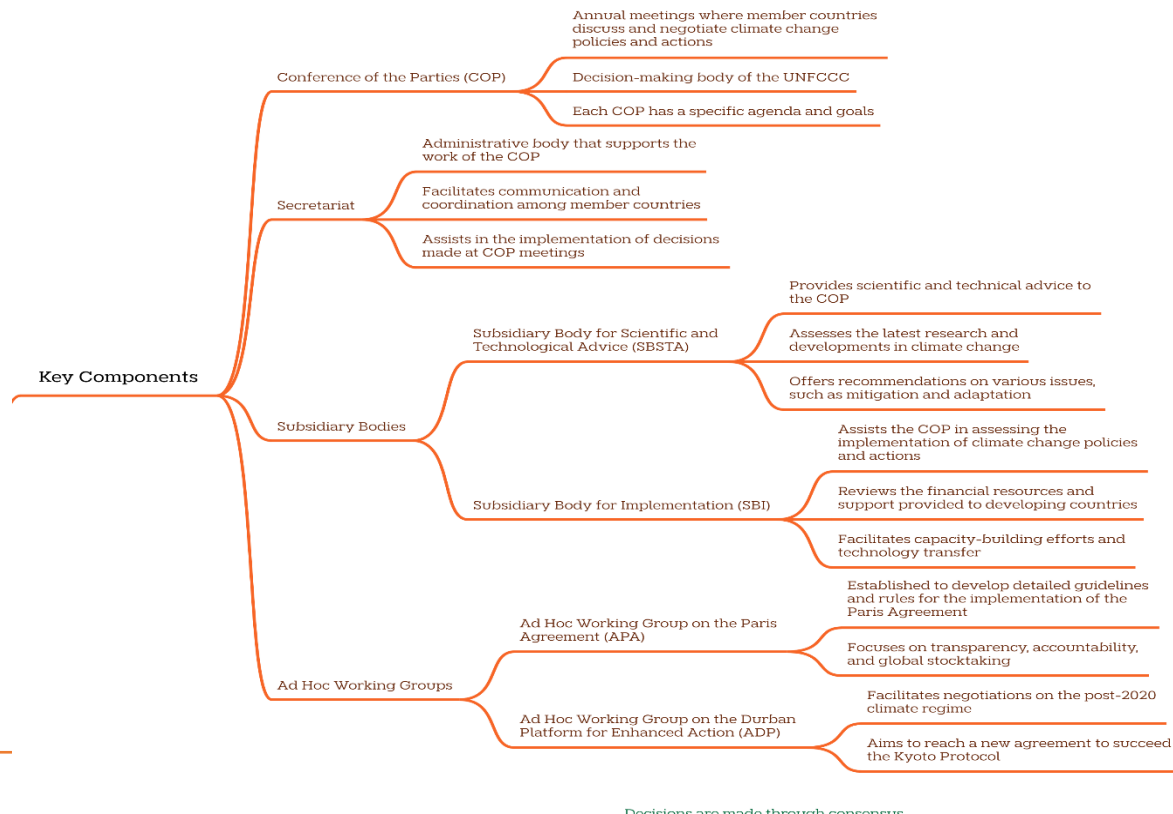
**Adoption:** At the Rio de Janeiro, Brazil, Earth Summit on May 9, 1992, the UNFCCC was ratified. It became operative on March 21, 1994.

**Overarching Goal:** Preventing hazardous human interference with the climate system is the main goal of the UNFCCC, which aims to stabilize greenhouse gas concentrations in the atmosphere.

The common but differentiated responsibilities and respective capacities (CBDR-RC) principle, which recognizes the various historical and contemporary contributions of nations to climate change, is one of the guiding principles of the treaty.

#### Key Elements:

- **Conference of the Parties, (COP):** Every year, officials from member nations convene at the COP, the UNFCCC's highest decision-making body, to evaluate the state of affairs, negotiate agreements, and make decisions on a range of climate change-related issues.
- **Subsidiary Bodies:** The Subsidiary Body for Scientific and Technological Advice (SBSTA) and the Subsidiary Body for Implementation (SBI) are the two subsidiary bodies of the UNFCCC. Regarding implementation and scientific issues, these entities advise and counsel the COP.
- **Secretariat:** The COP and its affiliated entities receive support from the Secretariat, which is situated in Bonn, Germany. It offers administrative help, meeting coordination, and communication facilitation.



### 1.7.2 Role of UNFCCC in combatting climate change

- **Creating the Framework:** The fundamental international agreement that establishes the general framework for tackling climate change is the UNFCCC. It lays out the agreements and guiding principles for international collaboration on climate-related challenges.
- **Conference of the Parties (COP) Meetings:** Every year, representatives from almost all nations convene at the COP meetings to evaluate the state of affairs, negotiate new accords, and decide on important matters like adaptation, mitigation, financing, and technology transfer concerning climate change.
- **Implementation of the Paris Agreement:** The 2015 Paris Agreement was negotiated and adopted with significant assistance from the UNFCCC. Under the auspices of the UNFCCC, the agreement establishes a framework for nations to submit nationally determined contributions (NDCs) detailing their climate action plans and specifies specific targets to limit the rise in global temperatures.
- **Nationally Determined Contributions (NDCs):** Member nations' NDC submission and evaluation are facilitated by the UNFCCC. Each nation's unique goals and strategies for reducing greenhouse gas emissions and preparing for the effects of climate change are outlined in these contributions.
- **Global Goals for Emission Reduction:** Global emission reduction objectives are collectively defined by countries through UNFCCC negotiations and agreements. As stated in the Paris Agreement, keeping global warming well below 2 degrees Celsius over pre-industrial levels is the long-term objective.
- **Financial Mechanisms:** The Green Climate Fund (GCF), which aims to assist developing nations in their efforts to reduce and adapt to climate change, is one of the financial mechanisms under the UNFCCC's supervision. Financial resources for climate-related projects are mobilized and allocated with assistance from the UNFCCC.
- **Technology Transfer:** To help developing nations embrace greener and more sustainable

practices, the UNFCCC helps transfer ecologically sound technologies to those nations. Transferring technology is crucial to combating climate change worldwide<sup>10</sup>.

- **Resilience and Adaptation:** The UNFCCC takes steps to foster resilience and adapt to the effects of climate change. To improve adaptable capacity, this involves the creation of the Adaptation Committee and the National Adaptation Plans (NAPs).
- **Capacity-Building:** To improve developing nations' abilities to take part in climate negotiations, carry out climate actions, and obtain climate finance, the UNFCCC supports capacity-building initiatives in these nations.
- **Research and Assessment:** By assisting with studies and evaluations related to climate change science, the UNFCCC contributes to the reports developed by the Intergovernmental Panel on Climate Change (IPCC). Policy formulation and decision-making are guided by this scientific foundation.

The United Nations Framework Convention on Climate Change (UNFCCC) provides support to developing nations' efforts to strengthen their capacity to lead climate discussions, carry out climate policies, and secure climate finance.

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10 <https://www.weforum.org/agenda/2023/02/technology-climate-change-adaptation>

## Exercise



1. What is the overarching goal of the UNFCCC, and when did it come into effect?
2. What is the principle of common but differentiated responsibilities and respective capacities (CBDR-RC), and how does it influence the UNFCCC?
3. What are the key roles and functions of the Conference of the Parties (COP) within the UNFCCC framework?
4. How does the UNFCCC facilitate the implementation of the Paris Agreement and the submission of Nationally Determined Contributions (NDCs)?
5. What financial mechanisms does the UNFCCC oversee to support climate change mitigation and adaptation in developing countries?
6. How does the UNFCCC contribute to the global transfer of environmentally sound technologies and the promotion of resilience and adaptation to climate change effects?

## Unit 1.8. Climate Risk Assessment

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. What is Climate Risk Assessment?
2. What are the key elements involved in Risk Assessment?

### 1.8.1 What is Risk Assessment?

A methodical analysis of the possible negative effects and vulnerabilities related to climate change is known as climate risk assessment. The analysis focuses on the potential effects that diverse sectors, ecosystems, and communities may face as a result of alterations in climate conditions, including temperature, precipitation, sea level rise, and extreme weather events. The objective is to recognize and comprehend the dangers that climate change poses to facilitate the creation of sensible adaptation plans and well-informed decision-making.



Fig 1.8.1 shows the different steps in climate risk Assessment,

### 1.8.2 Important elements of the assessment of climate risk comprise

1. Identification of Climate Hazards: Acknowledging some climate-related hazards that can affect a region or industry, such as heatwaves, droughts, floods, storms, and sea level rise.
2. Exposure Assessment: Assessing the degree to which assets, communities, ecosystems, and infrastructure are vulnerable to climate hazards to identify what is at risk and who is at risk.
3. Vulnerability analysis: The process of determining how sensitive and adaptable exposed elements are to changes in the climate to determine how vulnerable they are.
4. Impact assessment: The process of determining how possible climate risks may affect various industries, such as infrastructure, agriculture, water resources, ecosystems, and health.
5. Risk characterization: The process of combining exposure, vulnerability, and influence data to express and measure the degree of risk related to particular climatic scenarios.
6. Scenario planning: Taking into account various climatic scenarios and the hazards they pose in order to guide the creation of policies and plans for adaptation.

In order to reduce the negative effects of climate change, governments, corporations, and communities must adopt resilient and adaptive solutions. This requires the completion of climate risk assessments. In light of a changing climate, this approach aids in ensuring sustainable development, safeguarding people at risk, and advancing the long-term health of both natural and human systems.



Scan to learn more about  
India's Long-Term Low-Carbon Development Strategy

## Exercise



1. What is the primary objective of a climate risk assessment, and what elements are analyzed during this process?
2. How does the identification of climate hazards contribute to the overall climate risk assessment?
3. What role does exposure assessment play in determining the risks associated with climate change?
4. How is vulnerability analysis used to understand the sensitivity and adaptability of elements exposed to climate change?
5. What is the purpose of impact assessment in the context of climate risk assessment, and which sectors are typically analyzed?
6. Why is scenario planning important in the process of climate risk assessment, and how does it guide policy and adaptation strategy development?

## Unit 1.9. Principle challenges and opportunities for climate change action and the significance of low-carbon development

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. The complexity and dynamics of climate change action
2. What are the importance of low-carbon development?

### 1.9.1 Challenges and opportunities for climate changes action

The following are some fundamental ideas, difficulties, and chances related to addressing climate change:

#### Challenges

- **International Cooperation and Coordination:** International cooperation is needed to address the global issue of climate change. Diverse interests and agendas might make it difficult to agree on objectives, promises, and courses of action.
- **Regulatory and Policy Barriers:** The swift adoption of sustainable practices is impeded by inconsistent or inadequate climate policies and regulations. Effective climate action can occasionally be hampered by political and economic factors.
- **Budgetary Obstacles:** Making the shift to a low-carbon economy will cost a lot of money in infrastructure, renewable energy, and adaption strategies. Getting sufficient funds together and getting past financial obstacles are still major obstacles.
- **Innovation in Technology:** It's critical to develop and implement advanced technologies for sustainable agriculture, carbon capture, and clean energy. The urgency of climate change may not necessarily coincide with the rate of technological advancement, though.
- **Changes in Social and Behavioral Domains:** It can be difficult to promote the broad adoption of sustainable habits and lifestyles. Advancement may be hampered by societal inertia, resistance to change, and ignorance.
- **Climate Justice and Equality:** Since vulnerable populations are disproportionately affected by climate change, justice and equality must be given top priority in climate policy. One of the ongoing challenges is striking a balance between the duties of developed and developing countries.
- **Low Degree of Public Awareness:** A lack of public knowledge about climate change issues may make support for climate action wane. To mobilize public opinion and engagement, effective communication and education are important

#### Opportunities

- **Transition to Renewable Energy: Opportunity:** There is an excellent opportunity to cut greenhouse gas emissions as a result of the world's transition to renewable energy. Future energy sustainability is facilitated by developments in clean energy technologies such as wind, solar, and others<sup>11</sup>.
- **Policy Innovation:** Climate action can be sparked by creative policies like carbon pricing, incentives for renewable energy, and subsidies for sustainable agriculture. Governments have the authority to develop and carry out progressive policies.
- **Engagement of the Private Sector:** Companies may make a significant contribution to addressing climate change through implementing sustainable practices, purchasing clean technology, and incorporating environmental factors into their business plans.
- **Technological Progress:** The continuous progress in technology, encompassing energy storage, carbon capture, and sustainable agricultural methods, presents novel approaches for reducing the effects of climate change and adjusting to it.
- **Community Initiatives:** Community-based and local efforts have the power to start change from the ground up. Resilience and sustainability are promoted when communities are empowered to take climate action.
- **Global Cooperation:** Cooperation between states, global institutions, and interested parties can improve the efficacy of climate initiatives. An enhanced worldwide response is facilitated by the pooling of information, assets, and experiences.
- **Adolescence and Political Engagement:** Advocating for immediate action and drawing attention to climate challenges are youth-led movements and activism. Using the zeal and vigor of the younger generation to drive change can be quite effective.

### 1.9.2 Significance of low – Carbon Development

A calculated approach to economic expansion, low-carbon development seeks to decrease greenhouse gas (GHG) emissions and lower the total carbon footprint resulting from human activity.

Some key aspects highlighting the significance of low-carbon development

- **Climate Change Mitigation:** The main objective of low-carbon development is to lessen greenhouse gas emissions to slow down global warming. This is essential to achieving global climate goals and averting the worst effects of climate change.
- **Transition to Renewable Energy:** The use of renewable energy sources including solar, wind, hydro, and geothermal power is encouraged by low-carbon development. Switching away from fossil fuels lowers emissions related to energy production and lessens reliance on finite resources.
- **Energy Efficiency:** Enhancing the energy efficiency of transportation, buildings, and industries is a crucial component of low-carbon development. This raises economic productivity and competitiveness while lowering carbon emissions.
- **Green Employment Creation:** Green employment is produced as a result of the shift to a low-carbon economy, which opens up new business and technological prospects. Roles in sustainable agriculture, energy efficiency, renewable energy, and environmental preservation fall under this category.
- **resilient Infrastructure:** Building robust and sustainable infrastructure is a key component of low-carbon development. This includes resilient infrastructure against climate change, such as green urban design, efficient public transit, and climate-resilient structures.
- **Conservation of Biodiversity:** Low-carbon development aids in the preservation of biodiversity by reducing environmental damage and protecting natural ecosystems. Maintaining ecological resilience and balance requires ecosystem protection.
- **Equity in Social and Economic Domains:** Low-carbon development encourages

11 <https://www.un.org/sustainabledevelopment/energy>

equitable and sustainable growth intending to address social and economic inequality. It highlights how crucial it is to guarantee that the advantages of progress are distributed fairly among various societal groups.

- **Global Collaboration:** International collaboration is necessary to make the shift to low-carbon development. Countries, corporations, and associations work together on technology transfer, policy creation, and research to hasten the global adoption of sustainable practices.
- **Adapting to Climate change:** Adaptation techniques for the inevitable effects of climate change are frequently part of low-carbon development programs. This entails strengthening the resilience of sectors and communities that are susceptible to extreme weather events and other climate-related problems.
- **Competitiveness in the Market:** Companies can increase their competitiveness in a shifting global market by implementing low-carbon strategies. Sustainable and ecologically conscious company practices are becoming more and more valued by stakeholders, investors, and customers.

A comprehensive and forward-thinking strategy, low-carbon development addresses the interconnected problems of environmental sustainability, socioeconomic well-being, and climate change. Societies may create a future that is more fair, resilient, and sustainable by adopting this approach.

## Exercise



1. What are some of the main challenges in achieving international cooperation and coordination on climate change?
2. How do regulatory and policy barriers impede the swift adoption of sustainable practices?
3. In what ways can budgetary obstacles affect the transition to a low-carbon economy, and what are some potential solutions?
4. What role does public awareness play in climate action, and why is it crucial for mobilizing support for climate policies?
5. How can technological progress, such as advancements in energy storage and carbon capture, contribute to mitigating the effects of climate change?
6. What are the key aspects of low-carbon development, and how do they contribute to climate change mitigation?
7. How does the transition to renewable energy support the goals of low-carbon development and reduce greenhouse gas emissions?
8. In what ways does low-carbon development promote energy efficiency in transportation, buildings, and industries?
9. How does low-carbon development contribute to the creation of green employment and new business opportunities?
10. Why is international collaboration essential for the transition to low-carbon development, and what are some areas where countries can work together?

## Unit 1.10 International Mechanism that supports Low-Carbon Development

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. Understanding International Agreements and Treaties
2. Exploring Climate Change Mitigation Strategies
3. Analysing Carbon Pricing Mechanisms

Low-carbon development is aided by a number of significant international mechanisms that facilitate technology transfer, offer financial support, and encourage sustainable behaviors. These systems are essential for allocating funds and aiding nations in their efforts to shift to low-carbon and climate-resilient development paths. The following are a few noteworthy global mechanisms:

- **The Green Climate Fund (GCF):** Under the United Nations Framework Convention on Climate Change (UNFCCC), the GCF is a financing mechanism. Its goal is to aid poor nations in their attempts to mitigate the effects of and prepare for climate change. To finance initiatives and programs that support resilience and low-carbon development, the GCF offers grants, equity, and concessional loans.
- **The Global Environment Facility (GEF):** The Global Environment Fund (GEF) is an alliance of numerous worldwide organizations that offers funding for initiatives on land degradation, sustainable development, mitigation and adaptation of climate change, and biodiversity protection. It serves as a funding source for several environmental agreements, such as the UNFCCC.
- **World Bank Climate Finance:** The World Bank allocates considerable amounts of funds to global climate-related programs<sup>12</sup>. It offers grants, loans, and technical support to help nations execute adaptation and mitigation strategies for climate change. The World Bank Group member International Finance Corporation (IFC) is primarily concerned with private sector participation in climate finance.
- **The Clean Development Mechanism (CDM):** The CDM was crucial in advancing low-carbon development, even if it is now primarily linked to the Kyoto Protocol. Through the Certified Emission Reductions (CERs), which are essentially carbon credits, wealthy nations were able to fund emission reduction initiatives in developing countries as a result to the CDM.
- **The Global Innovation Lab for Climate Finance:** To create and test innovative financial tools and strategies that might encourage private investment for climate-related initiatives, this effort brings together stakeholders from the public and commercial sectors. It seeks to open up new funding sources and close financial shortfalls.
- **Climate Investment Funds (CIF):** The Clean Technology Fund (CTF) and the Scaling up Renewable Energy Program (SREP) are two of the funding programs that make up the Climate Investment Funds (CIF). Through encouraging the use of clean technology and the growth of renewable energy sources, these grants seek to promote climate action in poor nations.

<sup>12</sup> <https://www.worldbank.org/en/news/press-release/2023/12/01/world-bank-group-doubles-down-on-financial-ambition-to-drive-climate-action-and-build-resilience>

## Exercise



1. What is the primary goal of the Green Climate Fund (GCF), and how does it support low-carbon and climate-resilient development in poor nations?
2. How does the Global Environment Facility (GEF) contribute to addressing land degradation, climate change, and biodiversity protection?
3. In what ways does the World Bank Climate Finance support countries in implementing adaptation and mitigation strategies for climate change?
4. What role did the Clean Development Mechanism (CDM) play in low-carbon development, and how did it facilitate emission reduction efforts in developing countries?
5. How does the Global Innovation Lab for Climate Finance aim to enhance private investment in climate-related initiatives, and what types of financial tools and strategies does it develop?

## Unit 1.11. The role and landscape of climate finance

### Unit objectives



After reading this unit, you will be able

1. Understanding Climate Finance
2. Exploring Funding Sources
3. Examining International Frameworks
4. Assessing Challenges and Barriers

Climate finance is crucial in solving the worldwide difficulties caused by climate change. It entails raising funds and allocating it to programs and projects that help with resilience-building, adaptation, and climate mitigation. Climate financing is a multifaceted field with a wide range of sources, procedures, and institutions. An outline of the climate finance landscape and its role is provided below

### 1.11.1 The role of climate finance

- **Mitigation:** Initiatives and programs targeted at lowering or avoiding greenhouse gas emissions are supported by climate funding. This covers spending on energy-efficient technologies, sustainable transportation, renewable energy, and other climate change mitigation strategies.
- **Adaptation:** Resources are allocated to initiatives that improve the ecosystem and communities' capacity to adjust to the effects of climate change. This could entail planning for changing weather patterns, managing water resources, and building infrastructure.
- **Building Resilience:** Climate financing helps communities and vulnerable areas become more resilient. This covers things like robust infrastructure, early warning systems, and plans for handling extreme weather.
- **Building Capacity:** Projects aimed at enhancing institutional strength, education, and training receive financial support. This facilitates the efficient use of climate funds and improves national and regional actions regarding climate-related issues.

### 1.11.2 Landscape of climate finance

The intricate and dynamic landscape of financial resources and climate change mitigation strategies. It entails raising funds and utilizing it to support initiatives that reduce greenhouse gas emissions, build resilience, and make it easier to adapt to the effects of climate change. The landscape of climate finance is made up of several players, tools, and projects operating at all scales, from regional to international.

- **Public financing:** Governments in both wealthy and developing countries contribute public funds to climate financing<sup>13</sup>. This covers payments to climate funds and organizations, as well as direct budgetary allotments and foreign aid.

<sup>13</sup> <https://climatepromise.undp.org/news-and-stories/what-climate-finance-and-why-do-we-need-more-it>

- **Private Finance:** By funding climate-related projects, the private sector contributes significantly to climate finance. This covers financial support for energy-efficient technology, sustainable corporate practices, and renewable energy initiatives.
- **Multilateral Development Banks (MDBs):** Organizations that fund climate-related projects financially include the World Bank, Asian Development Bank, and African Development Bank. They mobilize assets from global financial markets and different nations.
- **Bilateral Institutions:** Through bilateral organizations like national development banks and development cooperation agencies, nations contribute to climate funding. They could have certain geopolitical choices and thematic priorities.
- **Climate Funds:** The purpose of specialized climate funds is to route and manage climate financing. Examples of these funds are the Adaptation Fund and the Green Climate Fund (GCF). Both adaptation and mitigation projects are supported by this funding.
- **Carbon Markets:** The buying and selling of emissions reductions is made possible by mechanisms such as carbon trading and offsetting. This market-based strategy offers a source of climate funds as well as incentives for carbon reduction initiatives.
- **Global Climate Finance Architecture:** The Paris Agreement, the UNFCCC Financial Mechanism, and other international collaborations are only a few of the organizations, agreements, and projects that make up the global climate finance system.
- **Non-Governmental Organizations (NGOs):** By utilizing climate finance to assist local communities and achieve sustainable development objectives, NGOs frequently play a critical role in putting climate projects into action on the ground.

### 1.11.3 Limitations of climate finance

While climate finance is critical to solving climate change concerns, the current climate finance ecosystem has significant limitations and challenges. Several of these restrictions consist of:

- **Inadequate Funds:** One of the main obstacles is the shortage of funding for programs related to climate change. The amount of funds needed to properly mitigate and adapt to climate change is greater than what is now available.
- **Absence of Long-Term and Predictable Finance:** For many climate-related projects to be successful, ongoing, long-term funding is necessary. Unfortunately, it is frequently difficult to find stable, long-term funding, which makes it difficult to design and carry out projects that will have a lasting effect.
- **Obtaining Finance:** Access to climate funding may be impeded for developing nations, especially the least developed ones, by a number of obstacles, such as inadequate institutional capacity, convoluted application procedures, and strict qualifying standards.
- **Risk aversion:** Because of the perceived risks associated with uncertainty in policy, technology, and market conditions, private sector institutions, like banks and investors, might be hesitant to participate in climate-related projects.
- **Fragmented and Complex Financing Architecture:** Numerous funds, mechanisms, and projects with various goals, standards, and protocols make up the climate financing landscape. Complexity like this can make it difficult to coordinate, align, and be effective.
- **Restricted Resources for Adaptation:** The majority of climate money is frequently channelled toward mitigation programs, leaving adaptation efforts, particularly in sensitive areas, with little resources. The efforts made to increase resilience may be hampered by this imbalance.
- **High Transaction Costs:** There may be significant transaction costs associated with obtaining and allocating climate funds, such as transaction fees, project planning costs, and administrative expenditures. Potential donors may be discouraged by these

expenses, particularly for smaller projects.

- **Inadequate Reporting and Measurement:** Difficulties in precisely calculating and disclosing the effects of initiatives that get funding may make it difficult to evaluate the efficacy of climate finance. Mechanisms for reporting that are transparent and standardized are necessary for learning and accountability.

### LANDSCAPE OF CLIMATE FINANCE IN 2021/2022

Global climate finance flows along their life cycle in 2021 and 2022. Values are averages of two years' data to smooth out fluctuations, in USD billions.

#### SOURCES AND INTERMEDIARIES

Which type of organizations are sources or intermediaries of capital for climate finance?

Governments \$100  
National DFIs \$238  
Bilateral DFIs \$33  
Multilateral DFIs \$93  
Multilateral climate funds \$3  
SOEs \$110  
State-owned FIs \$61  
Other \$15

Commercial FIs \$235

Households/individuals \$184

Corporations \$192

Public Private

#### INSTRUMENTS

What mix of financial instruments is used?

Grants \$69

Low-cost project debt \$76

Project-level market rate debt \$561

Project-level equity \$54

Unknown \$7

Debt \$129

Balance sheet financing \$498

Equity \$368

1.27 TRILLION USD ANNUAL AVERAGE

#### USES

What types of activities are financed?

Adaptation \$63

Dual benefits \$51

Mitigation \$1,150

#### SECTORS

What is the finance used for?

Water & waste \$67

Industry \$9

Buildings & infrastructure \$240

Others & cross-sectoral \$50

AFOLU \$43

Energy systems \$515

Transport \$336

"Other" public sources include export credit agencies and unknown public funds

"Other" private sources include institutional investors, funds, and unknown

"AFOLU" stands for agriculture, forestry, other land use, and fisheries. "Others & cross-sectoral" includes \$6bn unknown

Source: Climate Policy Initiative

• **Political and Economic Challenges:** The availability and sustainability of climate funding may be impacted by political and economic unpredictability, shifts in governmental objectives, and conflicting national interests.

## Exercise



1. What are the primary roles of climate finance in addressing climate change, and how does it contribute to mitigation, adaptation, and resilience-building?
2. How does the landscape of climate finance differ between public financing, private finance, and multilateral development banks?
3. What are some of the limitations of climate finance, including issues related to funding adequacy, access, and transaction costs?
4. How do specialized climate funds, such as the Green Climate Fund (GCF) and the Adaptation Fund, contribute to managing climate finance?
5. What challenges might developing countries face in accessing climate finance, and how can these challenges be addressed?
6. In what ways do carbon markets and NGOs play a role in the global climate finance architecture?

## Unit 1.12 Important sectors and industries for adopting low carbon action and various measures

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. Which sector can adopt low-carbon development?
2. How the implementations are possible?

Adopting low-carbon practices to reduce climate change is a critical responsibility for several sectors and industries. The following significant industries and the several steps they can take to lessen their carbon footprint are listed:

#### Energy sector

- **Utilization of Renewable Energy:** Boost the proportion of renewable energy sources in the energy mix, such as geothermal, hydro, wind, and solar.
- **Energy efficiency:** Use techniques and technology to improve the energy efficiency of power production, transmission, and consumption.
- **Transition from Fossil fuel:** gradual substitution of greener fuels for coal and other fossil fuels with significant emissions.

#### Transportation Sector

- **Electrification of Vehicles:** To decrease the reliance on conventional fossil fuel vehicles, encourage the use of electric vehicles (EVs) and make investments in the infrastructure needed for charging them.
- **Public Transportation and Active Mobility:** Encourage the use of public transportation, cycling, and walking to reduce the carbon footprint of commuting.
- **Fuel Efficiency Standards:** Implement and enforce fuel efficiency standards for vehicles to reduce emissions from the transportation sector.

#### Industry Sector

- **Energy Efficiency Measures:** Improve energy efficiency in industrial processes through technological upgrades, waste heat recovery, and optimized production methods.
- **Carbon Capture and Storage (CCS):** Implement CCS technologies to capture and store carbon emissions from industrial processes.
- **Transition to Low-Carbon Materials:** Develop and use low-carbon materials and sustainable manufacturing practices.

#### Agriculture and forestry

- **Sustainable Agriculture:** To cut emissions from agriculture, promote agroforestry, precision farming, and sustainable land management techniques.
- **Reforestation and afforestation:** In order to absorb carbon dioxide from the atmosphere, reforestation and afforestation should be prioritized.
- **Manure Management:** Adopt better methods for managing manure to cut down on methane emissions from cattle, and enforce fuel economy requirements for cars to cut down on emissions from the transportation sector.

**Buildings and Construction**

- Energy-Efficient Buildings: Retrofit existing buildings and use energy-efficient design concepts and technologies for new construction.
- Renewable Energy Integration: To produce clean energy, install on-site renewable energy technologies, such as solar panels on buildings.
- Green Building requirements: To guarantee sustainability in construction, adhere to and put into practice green building certification requirements.

**Waste Management**

- Waste Reduction and Recycling: To reduce the amount of greenhouse gas emissions from landfills, put waste reduction measures into place and improve recycling operations.
- Trash-to-Energy: To lessen dependency on fossil fuels, investigate and invest in technologies that turn trash into energy.

**Financial Sector**

- Sustainable Finance: Promote ethical investing practices, green bonds, and sustainable loans as means of attracting capital to low-carbon and sustainable initiatives.
- Climate Risk Assessment: To encourage resilience and sustainable investment, incorporate climate risk assessments into financial decision-making.

**Healthcare Industry**

- Green healthcare facilities are those that are built and run with an emphasis on sustainable practices, waste minimization, and energy efficiency.
- Adopt sustainable buying techniques to reduce the negative environmental effects of medical equipment and supplies.

## Exercise



1. What are some key strategies for reducing the carbon footprint in the energy sector, and how do they contribute to climate change mitigation?
2. How can the transportation sector contribute to lowering carbon emissions through the adoption of electric vehicles and improvements in public transportation?
3. What measures can the industry sector take to enhance energy efficiency and reduce carbon emissions, including the use of carbon capture and storage (CCS)?
4. In what ways can agriculture and forestry practices, such as reforestation and sustainable agriculture, help in reducing greenhouse gas emissions?
5. How can the buildings and construction sector incorporate energy-efficient technologies and renewable energy integration to minimize its environmental impact?

## Unit 1.13. The necessity of incorporating climate change action into national planning procedure

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. How nations implement an action plan
2. How the implementation is possible

Considering the interdependence of climate change with several facets of society, the economy, and the environment, climate change must be taken into account in national planning procedures for multiple reasons. The following is a summary of why this integration is necessary:

- **Preparedness and Risk Reduction:** Storms, floods, and wildfires are examples of extreme weather occurrences that are becoming more often and intense due to climate change. By including climate change in national planning, governments can better prepare for disasters by identifying risk areas and putting risk-reduction measures in place.
- **Stability and Resilience of the Economy:** Ecosystem disruption can result from climate-related factors such as water scarcity, crop failures, and extreme weather damage. Planning with climate change in mind promotes stability and reduces the financial impact of climate-related disasters by assisting countries in creating resilient economic plans.
- **Planning and Developing Infrastructure:** Increased temperatures, altered precipitation patterns, and increasing sea levels are all effects of climate change on infrastructure. To develop and build resilient infrastructure that can survive the effects of a changing climate, lowering the risk of damage and guaranteeing lifespan, national planning must take climate change into account.
- **Management of Resources:** The distribution and accessibility of natural resources, such as water, energy, and agricultural land, can be impacted by changes in climatic trends. To ensure that these essential resources are used effectively and preserved for future generations, national planning must consider sustainable resource management.
- **Preservation of Biodiversity:** Threats to biodiversity arise from climate change, which affects ecosystems and species worldwide. Planning at the national level should take biodiversity preservation and protection meticulously, as it is crucial for food security, ecosystem services, and general environmental health.
- **International Agreements and Cooperation:** Global action is needed to address the issue of climate change. Planning at the national level that is in line with international accords, like the Paris Agreement, shows a dedication to tackling climate change globally and encourages collaboration with other countries to accomplish common climate objectives.
- **Long-Term Vision and Policy Consistency:** Consistency in policies and regulations is ensured by including climate change in national planning. It assists in establishing a long-term vision for sustainable development by coordinating immediate objectives with the more general goal of reducing the effects of and adjusting to climate change.
- **Social Equity and Vulnerable Communities:** Vulnerable groups are frequently

disproportionately affected by climate change. Planning at the national level should take social equity issues into account by taking into account how they will affect marginalized groups and making sure that policies for adaptation and mitigation put everyone's welfare first.

- **Business and Investor Confidence:** Risks associated with climate change are being taken into account by businesses and investors more often when making decisions. Climate-resilient national planning fosters investor and company confidence in sustainable and climate-resilient projects by creating a stable and predictable regulatory framework.

Building resilience, advancing sustainable development, and ensuring the welfare of present and future generations in the face of climate change all depend on integrating climate change into national planning processes. It is a comprehensive strategy that tackles the various issues raised by climate change and promotes a more resilient and sustainable future.

## Exercise



1. How does integrating climate change into national planning help improve preparedness and risk reduction for extreme weather events?
2. In what ways can national planning that considers climate change contribute to the stability and resilience of the economy?
3. Why is it important to include climate change considerations in the planning and development of infrastructure?
4. How does climate change affect resource management, and why should this be addressed in national planning?
5. What role does national planning play in aligning with international agreements like the Paris Agreement, and how does this impact global cooperation?

## Unit 1.14 Incorporating climate change into national planning procedures

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. Recognizing the Importance of National Planning
2. Analysing Policy and Legal Frameworks
3. Integrating Climate Change Considerations

Planning and action related to climate change in India are carried out by local, regional, and national agencies in different ways. Below are specific examples of the roles that are undertaken by the various levels of governance.

#### Local Organizations: Municipal corporations.

- **Urban Planning for Climate Resilience:** To make sure that cities are planned with climate resilience in mind, municipalities are active in urban planning. This includes initiatives such as green spaces, sustainable waste management, and climate-responsive infrastructure.
- **Public Awareness Campaigns:** Local organizations run campaigns to educate the public about climate change and motivate them to adopt sustainable practices and lower their carbon emissions.

#### Local Self-Government: Panchayats

- **Encouraging Climate-Resilient Agriculture:** Panchayats strive to support sustainable farming practices, encourage water conservation, and put climate-resilient agricultural techniques into action locally.
- **Disaster Preparedness and Response:** To handle climate-related disasters, local entities play a critical role in the development and execution of strategies for disaster preparedness and response.

#### State government's regional institutions:

- **State environmental departments: Development and Implementation of Policies:** State environmental agencies develop and carry out national goals-aligned climate change policies that are adapted to local demands. For instance, specific initiatives for climate resilience are outlined in the State Action Plans on Climate Change (SAPCCs) of several Indian states.
- **Natural Resource Management:** To lessen the effects of climate change, states must manage and protect their natural resources, such as forests and waterways.
- **State Electricity Regulatory Commissions: Encouraging Renewable Energy:** State electricity regulatory commissions are involved in establishing rates and guidelines that encourage the fusion of renewable energy sources with conventional power sources. Supporting wind and solar energy initiatives is part of this.

**National Institutions:****The Ministry of Environment, Forests, and Climate Change (MoEFCC):**

- **Legislation and Policy Framework:** The MoEFCC is in charge of creating national climate change policies and legislation. India's approach to combating climate change is outlined in the National Action Plan on Climate Change (NAPCC), which addresses issues related to energy, water, and agriculture.
- **International Negotiations:** The ministry participates in globally coordinated initiatives by representing India in international climate change negotiations. India is a party to the Paris Agreement, and the MoEFCC is essential to the country meeting its obligations.

**National Disaster Management Authority, (NDMA)**

- **Disaster Risk Reduction:** NDMA focuses on disaster risk management, which is directly related to adapting to climate change. It develops guidelines and strategies aimed at strengthening the nation's ability to withstand calamities caused by climate change.

**Indian Council of Agricultural Research, (ICAR)**

- **Research and Innovation:** ICAR carries out studies to create crop types that are robust to climate change as well as sustainable farming methods. This helps increase the agriculture sector's resistance to the effects of climate change.

**The Solar Energy Corporation of India (SECI) and the National Institute of Wind Energy (NIWE):**

- **Development of Renewable Energy:** The creation and advancement of renewable energy sources is the main emphasis of these organizations. While SECI supports solar power projects, NIWE focuses on wind energy, supporting India's attempts to become less dependent on fossil fuels.

## Exercise



1. What are some specific climate resilience measures that municipal corporations in India undertake in their urban planning?
2. How do Panchayats contribute to climate-resilient agriculture and disaster preparedness at the local level?
3. What roles do state environmental departments play in managing natural resources and implementing climate change policies?



## 2. Greening Businesses through Aligning Policy and Action.

- UNIT 2.1 Identifying global programs that assist nations in preparing for climate change.
- UNIT 2.2 Important components of the 2030 Agenda for Sustainable Development and the Paris Agreement on Climate Change, and understand how climate action fits into the larger context of sustainable development.
- UNIT 2.3 Greenhouse Gas (GHG) emissions accounting and identifying the data needed to conduct a greenhouse gas inventory for various sectors, such as corporate, finance, renewable energy, etc.
- UNIT 2.4 The factors that make data sources suitable for the inventory, include accuracy, reliability, and relevance.
- UNIT 2.5 Approaches for gathering existing data, generating new data, and adapting data to meet the needs of the inventory





## Chapter Highlights

**After reading this chapter, you will be able to:**

1. The global landscape of climate change preparedness programs.
2. The significance of the 2030 Agenda for Sustainable Development and the Paris Agreement.
3. The practical aspects of GHG emissions accounting and inventory creation in diverse sectors.
4. Criteria for selecting suitable and reliable data sources for emissions assessments.
5. Various approaches for collecting, generating, and adapting data

## Unit 2.1 Identifying global programs that assist nations in preparing for climate change.

### Unit objectives

After reading this unit, you will be able to:



1. Demonstrate Knowledge of Key Climate Change Preparedness Programs
2. Analyse the Impact and Effectiveness of Climate Change Programs
3. Assess the Importance of International Collaboration
4. Critically Evaluate Challenges and Opportunities
5. Synthesize Information for Informed Decision-Making
6. Connect Global Programs to National Climate Policies

### 2.1.1 What are the global Programs to assist nations?

Understanding and recognizing worldwide initiatives and activities aimed at assisting countries in preparing for the problems posed by climate change. It involves exploring and recognizing numerous worldwide initiatives, agreements, and collaborations that give nations with support, resources, and frameworks for dealing with and adapting to the effects of climate change.

#### Nature-Based Solutions for Climate Resilience

- Potential components of nature-based solutions include ecosystem restoration, green infrastructure, biodiversity conservation, natural carbon sequestration, climate-resilient agriculture, coastal protection and restoration, community-based adaptation, policy and governance, and natural ecosystems to address environmental challenges, including climate change.
- Research and assessment, partnerships and collaboration, capacity building, monitoring and evaluation, climate funding, and knowledge sharing are some of the implementation tactics. By strengthening adaptation and mitigation efforts, these strategies seek to increase resilience to the effects of climate change.
- The incorporation of these solutions into policies, practices, and landscapes can enhance the resilience of communities and ecosystems against the adverse effects of climate change.

#### Supporting Agency NBS

- **UNDP:** Assists nations in adopting plans for mitigating and adapting to climate change that rely on natural solutions.
- **UNEP:** Strengthens nations' ability to embrace natural solutions, especially for biodiversity preservation and ecosystem restoration.
- **GEF:** Offers funding to initiatives that use natural solutions to address environmental sustainability and climate resilience.
- **World Bank Group:** Assists nations in incorporating environmentally friendly solutions into their development initiatives.
- **The Green Climate Fund (GCF):** provides funding for initiatives that prioritize mitigation and adaptation strategies for climate resilience, with a focus on nature-based solutions.
- **The International Union for Conservation of Nature (IUCN):** promotes the conservation and restoration of ecosystems and encourages the inclusion of nature-based solutions in

- climate plans.
- **World Wildlife Fund (WWF):** Focusing on sustainable land use and conservation methods, the WWF collaborates with nations to develop and implement nature-based solutions.
- **United Nations Food and Agriculture Organization (FAO):** Helps countries embrace natural solutions for forestry and agriculture.
- **The Nature Conservancy:** Assists nations in creating and putting into practice environmentally friendly solutions for climate resilience.

### The Global Climate Resilience Program

Enhances ecological and community resilience to the effects of global climate change.

#### The Key components are

- Ecosystem Restoration: Consists of planting new trees, afforestation, and sustainable land use.
- Green infrastructure: Integrates landscape features into infrastructure and urban design<sup>1</sup>.
- Conservation of Biodiversity: Preserves and expands biodiversity.
- Natural Carbon Sequestration: Strengthens carbon dioxide absorption and storage capabilities of natural systems.
- Climate-Resilient Agriculture: Encourages environmentally friendly farming methods.
- Coastal Protection and Restoration: To protect against storm surges and sea level rise, natural coastal ecosystems are tapped into.
- Community-Based Adaptation: Promotes sustainable use of natural resources to increase community resilience.
- Governance and Policy: Incorporates environmentally friendly solutions into regional and national policies.

#### Implementation Techniques

- Research and Evaluation: carries out in-depth evaluations.
- Capacity Building: Increases the ability of organizations, governments, and local communities.
- Partnerships and Collaboration: Promotes cooperation amongst local communities, businesses, NGOs, and governments.
- Monitoring and Evaluation: Creates strong systems for monitoring and evaluating.
- Climate Finance: Arrange financial resources from global sources.

#### Agency's supporting Climate change Mitigation

- Global Initiative for Climate Change (IPAC): Facilitates nations' efforts to achieve net-zero greenhouse gas emissions and a more robust economy by 2050.
- CGI: A membership-based organization, the Global Climate Resilience Fund (CGI) unites executives from the public, private, and nonprofit domains to promote action on global issues.
- IFRC: For many years, the organization has helped people all around the world prevent and lessen the effects of climate-related disasters.<sup>2</sup>

The goal of strengthening capacity for effective adaptation is supported by a series of initiatives enveloped by the International Panel on Deltas and Coastal Areas (IPDC).

<sup>1</sup> <https://citygreen.com/green-infrastructure-in-urban-planning>

<sup>2</sup> <https://www.ifrc.org/our-work/disasters-climate-and-crises/climate-smart-disaster-risk-reduction/global-climate>

### **International Program for Action on Climate (IPAC)**

The International Program for Action on Climate (IPAC) was launched at the OECD Ministerial Council Meeting in May 2021 to support countries' efforts towards net-zero GHG emissions and a more resilient economy by 2050. The IPAC provides fact-based operational tools with policy advice and indicators as a cornerstone. The IPAC Technical Expert Group focuses on the conceptual framework, selection of IPAC indicators, guidance on the use and interpretation of indicators, and methodological and measurement priorities.

The IPAC has selection criteria that are flexible, pragmatic, balanced, representative, clear, easy to communicate, actionable, and cost-effective. The IPAC also offers a preliminary IPAC dashboard, which provides an overview of countries' progress and trajectories towards carbon neutrality. The dashboard is linked to benchmarks and targets when possible, and the IPAC is also working on proposals for indicator development.

The IPAC is open to OECD members and selected non-members and is part of the Horizontal Project on Building Climate and Economic Resilience.<sup>3</sup>

### **The United Nations Environment Program (UNEP)**

Through its numerous partnerships, programs, and initiatives, the United Nations Environment Program (UNEP) plays a significant role in international efforts to combat climate change. It backs climate funding, regular Global Environment Outlook (GEO) reports, capacity building, ecosystem management and biodiversity protection, climate science and research, and the implementation of international environmental accords. To promote alliances and cooperative projects, it additionally works with governments, NGOs, international organizations, and the private sector. Recognizing the connection between climate change and heightened vulnerability to natural disasters, UNEP also tackles disaster risk reduction. The organization's responsibilities also include catastrophe risk reduction, partnerships, and climate finance in addition to the "Global Program on Climate Change".

### **International Panel on Deltas and Coastal Areas (IPDC)**

The UN has called for coordinated actions for global climate resilience, particularly in deltas, coastal areas, and small islands. The Netherlands Ministry of Infrastructure and Water Management, Deltares, the Global Center on Adaptation, and the Delta Alliance have launched the International Panel for Deltas, Coastal Areas and Islands (IPDC) to support countries and islands in developing climate adaptation strategies and plans. The IPDC aims to leverage multi-actor collaboration by connecting policy, finance, science, and implementation layers. The expected impact is increased engagement, tailored knowledge agenda, accelerated financing, and increased capacity for planning and implementation. This will lead to a resilient society and a sustainable environment.

### **Climate Change Program (CCP)**

A vital part of UNITAR's efforts to combat climate change is the Climate Change Program (CCP), which aims to increase capacity to deal with the effects of climate change at the national and regional levels. To create resource materials such as training modules and climate screening tools, networks are established with international organizations and centers of excellence. Workshops and fellowships subsequently serve to circulate these outputs among the various countries. The partners of the program collaborate at the national level to enhance and develop capacity for climate change that is reflected in the CCP network. The Global Climate Change Adaptation network, led by the United Nations Environment Program (UNEP), is an illustration of a worldwide network that strives to improve institutional, scientific, and technical capacity for adaptation in a coordinated and cohesive way.

<sup>3</sup> <https://www.oecd.org/en/about/programmes/net-zero-climate-and-economic-resilience-in-a-changing-world.html>

### Exercise



1. What are some of the key components of nature-based solutions for climate resilience?
2. Which organizations support the implementation of nature-based solutions, and what are their specific roles?
3. What are the main components of the Global Climate Resilience Program, and how do they contribute to enhancing climate resilience?
4. What is the purpose of the International Program for Action on Climate (IPAC), and what tools and indicators does it provide to support countries' climate goals?
5. How does the International Panel on Deltas and Coastal Areas (IPDC) aim to support countries in developing climate adaptation strategies for vulnerable regions?

## Unit 2.2 The 2030 Agenda for Sustainable Development and the Paris Agreement on Climate Change

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Will equip with 2030 Agenda and Paris Agreement Components
2. Identify and articulate key components.
3. Demonstrate an understanding of climate action's integration into sustainable development.
4. Foster a holistic perspective on global environmental and developmental challenges.

### 2.2.1 Important component of the 2030 agenda for sustainable development

In September 2015, all United Nations Member States endorsed the global program known as the 2030 Agenda for Sustainable Development. It lays out a thorough and challenging plan for tackling global issues and promoting sustainable development in all of its forms—economic, social, and environmental. The Sustainable Development Goals (SDGs), which are composed of 17 goals and 169 targets and are intended to be accomplished by 2030, are the focal point of the agenda<sup>4</sup>.

The following are some of the 2030 Agenda for Sustainable Development's main attributes:

Sustainable Development Goals (SDGs)

The 17 Sustainable Development Goals (SDGs) on the agenda each focus on a different global issue, such as gender equality, clean water, poverty, hunger, health, education, and climate action.



2030 Agenda for Sustainable Development Fig 2.1 1

image source <https://unosd.un.org>

<sup>4</sup> <https://sdgs.un.org/2030agenda>

### **Universal Application**

The objectives are global, encompassing all nations, irrespective of their degree of development. It acknowledges that issues related to sustainable development are interrelated and impact both industrialized and developing countries.

### **Leave No One Behind**

The agenda's core tenet, "leave no one behind," emphasizes the commitment to making sure that the advantages of development benefit all groups in society, particularly the most marginalized and disadvantaged members<sup>5</sup>.

### **Integrated Approach**

The agenda acknowledges how several facets of sustainable development are interconnected. It encourages an integrated strategy that views the social, cultural, and environmental facets as interrelated and supportive of one another.

### **Nationally Determined**

The agenda is voluntary to implement, and each nation is invited to customize its aims and tactics to fit the global goals into its unique national environment. The objectives are made to be context- and country-specific.

### **Collaboration & Partnerships**

The significance of partnerships and collaborations at the national, regional, and international levels is emphasized in Goal 17. To achieve sustainable development, it promotes collaboration between governments, the corporate sector, civil society, and international organizations.

### **Periodic Reporting and Review**

The agenda stipulates that progress toward reaching the SDGs at the national and international levels must be regularly reviewed and reported on. This facilitates the evaluation of the efficacy of tactics and interventions and permits modifications as necessary.



Scan to learn more about SDG 2030

<sup>5</sup> <https://www.undp.org/sites/g/files/zskgke326/files/publications/Ageing%20and%20the%20SDGs%20Issue%20Brief.pdf>

### 2.2.2 The Paris Agreement on climate change

To combat climate change and slow global warming, 195 (out of 198) member nations came together to form the historic Paris Agreement. It was approved on December 12, 2015, during the United Nations Framework Convention on Climate Change (UNFCCC) 21st Conference of the Parties (COP 21), which took place in Paris, France. On November 4, 2016, the agreement was put into effect.

#### Important elements of the Paris Agreement consist of

- **Limiting Global Warming:** The main goal is to keep the rise in the average global temperature well below 2 degrees Celsius over pre-industrial levels, with efforts focused on keeping the rise to 1.5 degrees Celsius. Reducing the likelihood of severe climate impacts is the goal of this more aggressive target.<sup>6</sup>
- **Nationally Determined Contributions (NDCs):** Each participating nation provides an NDC detailing its adaptation plans, mitigation objectives, and climate action plans. The determination of these contributions is meant to be at the national level, taking into account the unique conditions and capacities of each nation.
- **Global Stocktake:** A system for periodically evaluating the group's progress toward the objectives of the agreement. This entails assessing the NDCs of each nation as well as the overall success of international efforts to combat climate change.
- **Financial Assistance:** Developed nations are urged to contribute financial resources to assist developing nations in adjusting to the effects of climate change and making the transition to low-carbon, resilient economies. By 2020, the target is to raise \$100 billion yearly, and beyond that time, more funding will be provided.
- **Adaptation, Loss, and Damage:** Acknowledging the necessity of addressing loss and damage brought on by the effects of climate change, especially in vulnerable nations, as well as strengthening adaptive ability and resilience.
- **Accountability and Transparency:** To guarantee that nations are held responsible for their climate-related actions, a framework for accountability, reporting, and transparency has to be established.
- **Transfer of Technology:** Encouraging the creation and dissemination of environmentally friendly technologies to assist nations in making the shift to sustainable development paths.
- **Capacity building:** Developing nations' ability to take part in climate action and adjust to its effects.
- **Market Mechanisms:** Creating a system for sustainable development and using market-driven strategies, such as carbon trading, to aid in mitigation efforts.

### 2.2.3 How climate action fits into the larger context of sustainable development.

Since both sustainable development and climate action have as their main objectives improving human well-being, preserving the environment, and guaranteeing prosperity for present and future generations, they are inextricably linked. The following is how addressing climate change fits within the larger concept of sustainable development:

#### Linked Objectives:

Climate action and the Sustainable Development Goals (SDGs) are related and reinforce overlap. Several SDGs, including those about poverty, health, education, and economic growth, may be more difficult to achieve as a result of climate change effects, such as extreme weather events and increasing sea levels.

#### Sustainability of the Environment:

With the reduction of greenhouse gas emissions, the advancement of renewable energy, and the preservation of ecosystems, climate action tackles the environmental aspect of sustainability. In line with the SDGs relevant to life both land and below water, this enhances the planet's general resilience and health.

<sup>6</sup> <https://unfccc.int/process-and-meetings/the-paris-agreement>

### **Social Justice and Inclusivity**

Social justice and inclusivity are important, and climate action highlights this. To ensure that no one is left behind, vulnerable communities—which are frequently disproportionately affected by climate change—are the focus of initiatives related to both sustainable development and climate resilience (SDG 10: Reduced Inequalities).

### **Financial Sturdiness**

SDG 8: Decent Work and Economic Growth states that implementing sustainable practices and constructing climate resilience enhances economic resilience. Energy-efficient, green technology and renewable energy investments boost employment and the economy while lowering reliance on fossil fuels.

### **Well-being and Health**

By lowering air pollution, stopping the development of diseases caused by climate variables, and promoting improved living conditions, climate action has a beneficial impact on human health (SDG 3: Good Health and Well-Being).

### **Transition to Clean Energy**

A crucial component of both climate action and sustainable development is the switch to clean and sustainable energy sources. For economic growth, environmental sustainability, and poverty reduction, access to inexpensive and clean energy is essential (SDG 7).

### **Conscientious Production and Consumption**

By encouraging eco-friendly practices in businesses and communities, decreasing waste, and promoting sustainable resource management, climate action is consistent with the objective of responsible consumption and production (SDG 12).

### **Global Collaborations**

Global collaboration and partnerships are critical to both sustainable development and climate action (SDG 17). Sharing information, resources, and technology at the local, national, and international levels is necessary to combat climate change.

### **Adaptability and resilience**

Adapting to the effects of climate change and enhancing ecological and community resilience are two aspects of climate action. The SDG objective of building sustainable cities and communities (SDG 11) is in line with this resilience.

### **Knowledge and Consciousness**

Education and awareness are given priority in sustainable development and climate action (SDG 4). A deeper comprehension of the connections between the environmental, social, and economic systems is fostered via climate education.



Scan to learn more about the Paris Agreement 2015

## Exercise



1. How does the 2030 Agenda for Sustainable Development ensure that no one is left behind, and what role does this principle play in the implementation of the Sustainable Development Goals (SDGs)?
2. What are the primary goals of the Paris Agreement, and how do Nationally Determined Contributions (NDCs) contribute to these goals?
3. In what ways does climate action intersect with and support the achievement of various Sustainable Development Goals (SDGs), such as those related to health, economic growth, and social justice?
4. How does the Paris Agreement's focus on financial assistance and technology transfer support developing nations in their climate action efforts?
5. What are some specific ways in which climate action contributes to the sustainability of the environment, and how does this align with the objectives of the SDGs related to ecological health and clean energy?

## Unit 2.3 Introduction to Greenhouse Gas Emission Accounting (GHG accounting or Carbon Accounting)

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Understand the Concept of GHG Emissions Accounting
2. Identify GHG Emission Sources Across Different Sectors
3. Learn GHG Inventory Frameworks and Standards
4. Gather Data for GHG Inventory
5. Analyze Data and Calculate GHG Emissions

### 2.3.1 Introduction to Greenhouse Gas Emission Accounting (GHG Accounting)

Reducing greenhouse gas (GHG) emissions is essential to limiting the effects of climate change, which is a major worldwide concern. A methodical technique for quantifying, tracking, and controlling emissions across multiple sectors and industries is known as greenhouse gas emission accounting. It entails calculating and recording emissions from a range of sources, including waste management, manufacturing, transportation, agriculture, and energy generation. This aids in the understanding of the factors causing climate change and the development of focused emission reduction plans by organizations and policymakers.

To create efficient mitigation plans and monitor the advancement of emission reduction objectives, baseline emissions must be established, along with reduction targets. Through the provision of verifiable information about an organization's emissions performance to stakeholders, greenhouse gas emission accounting promotes accountability and transparency. To ensure regulatory compliance, it also supports reporting obligations and regulatory compliance.

Organizations may minimize costs and increase their competitiveness by employing GHG emission accounting to find areas for improvement in efficiency and cost savings. Because customers, investors, and other stakeholders place a higher value on sustainability and environmental responsibility, it also improves the sustainability and reputation of the company.

#### Importance of GHG Accounting

To combat climate change and advance sustainable development, GHG (Greenhouse Gas) accounting is crucial for the following reasons:

- **Understanding the impacts of climate change**

Comprehending the effects of climate change is essential for tackling its obstacles and putting into practice efficient adaptation and mitigation plans. With far-reaching effects on ecosystems, human health, infrastructure, and livelihoods, climate change has an impact on many facets of the environment, society, and economy. When analyzing the effects of climate change, keep the following points in mind:

**Ecosystems:** Temperature and precipitation patterns are altered by climate change, which causes habitat degradation, wildfires, and a loss of ecosystem services.

**Extreme Weather Events:** As a result of climate change, there is an increased risk of property damage, fatalities, and population displacement from extreme weather events.

**Sea Level Rise:** Sea levels rise as a result of melting polar ice caps and glaciers due to the thermal expansion of seawater brought on by rising global temperatures.

**Water Resources:** Droughts, decreased agricultural production, water shortages, and disputes over water resources are all consequences of climate change's impact on water availability and quality.

**Food Security:** By changing growing conditions, upsetting crop cycles, and increasing pests and diseases, climate change has an impact on agricultural output and food security.

**Human Health:** Air pollution, vector-borne infections, heat-related disorders, and other health concerns are made worse by climate change.

**Economic impact:** The economy is greatly affected by climate change, as seen by the destruction of infrastructure, interruptions in the supply chain, higher insurance premiums, and decreased productivity in key sectors.

- **Driving policy and organizational strategy**

Achieving emission reduction targets, promoting sustainability, and successfully addressing climate change all depend on organizational strategy and policy-driven GHG (Greenhouse Gas) accounting. The following are important factors that influence organizational strategy and policy in GHG accounting:

### **Policy Development and Implementation**

1. By enacting laws, rules, and standards, governments have a major influence on GHG accounting.
2. Mandatory reporting requirements, carbon pricing mechanisms, emissions trading schemes, and incentives for emissions reduction projects are a few examples of policies.
3. International GHG accounting standards can be encouraged to be adopted by governments.

### **Setting Emission Reduction Targets**

1. At the national, regional, or sectoral levels, policy frameworks might set aggressive emission reduction targets.
2. Subsidies, tax breaks, or financial initiatives are some of the ways that governments might encourage carbon reduction efforts.
3. Governments and organizations have the option to allocate resources towards capacity building and training programs to augment their proficiency in greenhouse gas accounting techniques, data gathering, and reporting.

### **Stakeholder Engagement and Transparency**

1. Cooperation and involvement with stakeholders are necessary for effective GHG accounting.
2. Transparency in the disclosure of emissions data enhances confidence and trust.

### **Integration with Sustainability Strategies**

1. GHG accounting should be incorporated into corporate responsibility programs and more comprehensive sustainability strategies.

### **Innovation and Continuous Improvement**

2. GHG accounting needs to be seen as a method of iteration that promotes innovation and continuous improvement.

- **Enhancing corporate responsibility**

Organizations must exhibit their dedication to sustainability, environmental stewardship, and social

responsibility by enhancing corporate responsibility through GHG accounting. The following are some significant ways that GHG accounting might improve corporate responsibility

**Measuring and Managing Carbon Footprint:** By quantifying emissions from a variety of sources, including energy consumption, transportation, waste generation, and supply chain activities, greenhouse gas (GHG) accounting enables enterprises to properly evaluate their carbon footprint. Organizations can find possibilities to reduce emissions and decrease their carbon footprint by implementing measures to analyze their influence on the environment.

**Setting Emission Reduction Targets:** Organizations can set research-based emission reduction objectives in line with international agreements like the Paris Agreement and climate science with the help of GHG accounting. Organizations can show that they are proactively tackling climate change and supporting international efforts to keep global warming well below 2°C by pledging to reduce emissions.

**Transparent Reporting and Disclosure:** Corporate accountability is improved and stakeholder trust is cultivated when GHG emissions data is reported transparently, involving consumers, workers, investors, and the general public. Organizations can exhibit transparency and a commitment to corporate responsibility by making emissions data publicly available through voluntary reporting efforts, annual filings, or sustainability reports.

**Enhancing the Sustainability of the Supply Chain:** Greenhouse gas emissions from suppliers and partners are included in the emissions from the full value chain, which transcends organizational boundaries. Organizations can find opportunities to reduce emissions throughout the supply chain and promote sustainable procurement practices by assessing and reporting Scope 3 emissions, or indirect emissions related to the organization's activities, such as purchased goods and services, transportation, and waste disposal.

**Encouraging Innovation and Efficiency:** To lower emissions, GHG accounting promotes innovation as well as the adoption of energy-efficient practices and technologies. Organizations can achieve cost savings, operational efficiency, and competitive advantage by investing in renewable energy sources, optimizing processes, and implementing energy-saving measures by evaluating emissions data and detecting emission hotspots.

**Engaging Stakeholders:** GHG accounting offers a forum for involving stakeholders in environmental projects and encouraging discussion of the potential hazards associated with climate change. Organizations may raise awareness, rally support, and promote a sustainable and accountable culture by integrating staff members, clients, investors, and communities in GHG accounting initiatives.

**Adapting to Regulatory and Market Trends:** GHG accounting assists businesses in staying ahead of market and regulatory developments associated with sustainability and climate change. Organizations can lower compliance risks and improve long-term resilience by proactively measuring and regulating emissions to adjust to changing legislation, investor expectations, and consumer preferences.

### 2.3.2. Principles of GHG accounting

A framework for precisely measuring, reporting, and managing emissions is provided by the principles of greenhouse gas accounting. These guidelines guarantee GHG accounting procedures are reliable, transparent, and consistent. The following are the fundamentals of GHG accounting:

**Relevance:** Emissions that significantly contribute to climate change should be the main focus of GHG accounting. This covers greenhouse gases, including nitrous oxide (N<sub>2</sub>O), carbon dioxide (CO<sub>2</sub>), methane (CH<sub>4</sub>), and fluorinated gases (SF<sub>6</sub>, HFCs, and PFCs). The extent of emissions must be appropriate given the operations, pursuits, and limitations of the organization.

**Completeness:** Within the operational boundaries of the organization, GHG accounting should take into account every relevant emission source and activity. This includes all scopes (1, 2, and 3).

**Accuracy:** To quantify emissions for GHG accounting, accurate and trustworthy data should be used. This entails making sure that emissions calculations are accurate by utilizing the proper emission factors, activity data, and measuring techniques. Procedures for data quality assurance should be put in

place to reduce mistakes and ambiguities in reporting.

**Consistency:** To guarantee the comparability and reliability of emissions data over time and between businesses, GHG accounting should adhere to uniform methodology, protocols, and standards. This involves following globally accepted GHG accounting standards, including the ones created by the ISO 14064 series or the Greenhouse Gas Protocol (GHGP).

**Transparency:** Stakeholders should be informed openly and transparently about GHG accounting. This entails revealing the methodology, assumptions, emissions statistics, and any restrictions or unknowns related to the reporting procedure. Transparency facilitates informed decision-making and increases confidence and trust among stakeholders.

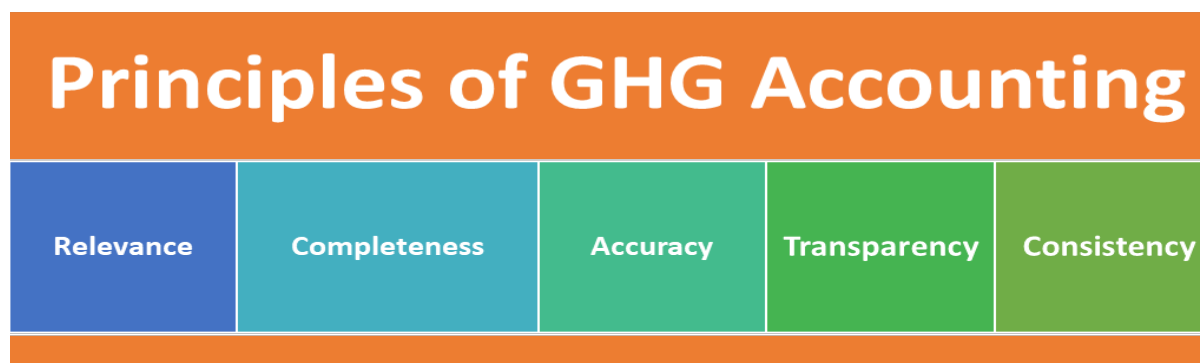


Fig 2.3.1 1 Principles of GHG Accounting

### 2.3.3 Scopes of GHG Accounting

Quantifying and reporting greenhouse gas emissions from diverse operations, activities, or entities is the focus of greenhouse gas (GHG) accounting. Typically, the boundaries and broadness of what is measured and reported are referred to as the scope of GHG accounting.

There are three widely accepted GHG accounting scopes, which are also known as Scope 1, Scope 2, and Scope 3 emissions.

#### Scope 1 Emission



Emissions falling under scope 1 refer to the direct release of greenhouse gases (GHGs) from sources under the ownership or control of the reporting entity. Usually, combustion or chemical reactions that take place on the site produce these pollutants. These emissions are typically measured and reported as part of an organization's GHG inventory, enabling them to understand their direct impact on climate change and identify opportunities for emission reduction and mitigation strategies.

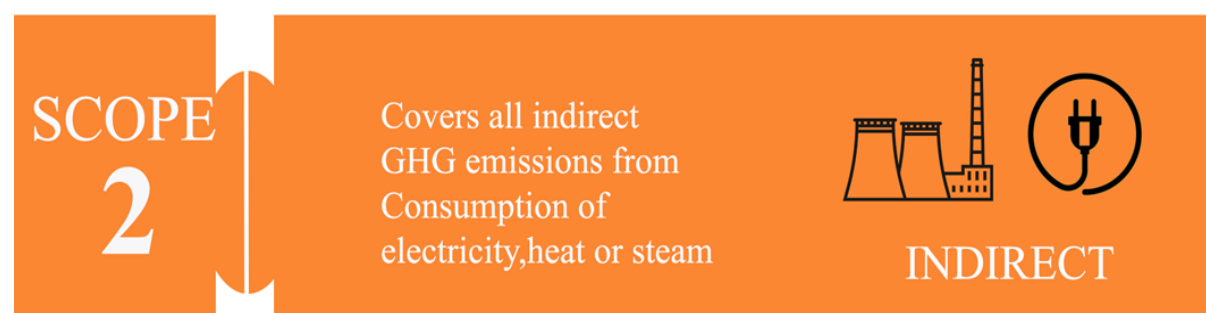
### Typical sources of emissions covered by Scope 1 are:

**Fossil fuel combustion:** This refers to the emissions produced when fuels such as coal, oil, natural gas, diesel, gasoline, and coal are burned to generate electricity, heat, and run on-site machinery or cars.

**Process Emissions:** Some industrial processes have the potential to emit greenhouse gases into the atmosphere. Chemical reactions that result in the creation of chemicals, steel, or cement could be the source of these emissions.

For example, in its Scope 1 emissions inventory, a manufacturing facility would list emissions from its vehicles, furnaces, boilers, and other combustion sources on site. The facility can reduce its environmental effect by investing in technologies to lower emissions, switching to cleaner fuels, or improving energy efficiency by measuring and monitoring these emissions over time.

## Scope 2 Emissions

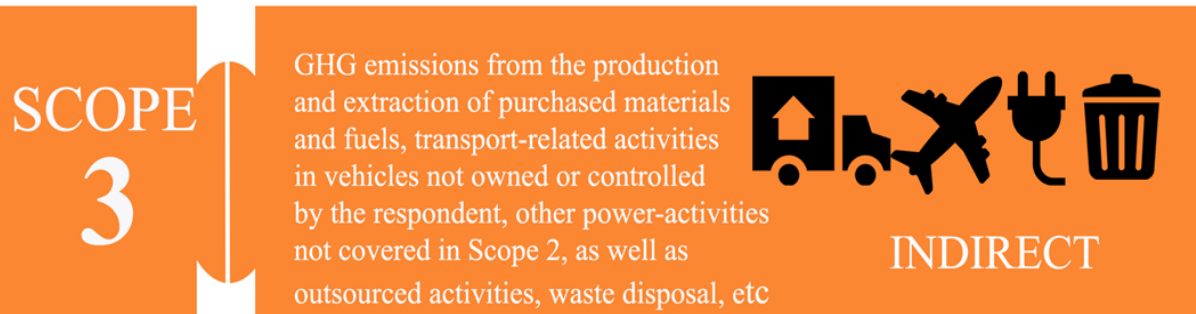


Indirect greenhouse gas (GHG) emissions related to the reporting entity's use of purchased or acquired energy, heat, steam, or cooling are referred to as scope 2 emissions. While Scope 2 emissions originate from energy generated off-site, they are nonetheless related to the organization's operations, in contrast to Scope 1 emissions, which are directly produced from sources that the organization owns or controls. Organizations can better understand the indirect impact of their energy use and find ways to cut emissions through energy-efficient practices, the purchase of renewable energy, or other sustainability projects by accounting for Scope 2 emissions. This supports organizations' attempts to mitigate climate change and address their carbon impact sustainably.

**Energy Use:** Using electricity is the most frequent cause of emissions covered by Scope 2. The emissions involved in producing the electricity that an organization purchases from the grid are produced at the power plants that feed the grid, which are usually situated off-site. These emissions are regarded as indirect as the organization's energy use causes them rather than the organization producing them directly.

**Heat, Steam, and Cooling:** Emissions related to the use of other energy sources, such as heat, steam, or cooling, may be included in Scope 2 emissions in addition to emissions related to electricity. These could be the steam produced off-site and brought to the company, the emissions from district heating systems, or the chilled water used in air conditioning.

## Scope 3 Emission



All additional indirect greenhouse gas (GHG) emissions that arise from the reporting entity's operations but are not under its direct ownership or control are included in Scope 3 emissions. These emissions happen at every stage of the organization's value chain, from upstream to downstream operations. Comprehending and mitigating Scope 3 emissions is vital for firms to conduct a thorough evaluation and oversee their environmental impact and carbon footprint. Organizations can improve their sustainability performance, reduce their environmental impact, fortify the resilience of their supply chain, and satisfy stakeholder expectations for transparency and accountability in the fight against climate change by measuring, disclosing, and actively controlling their Scope 3 emissions.

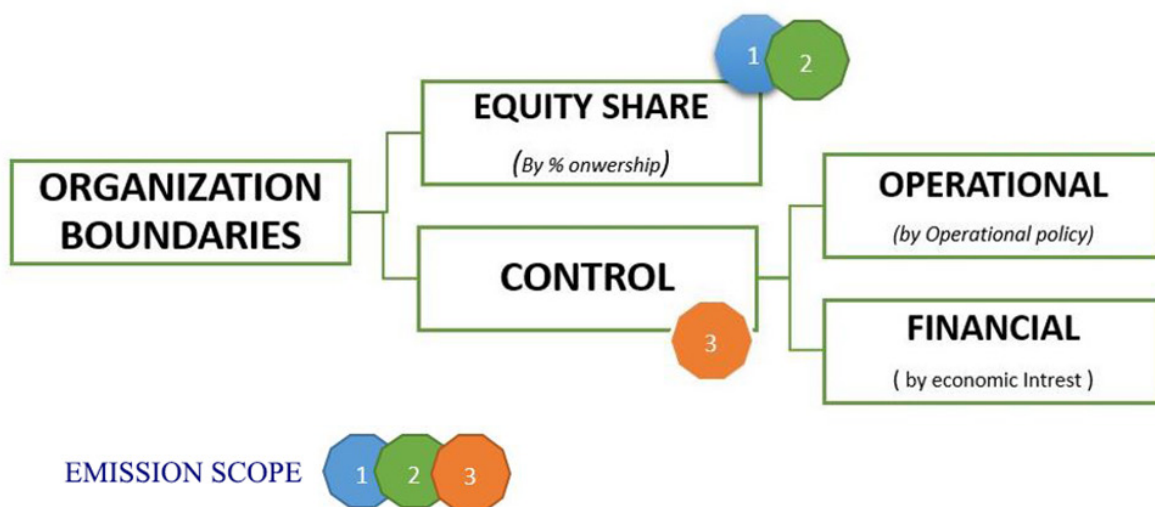
To better understand and manage scope 3 emissions, they are frequently categorized into several categories or subcategories. There are 15 types of Scope 3 emissions according to the Greenhouse Gas Protocol, a commonly used framework for GHG accounting. They are

- Purchased goods and services
- Capital goods
- Fuel- and energy-related activities not included in Scope 1 or Scope 2
- Upstream transportation and distribution
- Waste generated in operations
- Business travel
- Employee commuting
- Transportation and distribution (downstream)
- Processing of sold products
- Use of sold products
- End-of-life treatment of sold products
- Franchises
- Leased assets
- Investments
- Downstream leased assets

Because of their larger scope and the involvement of several stakeholders throughout the value chain, scope 3 emissions can be more difficult to quantify and manage than Scope 1 and Scope 2 emissions. It can be challenging for organizations to gather information, set limits, and assign blame for emissions that are outside of their direct control.

### 2.3.4 GHG accounting methods

Greenhouse gas (GHG) accounting methods are frameworks and procedures that businesses use to measure, monitor, and control their greenhouse gas emissions. These techniques offer a methodical approach to tracking environmental objective progress, baseline setting, reduction target setting, and emission measurement. The choice of GHG accounting technique is influenced by several variables, including stakeholder expectations, industry standards, legal requirements, and organizational goals. Organizations can improve environmental performance, increase transparency, and support international efforts to reduce climate change by putting strong GHG accounting processes into practice.



## Organization Boundaries

Determining organizational boundaries is a crucial stage in greenhouse gas (GHG) accounting since it determines the range of emissions that must be included in the inventory. The boundaries specify which emissions sources and activities are included in the accounting process because they are considered to be under the organization's control or influence. Generally speaking, organizational boundaries fall into one of three categories

### 1. Operational Control Approach

The ability of an organization to plan and oversee its operations, which may lead to the production of greenhouse gas emissions, is referred to as operational control. It entails having the power to decide how resources are used, such as how energy and fossil fuels are used, how machinery and equipment are run, and whether or not to undertake procedures that could release greenhouse gases.

**Range:** Scope 1 emissions are GHG emissions from sources that are under the operational control boundaries of an organization. The organization owns or controls the equipment, activities, or processes that are the direct source of these emissions.

**Reporting:** Organizations that perform greenhouse gas accounting identify and measure Scope 1 emissions from sources directly under their operational control. Compute emissions using emission factors or other calculation approaches, this entails gathering data on fuel use, production activities, equipment performance, and other pertinent aspects.

**Responsibility:** Since the organization owns or controls the sources of Scope 1 emissions, its operations are directly responsible for them. This puts the responsibility of monitoring, controlling, and reducing emissions on the organization whenever it is practical. Common solutions for lowering Scope 1 emissions include putting energy efficiency measures into place, switching to greener energy sources, and streamlining operational procedures.

**Reliability:** Establishing precise operational control boundaries is crucial to guaranteeing the confidence, accuracy, and transparency of greenhouse gas emissions reporting. To ensure uniformity and comparability in reporting methods, organizations should set uniform standards for identifying which emissions sources are within their operational control boundaries and follow established norms and guidelines, such as the Greenhouse Gas Protocol.

### 2. Financial Control Approach <sup>7</sup>

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Greenhouse gas accounting guidelines October 2022 by SCGJ

A corporation is said to have financial control over its operations when it possesses the power to steer the operating and financial policies of its operations in a way that maximizes economic gains from its undertakings. For instance, a corporation is said to have financial control if it is entitled to the majority of the operation's benefits. These rights are transferable, nevertheless. Similar to this, if a company keeps taking up the majority of the risks and benefits related to asset ownership, it is considered to have financial control over an activity.

According to this standard, the company may have financial control over the operation even if it has less than a 50 interest in it since the economic nature of the relationship between the company and the operation is given priority over the legal ownership status.

### 3. Equity Share Approach

When using the equity share approach, GHG emissions from a company's activities are accounted for based on the share of equity in the operation. The economic interest, or the degree of a company's rights to the risks and benefits resulting from an operation, is reflected in the equity share. The proportion of an operation that the firm owns determines the number of economic risks and rewards, and the equity share will typically be equal to the ownership percentage. In other scenarios, the legal ownership form is always superseded by the economic substance of the relationship the firm has with the operation to guarantee that the equity share accurately reflects the percentage of economic interest. The idea of economic substance taking precedence over legal form. The idea that economic substance should take precedence over legal form is in line with accepted practices in international financial reporting. To verify that the correct equity share proportion is applied for each combined operation, the staff creating the inventory may need to speak with the company's accounting or legal personnel. This guarantees the application of the right equity share for each joint operation.

#### Consolidation at multiple levels

When all levels of an organization resort to the same emissions compilation policy, data consistency will be achieved when consolidating their GHG emissions data. To start with, the parent company will be required to choose the consolidation approach (i.e. either the equity share or the financial or operational control approach).

### 2.3.5. GHG Inventories

Detailed compilations of greenhouse gas emissions from particular entities, regions, or nations across predetermined periods are known as greenhouse gas inventories (GHGs). These inventories are crucial resources for comprehending, keeping track of, and controlling human influences on climate change.

Corporations and regulatory bodies also use inventories to create compliance records that include permissible emission rates. Inventories are a useful tool for businesses, the general public, and other interest groups to better understand the sources and trends of emissions.

#### Examples of Greenhouse Gas Inventory

- As required by the UNFCCC, Annex I nations publish their annual emissions and sinks of greenhouse gases.
- National governments are required to provide inventories of all anthropogenic greenhouse gas emissions under the Kyoto Protocol.
- The Kyoto Protocol adds more specifications for annual reviews, reporting, and national inventory systems.
- Under the Clean Development Mechanism, project developers create inventories as part of their project baselines.

### 1. Methodologies

- Internationally recognized techniques, such as those that are provided by the Greenhouse Gas Protocol and the Intergovernmental Panel on Climate Change (IPCC), are used to develop greenhouse gas inventories.
- Techniques offer standardized ways to estimate emissions for a range of industries and activities, guaranteeing data consistency and comparability.

## 2. Data collection

- Gathering accurate data is essential to creating trustworthy GHG inventories.
- Utility bills, fuel receipts, production records, industry databases, surveys, and emissions measurements are a few examples of data sources.
- For every emission source, activity data (like fuel consumption, and production output) and emission factors (like CO<sub>2</sub> emissions per unit of activity) are gathered.

## 3. Calculating Emissions

- Activity data is multiplied by matching emission factors to determine emissions.
- Depending on the emission source, calculation techniques may vary, ranging from basic multiplication to more intricate models or measurements.

## 4. Quality Control and Assurance (QA/QC)

- Strict QA/QC protocols guarantee data accuracy, consistency, completeness, and integrity.
- Crucial elements of QA/QC include documenting procedures and assumptions, verification of calculations, and validation of data sources.

## 5. Reporting and Transparency

- Governmental organizations, regulatory councils, investors, and the general public are among the stakeholders to whom GHG inventories are disclosed.
- Making informed decisions is facilitated and trust is increased by transparent reporting, which includes disclosing methods, presumptions, and uncertainties.

### 2.3.6. Reporting frameworks and standards

Frameworks and standards for greenhouse gas (GHG) reporting are important guidelines that give businesses an organized way to measure, record, and control their GHG emissions.

Let us discuss the frameworks in short

#### 1. Greenhouse Gas Protocol (GHGP)

An internationally acknowledged and recognized approach to accounting for greenhouse gas emissions is the Greenhouse Gas Protocol (GHGP).

- The World Business Council for Sustainable Development (WBCSD) and the World Resources Institute (WRI) collaborated along with a range of stakeholders to develop the GHG Protocol
- The key objective of the GHGP is to give organizations a uniform framework for precisely measuring, controlling, and disclosing their greenhouse gas emissions transparently and openly.

#### (A). GHGP scopes

- Scope 1: Emissions are direct emissions from owned or controlled sources, such as process emissions and fuel combustion.
- Scope 2: Indirect emissions from the entity's use of heat, steam, or purchased electricity.

- Scope 3: Indirect emissions connected to the entity's operations but originating from sources it does not own or control.

### **(B). Guidelines for Accounting and Reporting**

For measuring, computing, and reporting greenhouse gas emissions for each scope, GHGP offers comprehensive rules and methods. It outlines uniform procedures for gathering data, choosing emission factors, and performing calculations to guarantee the uniformity and comparability of emissions data.

### **(C). Industry-Specific Recommendations**

To provide guidance and best practices for emissions accounting, GHGP produces sector-specific guidance that is targeted to different businesses and sectors.

### **(D). Product Life Cycle Accounting and Reporting Standard**

In addition, the GHGP offers guidelines for tracking greenhouse gas emissions linked to products at every stage of their life cycle, from the extraction of raw materials to disposal.

### **(E). Implementation**

To measure, compute, and report greenhouse gas emissions, organizations employ the GHGP guidelines. They accomplish this by locating sources, gathering information, using suitable methods, and reporting using predetermined reporting formats.

## **2. ISO 14064<sup>8</sup>**

Guidelines for measuring, tracking, reporting, and verifying greenhouse gas (GHG) emissions and removals are provided by the International Organization for Standardization (ISO) in ISO 14064, a globally recognized standard.

The goal is to make it easier for organizations to measure their emissions, set reduction goals, and support international efforts to mitigate climate change by facilitating the establishment of reliable, transparent, and consistent greenhouse gas inventories.

### **Structure of ISO 14064**

#### ***Part 1: Specification with Guidance at the Organization Level for Quantification and Reporting of Greenhouse Gas Emissions and Removals***

In addition to providing assistance in identifying organizational boundaries, calculating emissions, and creating GHG reports, this section covers the concepts and requirements for organizations to build GHG inventories.

#### ***Part 2: Specification with Guidance at the Project Level for Quantification, Monitoring, and Reporting of Greenhouse Gas Emission Reductions or Removal Enhancements***

The rules for measuring, tracking, and reporting emission reductions or removal enhancements arising from particular activities or projects are provided in Part 2, particularly focusing on GHG projects.

#### ***Part 3: Specification with Guidance for the Validation and Verification of Greenhouse Gas Assertions***

To guarantee the precision, dependability, and validity of GHG inventories and project reports, Part 3 lays out the guidelines and conditions for the validation and verification of GHG statements.

## **3. The Carbon Disclosure Project (CDP)**

When it was first founded in 2000, the Carbon Disclosure Project (CDP) was dedicated to the reporting of carbon emissions only. Deforestation and water security have been added to its scope over time.

The main objective of CDP is to gather and share environmental data with investors, lawmakers, and the general public from businesses and other entities to promote accountability, transparency, and action on

<sup>8</sup> <https://www.iso.org/obp/ui/en/#iso:std:66453:en>

climate change and environmental sustainability.

### **Key role of CDP**

- Distributes yearly questionnaires about environmental impacts, hazards, and opportunities to businesses, cities, states, and regions.
- Customizes questionnaires to various industries, addressing issues such as deforestation, energy use, water management, greenhouse gas emissions, and climate change initiatives.
- Generate rankings and reports by assigning scores to participating entities based on their environmental performance and transparency policies.
- Offers an online platform where businesses can access reporting obligations and submit responses.
- Examines gathered information to generate reports on environmental performance, opportunities, and hazards.

### **CDP Participation Overview**

- Registration and completion of annual questionnaires.
- Collection and reporting of environmental data.
- Access to CDP's online platform for reporting requirements and best practices.

Monitoring, reporting, and management of greenhouse gas emissions by organizations are greatly aided by the usage of GHG reporting frameworks and standards. These frameworks support worldwide efforts to address climate change by promoting transparency, accountability, and consistency in environmental reporting through the provision of defined principles and methodology.

## **2.3.7 Verification and validation process**

For greenhouse gas (GHG) accounting procedures to be precise, reliable, and credible, reported emissions data must be validated and verified. These procedures are essential for evaluating how well companies are performing in terms of fulfilling their pledges and goals for reducing greenhouse gas emissions and for giving stakeholders transparency.

### **Verification**

Verification is the process of independently assessing GHG emissions data, calculations, and reporting systems to guarantee their correctness, consistency, and adherence to pertinent guidelines or standards.

The principal objective of verification is to furnish stakeholders with reassurance regarding the credibility and reliability of the stated emissions data.

### **GHG Emissions Data Verification Benefits**

- **Enhances Credibility:** Provides independent assurance of the accuracy and trustworthiness of GHG emissions data.
- **Ensures Compliance:** Verifies adherence to GHG accounting standards and regulatory requirements.
- **Identifies Errors and Inconsistencies:** Examines GHG emissions data, calculations, and reporting methodologies.
- **Validates Reduction Efforts:** Assesses the effectiveness of GHG reduction initiatives.
- **Facilitates Decision-Making:** Provides reliable information for informed decisions related to climate change mitigation, investment, procurement, and policy development.
- **Promotes Transparency and Accountability:** Fosters accountability for environmental impacts and emissions reduction efforts.
- **Improves Market Access and Reputation:** Demonstrates commitment to environmental responsibility and sustainability.

## Validation

The process of methodically evaluating and verifying the accuracy, dependability, and efficacy of GHG emission reduction projects or efforts is known as validation in greenhouse gas (GHG) reporting. It is essential to guarantee that reported reductions in emissions are legitimate and compliant with accepted practices, guidelines, and legal obligations.

### The Significance of Validating GHG Reports

- **Evaluation of Project Design:** Validation guarantees that GHG reduction initiatives are set up in a way that will reduce emissions. It involves assessing baseline emissions, boundaries, methodology, and plans complying with standards and procedures.
- **Verification of Implementation:** Validators check that projects are implemented as intended and in compliance with specified procedures.
- **Assessment of Emissions Reductions:** Validation evaluates the real emissions reductions attained by GHG reduction initiatives, guaranteeing the accuracy and correctness of reductions that are reported.
- **Compliance Review:** Validation ensures consistency and comparability by reviewing a project's adherence to standards, procedures, and legal requirements.
- **Independent Assurance:** Validation gives stakeholders independent reassurance on the accuracy and credibility of reported emissions reductions<sup>9</sup>.
- **Encouraging Market Mechanisms:** Verified Carbon Reductions reassure consumers and authorities that credits

## Stakeholder Reporting in GHG.

Organizations dedicated to environmental responsibility and sustainability must disclose greenhouse gas emissions. Fostering trust and promoting accountability and openness need effective reporting to stakeholders. The main parties engaged in GHG reporting are listed in this report, along with tips for communicating effectively with each group of stakeholders.

### 1. Internal stakeholders.

Executives, managers, employees, and investors are examples of internal stakeholders in an organization. The actions and activities of the organization immediately impact them or involve them.

To set goals, make informed decisions, and carry out mitigation plans, internal stakeholders need access to comprehensive data on GHG emissions.

Reporting: GHG Emissions Management

- **Executive Leadership:** Provides GHG emissions summaries for strategic decision-making.
- **Managers and Employees:** Offer training sessions and internal communications.
- **Shareholders:** Include GHG performance metrics in annual reports and meetings.

### 2. External stakeholders.

Customers, suppliers, investors, regulators, non-governmental organizations, and the general public are examples of external stakeholders who are not affiliated with the company. Their concerns are about the organization's social and environmental impact.

To evaluate the organization's environmental impact, regulatory compliance, sustainability efforts, and ethical standards, external stakeholders look for transparent and reliable GHG data.

### Reporting to:

<sup>9</sup> <https://floodlightinvest.com/understanding-limited-and-reasonable-assurance-when-reporting-ghg-emissions>

- Customers: Communicate GHG reduction efforts through marketing and engagement platforms.
- Suppliers: Collaborate with suppliers to collect GHG data and set emission reduction targets.
- Investors: Publish GHG reports aligned with internationally recognized standards.
- Regulators: Ensure compliance with GHG reporting regulations and advocate for supportive policies.
- NGOs and Public: Engage with stakeholders through sustainability reports, press releases, social media, and public events.

### 2.3.8 GHG Accounting Implementation Strategy

- Establish objectives: Clearly define your goals, including the need for sustainability reporting, stakeholder participation, regulatory compliance, and emission reduction targets.
- Define scope and boundaries: Indirect emissions from purchased energy, other indirect emissions, and direct emissions should all be included in the operational scope of GHG accounting.
- Adopt suitable standards and procedures: Select approaches such as the GHG Protocol, ISO 14064, or guidelines unique to your industry.
- Define procedures for gathering data: Accurately collect GHG emission data by using data management tools, metering devices, and automated systems.
- Evaluate baselines: Determine hotspots for emissions, set a baseline for GHG emissions, and rank the various emission reduction projects.
- Set in place processes for reporting and monitoring: Monitor greenhouse gas emissions regularly, examine patterns, and assess how well emission reduction strategies are working.
- Involve stakeholders: Encourage cooperation, responsibility, and transparency through talks, workshops,
- Continuous improvement and verification: Establish a process for continuous improvement and verification of GHG accounting practices.

### Government and Regulatory Policies in GHG Accounting

- Policies from the government and regulatory bodies influence GHG accounting procedures, guaranteeing transparency, accountability, and consistency.
- Measurement, reporting, and verification of greenhouse gas emissions and removals are done methodically through GHG accounting.
- Global collaboration on climate action is based on international agreements and mechanisms like the UNFCCC.
- National laws control greenhouse gas emissions and mandate that entities that surpass specific emission levels disclose.
- GHG emissions from industrial sectors are regulated by Emission Trading Schemes (ETS), which necessitate accurate GHG accounting for allocation, enforcement, and monitoring.
- Carbon pricing schemes that encourage emission reductions include carbon taxes and emissions trading. Accurate carbon pricing and distribution of revenue are guaranteed by strong GHG accounting procedures.
- Industry-specific laws control greenhouse gas emissions and reporting requirements. Examples include the International Maritime Organization's (IMO) ship emission restrictions and the International Civil Aviation Organization's (CORSIA) Carbon Offsetting and Reduction Scheme for International Aviation.
- Initiatives and standards for voluntary GHG accounting created by international organizations, NGOs, or business associations are supported by governments.
- Systems for monitoring, reporting, and verification (MRV) guarantee precision, consistency, and adherence to legal mandates.

### Innovation and Best Practices

**Technological developments include**

- Accurate emission measurement by the combination of satellite data and remote sensing.
- Internet of Things and sensor networks for monitoring and gathering data in real-time.
- Data recording that is transparent and unchangeable thanks to blockchain technology.
- Lifecycle Assessment (LEA)
- Extensive techniques to evaluate items and services' overall environmental impact.
- Locating hotspots and potential areas for emission reduction.
- Emissions accounting scope three includes a thorough analysis of the carbon footprint and potential for emission reduction.

**Harmonization and Standardization**

- Application of globally accepted GHG accounting guidelines and conventions.

**Data Transparency and Disclosure**

- Improving the GHG emissions data's transparency and disclosure.

**Sustainable Development Goals (SDGs) integration**

- GHG accounting procedures that are in line with the SDGs of the UN.

**Knowledge Sharing and Capacity Building**

- Investing in training, knowledge, and capacity building

## Exercise



1. What is the primary goal of greenhouse gas (GHG) emission accounting, and why is it crucial for organizations and policymakers?
2. How does understanding the impacts of climate change help in the development of adaptation and mitigation plans? Provide examples of how climate change affects ecosystems and human health.
3. What are the main differences between Scope 1, Scope 2, and Scope 3 emissions, and why is it important for organizations to account for all three scopes?
4. Describe the principles of GHG accounting. How do relevance, completeness, accuracy, consistency, and transparency contribute to effective GHG accounting?
5. Explain the different organizational boundary approaches in GHG accounting (Operational Control, Financial Control, and Equity Share). How do these approaches impact the measurement and reporting of emissions?
6. What is the primary purpose of greenhouse gas (GHG) inventories?
7. Name three key examples of greenhouse gas inventories mentioned in the unit.
8. What are the three scopes defined by the Greenhouse Gas Protocol (GHGP) for emissions accounting?
9. How does the ISO 14064 standard support organizations in GHG accounting?
10. What is the role of the Carbon Disclosure Project (CDP) in environmental reporting?
11. What are the main benefits of verifying GHG emissions data?
12. Why is validation important in the context of GHG emission reduction projects?
13. List three types of stakeholders mentioned in the passage who are involved in GHG reporting.
14. What are some of the technological innovations in GHG accounting highlighted in the passage?
15. How do government and regulatory policies impact GHG accounting practices?

## Unit 2.4 Selecting Suitable Data Sources for Inventory

### Unit objectives

After reading this unit, you will be able to understand the following points:



3. Understand the criteria for selecting data sources
4. Evaluating Data Accuracy
5. Assessing Data Reliability
6. Ensuring Data Relevance

To accurately measure, report, and manage emissions, data sources are essential to the management of greenhouse gas (GHG) inventories. The accuracy, reliability, and relevance of data sources are some of the aspects that make them suitable for greenhouse gas inventories. Below is a thorough explanation of these variables:

### 2.4.1. Accuracy

**Accuracy of Measurement:** To reduce uncertainty and estimating mistakes, data sources should offer accurate emissions measurements. The reliability of emission data is increased by precise measurement methods, such as validated models or direct emissions monitoring.

**Calibration and Quality Assurance:** Over time, the accuracy and consistency of emission data are guaranteed by the calibration of monitoring equipment and adherence to quality assurance protocols. Maintaining data correctness requires regular quality control procedures and calibration tests.

**Representativeness:** All pertinent emission sources, activities, and geographic regions should be represented in the data, which should be indicative of the emissions under evaluation. Within the inventory boundaries, variation and variability should be captured by sampling strategies and data-gathering methodologies.

### 2.4.2. Reliability

**Reliability and Consistency:** When independent sources or methodologies are used for replication or verification, consistent findings are obtained from dependable data sources. Emission estimates are reliable when measuring methods, data gathering processes, and data processing are consistent.

**Data Completeness:** All pertinent emission sources and sectors inside the inventory boundaries are fully covered by dependable data sources. The accuracy of the inventory could be jeopardized by incomplete data if emissions are understated or misrepresented about

**Data Validation and Verification:** Emission data is more credible and reliable when it has been independently validated and verified. Stakeholder confidence is boosted when third-party auditors or peer review procedures verify the precision and integrity of emission estimates.

### 2.4.3. Relevance

**Temporal Relevance:** Information must be current and represent actions and trends in emissions. Accurate data makes decision-making easier and makes it easier to monitor the achievement of emission reduction goals.

**Geographic Relevance:** Taking into account the geographical distribution of emission sources and their effects, emission statistics should be relevant to specific regions. To handle regional variability and capture localized emissions patterns, local or regional data may be required.

**Sectoral Relevance:** Within the inventory boundaries, all pertinent sectors and emission sources, including direct and indirect emissions, should be included in the data. Targeted mitigation plans and sectoral analysis to find potential for emission reduction are made possible by sector-specific data.

Accuracy, reliability, and relevance are characteristics of appropriate data sources for greenhouse gas inventories and provide a strong basis for well-informed decision-making and successful climate action. The incorporation of precise, dependable, and pertinent data sources augments the validity and soundness of greenhouse gas inventories, bolstering endeavors to alleviate climatic variability and advance sustainable growth.

## Exercise



1. What role does calibration and quality assurance play in ensuring the accuracy of emissions data?
2. How does data completeness contribute to the reliability of greenhouse gas inventories?
3. Why is temporal relevance important in the management of GHG inventories?
4. What are the key factors that ensure the reliability of data sources in greenhouse gas accounting?
5. How does sectoral relevance affect the effectiveness of emission reduction strategies?

## Unit 2.5. Data Collection and Adaptation Approaches

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Gathering existing data.
2. Generating new data.
3. Adapting data to inventory needs.

### 2.5.1 Gathering Existing Data for GHG Accounting

**Emission Inventories:** To compile information on greenhouse gas (GHG) emissions, current emission inventories at various levels (such as national, regional, and industry-specific) are reviewed. Governmental agencies, global organizations, and academic institutions frequently compile these inventories.

**Corporate Reports and Disclosures:** Information is gathered from environmental declarations, carbon accounting records, and company sustainability reports. As part of their sustainability activities, many businesses voluntarily report their GHG emissions, which provides important information for GHG accounting.

**Government Databases:** Getting access to databases maintained by the government that aggregate information on energy usage, industrial activity, greenhouse gas emissions, and land use changes. Databases kept up to date by statistical offices, energy departments, and environmental authorities are a few examples.

**International data sources and archives:** Making use of international databases and archives, such as the World Resources Institute (WRI) Greenhouse Gas Protocol Database and the Intergovernmental Panel on Climate Change (IPCC) Emission Factor Database. Standardized emission factors and calculation techniques are provided by these resources.

**Scientific Literature:** Examining peer-reviewed research publications to obtain information on GHG accounting best practices, methodology, and emission variables. Research reports, conference proceedings, and academic journals are all excellent places to find information.

### 2.5.2 Generating New Data for GHG Accounting

**Direct Measurement and Monitoring:** The process of measuring greenhouse gas emissions directly from sources including industrial facilities, transportation fleets, and agricultural operations involves installing sensors and monitoring equipment. Accurate accounting is made possible by real-time data from continuous monitoring.

**Emission Estimation Models:** Using or creating emission estimating models is a way to provide data on greenhouse gas emissions based on activity information and emission parameters. To estimate emissions, these models take into account variables including fuel usage, production output, and process data.

**Interviews and Surveys:** Interviewing or conducting surveys with stakeholders to obtain information on processes, activities, and emission sources that might not be obtained in any other way. This method works especially well for gathering information from farmers, small enterprises, and other organizations.

**Using satellite:** To estimate greenhouse gas emissions from deforestation, changes in land use, and other geographically distributed sources, GIS analysis, satellite images, and remote sensing techniques are used. Data from remote sensing offer important insights into changes in land use patterns, biomass, and vegetation cover.

**LCA and Product Life Cycle Analysis:** To calculate the embodied greenhouse gas emissions connected

to goods, services, and supply chains, life cycle assessments (LCAs) and product life cycle analyses are carried out. With this method, emissions are evaluated at every stage of a product's life cycle, from the extraction of raw materials to their disposal.

### 2.5.3 Adapting Data to Meet GHG Accounting Needs

**Normalization and Conversion:** To guarantee compatibility and comparability, normalize data to a standard unit of measurement (such as CO<sub>2</sub> equivalent) and convert data from various sources. This calls for the use of defined procedures, emission factors, and conversion factors.

**Quality Assurance and Quality Control (QA/QC):** It involves applying processes to verify the completeness, accuracy, and dependability of data. This could involve reconciliation with other datasets, error detection systems, and data validation tests.

**Temporal and Spatial Allocation:** The process of assigning emissions data to particular periods and geographical regions to precisely depict emission trends and spatial distribution is known as temporal and spatial allocation. Applying spatial allocation techniques, emission factors, and temporal characteristics are required for this.

**Uncertainty Analysis:** To measure the degree of uncertainty surrounding accounting procedures and data on greenhouse gas emissions, uncertainty analysis is performed. This makes it easier for stakeholders to comprehend the robustness and dependability of the inventory data and pinpoint areas in need of development.

**Benchmarking and Comparing:** To find potential for emissions reductions and performance enhancements, GHG emissions data are compared to industry benchmarks, historical trends, and peer performance. This makes setting goals and making decisions more informed.

**Stakeholder Engagement and Transparency:** To guarantee accountability, transparency, and stakeholder buy-in, it is important to involve stakeholders at every stage of the data adoption process. To build confidence and trust, this entails disclosing data limits, presumptions, and uncertainties.



### Exercise

1. What are the primary sources of existing data for GHG accounting?
2. How does direct measurement and monitoring contribute to GHG accounting?
3. What methods can be used to generate new data for GHG accounting?
4. Why is normalization and conversion important in adapting data for GHG accounting?
5. What is the role of uncertainty analysis in GHG accounting?
6. How can stakeholder engagement enhance the transparency and credibility of GHG data?



- [illegible]



## Chapter Highlights

After reading this chapter, you will be able to:

1. GHG Inventory management
2. Fundamentals of Emission reporting
3. Interpret emission data
4. Emission calculation methodology
5. Identify and assess emission reduction opportunities

## Unit 3.1. Identifying Various Sources of GHG Emissions and the Importance of Accurate Source Identification in GHG Accounting

### Unit objectives

After reading this unit, you will be able to understand the following points:

1. Understanding GHG Emission Sources
2. Identifying GHG Emission Sources
3. Assessing the Importance of Accurate Source Identification
4. Applying Source Identification Techniques



### 3.1.1 Identifying Various Sources of GHG emission

Categorizing activities, processes, and sectors such as energy consumption, industrial operations, agriculture, waste management, transportation, deforestation, and land use are necessary to identify different sources of greenhouse gases (GHGs). This aids in the development of plans by organizations for resource allocation, emission reduction, regulatory compliance, and enhanced sustainability performance. Sources of GHG emissions are listed below.

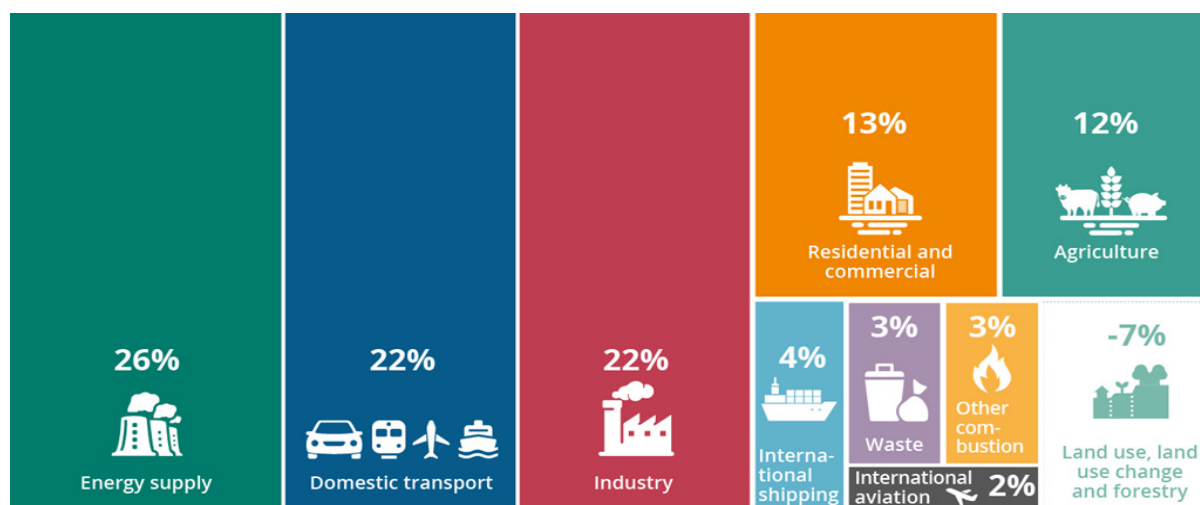


Fig 3.1.1 Source of GHG Emission (European Environment Agency)

**Energy consumption:** When fossil fuels are burned, nitrous oxide (N<sub>2</sub>O), methane (CH<sub>4</sub>), and carbon dioxide (CO<sub>2</sub>) are released in power plants, factories, vehicles, and residential structures.

**Industrial Processes:** By using fossil fuels and causing chemical reactions, industrial processes like cement manufacture, chemical manufacturing, and metal mining release CO<sub>2</sub> and other greenhouse gases into the atmosphere.

**Land Use and Agriculture:** While land-use changes and fertilizer application produce N<sub>2</sub>O and CO<sub>2</sub>, enteric fermentation in animals and rice paddies releases CH<sub>4</sub>.

**Management of Waste:** Organic waste decomposition in landfills releases CH<sub>4</sub>, whereas garbage incineration releases CO<sub>2</sub> and other greenhouse gases.

**Degradation of Land and Deforestation:** Urbanization and agricultural clearing of forests releases

stored carbon and lowers the ecosystem's ability to absorb CO<sub>2</sub>, which increases greenhouse gas emissions.

**Transport:** Nitrogen oxides (NO<sub>x</sub>), CH<sub>4</sub>, and CO<sub>2</sub> are released when fossil fuels are burned in automobiles and aircraft, which increases greenhouse gas emissions.

**Chemical and Industrial Processes:** Gases that are generated by chemical reactions in a variety of industrial processes include CO<sub>2</sub>, CH<sub>4</sub>, and fluorinated gases (SF<sub>6</sub>, HFCs, and PFCs).

**Treatment of Wastewater:** While nitrous oxide is emitted during treatment procedures, CH<sub>4</sub> is produced by the anaerobic breakdown of organic materials in wastewater treatment plants.

### 3.1.2 Understanding the Importance of Accurate Source Identification in GHG Accounting

#### Strategies for Reducing Emissions:

Establishing precise GHG source identification helps firms create focused emission reduction plans. Businesses can optimize the efficacy of mitigation strategies by concentrating their efforts on sources of high emissions.

#### Establishing Reduction Objectives:

Organizations can set realistic and attainable GHG reduction targets by having a clear understanding of the relative contributions of various emission sources. This guarantees that the largest contributors to emissions are the focus of mitigation efforts.

#### Allocation of Resources:

Organizations can effectively spend resources to mitigate emissions hotspots by accurately identifying the sources of emissions. By doing this, investments in energy-saving techniques, sustainable practices, and emission-reduction technologies are maximized.

#### Adherence to Policy:

Organizations must report their greenhouse gas emissions and adhere to legal regulations in many areas. Precise source identification guarantees adherence to reporting guidelines and expedites regulatory compliance.

#### Stakeholder Communication:

Accurate source identification and transparent reporting of greenhouse gas emissions are important for building credibility and trust with investors, consumers, and the public, among other stakeholders.

#### Risk Monitoring:

Understanding and identifying GHG sources aids in organizations' risk and vulnerability assessments connected to climate change. This makes it possible to make decisions that are more informed and take preventative action to lessen the risks brought on by climate change.

#### Enhancing Eco-Friendliness Achievement:

To demonstrate environmental stewardship and monitor progress toward sustainability targets, accurate source identification is crucial. It improves an organization's overall sustainability performance by enabling them to precisely measure, manage, and report on their GHG emissions.

Accurate source identification of greenhouse gas emissions is crucial for informed decision-making and effective climate action. It helps organizations develop strategies, allocate resources efficiently, ensure regulatory compliance, and enhance sustainability performance. Accurate source identification fosters stakeholder trust, supports risk management, and contributes to global climate change efforts.

### Exercise



1. Why is identifying various sources of GHG emissions important for organizations?
2. What are some specific greenhouse gases released from energy consumption and industrial processes?
3. How does accurate source identification contribute to establishing effective emission reduction strategies?
4. In what ways does precise GHG source identification support regulatory compliance and stakeholder communication?
5. How does understanding GHG emission sources aid in risk monitoring and enhancing sustainability performance?

## Unit 3.2 Carbon Footprint

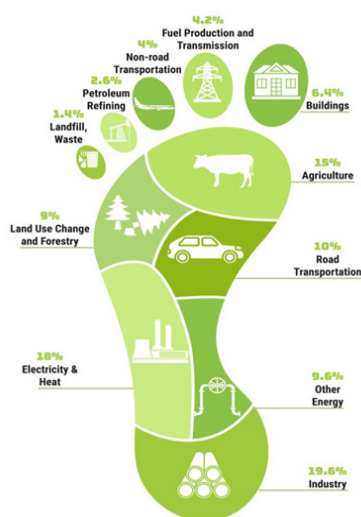
### Unit objectives

After reading this unit, you will be able to understand the following points:

1. Understanding the Concept of Carbon Footprinting
2. Exploring the Role of Carbon Footprinting
3. Examining the Carbon Footprinting Process



### 3.2.1 What is carbon footprint?



The phrase “carbon footprint” describes the total amount of greenhouse gases (GHGs) that a person, business, occasion, product, action, or event emits during a given period. GHGs are measured in terms of carbon dioxide equivalents (CO<sub>2</sub>e). It measures how much human activity affects the environment and how much it contributes to climate change.

Three major types of greenhouse gasses are mainly included in carbon footprints Carbon dioxide (CO<sub>2</sub>) Methane (CH<sub>4</sub>) and Nitrous oxide N<sub>2</sub>O

#### What are the different types of carbon footprints?

Depending upon the sector and activities there are many types of carbon footprint, they are

#### Product Carbon Footprint:

This kind of carbon footprint measures greenhouse gas emissions related to a product’s whole life cycle, from extraction and production of raw materials through distribution, use, and disposal. It helps customers make wise purchasing decisions by offering insightful information on how items affect the environment.

#### Corporate Carbon Footprint:

Organizational carbon footprints measure the overall greenhouse gas emissions—both direct and indirect—that are related to an organization’s operations. These evaluations offer a thorough picture of an organization’s environmental effect and are frequently used to establish goals for reducing emissions and monitor advancement over time

#### Individual carbon footprint:

The total quantity of greenhouse gas (GHG) emissions over a certain period that are directly and indirectly related to an individual’s activities and lifestyle choices is referred to as their “individual carbon footprint.” This amount is usually expressed in terms of carbon dioxide equivalent (CO<sub>2</sub>e).

#### How to calculate Carbon footprint?

A carbon footprint is a measure of the total amount of greenhouse gas emissions from various activities and lifestyle choices. It involves identifying the scope of the assessment, gathering data on activities and consumption patterns, converting the data into CO<sub>2</sub> equivalent, calculating emissions, aggregating results, normalizing data, interpreting results, and monitoring and reviewing emissions data. The process can vary in complexity depending on the data’s scope, detail, and accuracy. Consulting with sustainability experts or environmental consultants may be beneficial for more comprehensive assess-

ments or complex emission sources. Organizations may also choose to verify their carbon footprint calculations through independent third-party verification to ensure accuracy and credibility.

The carbon footprint calculation is derived from global guidelines such as ISO 14064 and the GHG Protocol. Depending on the fuel type, energy purpose, and planned activity, emissions are computed by multiplying the consumption statistics by the emission factor<sup>1</sup>. Consequently, the computation is summarized by the following formula:

$$\text{Carbon footprint} = \text{Activity data} \times \text{Emission factor}$$

### 3.2 Role of Carbon Footprinting

- **Evaluation of the Environmental Impact**
  - a. The number of greenhouse gases (GHGs), mostly carbon dioxide (CO<sub>2</sub>), either released directly or indirectly because of human activity is measured by carbon footprinting.
  - b. It offers a thorough evaluation of the environmental effects of different operations, goods, services, or establishments.
- **Mitigation of Climate Change**
  1. Carbon footprinting aids in locating the main contributors of greenhouse gas emissions and in setting priorities for emission reduction.
  2. People, companies, and governments can reduce their impact on climate change by implementing policies to reduce their carbon footprint.
- **Accountability and Sustainability Reporting**
  - a. A crucial part of sustainability reporting frameworks like the Carbon Disclosure Project (CDP) and the Global Reporting Initiative (GRI) is carbon footprinting.
  - b. It increases accountability and transparency by allowing stakeholders to monitor the status of emission reduction goals.
- **Support for Decision-Making and Risk Management**
  - a. Carbon footprinting offers useful information that helps stakeholders evaluate the environmental effects of various solutions.
  - b. It supports the development of adaptation plans and the identification of possible hazards related to the effects of climate change

### 3.3. Process of Carbon Footprinting

- **Define the Scope**
  - a. Establish the parameters and criteria for the carbon footprint assessment, including the sources and types of emissions that will be covered.
  - b. Take into account all indirect emissions from the value chain (Scope 3), in addition to indirect emissions from purchased heat, steam, and electricity (Scope 2).
- **Information Gathering**
  - a. Compile information on fuel usage, energy consumption, waste production, transportation, and other pertinent operations.
  - b. Use credible sources, such as utility bills, fuel invoices, and operating records, to ensure data correctness and dependability.
- **Emission Calculation**
  - a. Using emission factors relevant to each emission source, convert activity data into CO<sub>2</sub> equivalent emissions.
  - b. To calculate the overall carbon footprint, compute the emissions for each category and add them together.
- **Reporting and Communication**
  - a. Using a report or dashboard, make the carbon footprint statistics easily accessible.
  - b. Share findings with pertinent parties, such as management, staff, clients, investors, and law enforcement.

<sup>1</sup> <https://portfoliomanager.energystar.gov/pdf/reference/Emissions.pdf>

- **Development and Execution of Action Plans**
  - a. Using the emission hotspots and reduction possibilities that have been identified, develop a strategy for managing carbon emissions.
  - b. Put into practice strategies to cut emissions, such as supply chain optimization, adoption of renewable energy sources, enhancements in energy efficiency, and behavioral adjustments.
- **Monitoring and Verification**
  - a. Track progress toward emission reduction targets and continuously monitor emissions.
  - b. If necessary, confirm reduction accomplishments and data on carbon footprint by independent third-party verification.

Carbon footprinting is a crucial process for assessing environmental impact, guiding climate change mitigation, and promoting sustainability, enabling stakeholders to effectively manage and reduce their carbon footprint.

## Exercise



1. What are the main types of carbon footprints, and how do they differ from each other?
2. What is the formula used to calculate a carbon footprint, and what does each component represent?
3. How does carbon footprinting contribute to climate change mitigation and sustainability reporting?
4. What are the key steps in the carbon footprinting process, and what does each step involve?
5. Why is it important to include both direct and indirect emissions when calculating a carbon footprint?

## Unit 3.3 GHG emission calculations methodologies and implementation

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. GHG emissions and their role in climate change.
2. Exploring different methodologies for calculating GHG emissions.
3. Implementing GHG emission calculation methodologies

### 3.4.1 Methodologies

There are several approaches for figuring out greenhouse gas (GHG) emissions, and they are all geared toward different industries, pursuits, or goals. Comprehending these techniques is crucial to precisely evaluate emissions, execute mitigation strategies, and provide environmental performance reports. The various methods for calculating greenhouse gas emissions and their uses are examined in this document.

#### 1. Input-Output Analysis for GHG Emission Calculation

The fundamentals of input-output analysis are as follows

- Monitors the movement of products and services between economic sectors.
- Tracks the products and services that each industry purchases from other sectors.
- Quantify the inputs needed by each sector to produce a unit of output using the Leontief Inverse Matrix.

#### Steps to Calculate GHG Emissions Using Input-Output Analysis

- **Input Data Collection:** Compiles sector-specific transaction data from input-output databases or national statistical organizations.
- **Application of Emission Factors:** Emission factors are applied to the input-output tables' ultimate demand and intermediate input components.
- To calculate total emissions, multiply emission coefficients in the input-output tables by the appropriate quantities of demand or input.
- **Sensitivity Analysis and Scenario Modelling:** to determine how changes in input parameters affect emission estimates, sensitivity analysis is carried out.

#### Leontief Inverse Matrix

It is named after the economist Wassily Leontief, who pioneered input-output analysis. The Leontief Inverse Matrix is a mathematical tool that is used to evaluate the interdependencies across various economic sectors. The Leontief Inverse Matrix is a tool used to measure how changes in the end demand for products and services impact the production levels across different sectors.

#### Application of Input-Output Analysis in GHG Management

- **Carbon Footprint Assessment:** This tool makes it possible to measure carbon footprints at the sectoral, national, or regional levels.
- **Policy Analysis and Decision Support:** Provides information for policy formation by evaluating the effects on emissions-intensive industries of environmental laws, carbon pricing schemes, or technological advancements.

## 2. Economic Input-Output Life Cycle Assessment (EIO-LCA)

EIO-LCA is a methodology that integrates IOA and LCA principles to assess the environmental impact of economic activities. It uses economic data from input-output tables and life cycle inventories to evaluate the environmental footprint of goods and services.

The following are the elements of the Economic Input-Output Life Cycle Assessment

- **Analysis of Input-Output (IOA)**
  - IOA is a method of economic modeling that measures the movement of products and services between different economic sectors. The interdependencies between sectors are captured in input-output tables, which demonstrate how shifts in the final demand for products and services impact production levels throughout the economy.
  - Input-output tables are the cornerstone of EIO-LCA, providing a relationship between economic activity and the impact on the environment. They offer information on the material, energy, and emission fluxes connected to various industries and goods.
- **Life Cycle Evaluation (LCE)**
  - LCA is a procedure that evaluates how a product or process will affect the environment at every stage of its life, from the extraction of raw materials to the disposal of its end-of-life. It measures emissions, the use of resources, and possible effects on the environment, including eutrophication, acidification, and climate change.
  - Life cycle inventories are used in EIO-LCA to calculate the environmental costs of producing and consuming products and services. Data on energy use, material inputs, waste production, and emissions from different life cycle stages are included in this.
  - Integration of Environmental and Economic Data
  - To calculate the environmental effects of economic activity, EIO-LCA combines environmental and economic data. Input-output coefficients and life cycle inventory data are combined to quantify the environmental effects of economic activity changes, such as changes in technology adoption or consumption habits.
  - The methodology makes it possible to estimate environmental indicators linked to many industries and goods, including greenhouse gas emissions, energy and water consumption, and other effects of resource depletion.
- **Applications and Benefits**
  - By identifying hotspots and chances for environmental improvement, EIO-LCA can be used to evaluate the environmental performance of goods, sectors, areas, or policies.
  - It influences the formulation of environmental policies, sustainability programs, supply chain management, and product design, all of which aid in the decision-making process.
  - The method is a useful instrument for comprehending the environmental effects of economic systems and directing efforts towards sustainable development since it has benefits including scalability, comprehensiveness, and consistency.

## 3. Spend-Based Methodology

The spend-based approach, centers on examining expenditure trends to locate emission hotspots and carry out focused mitigation strategies.

To determine the carbon intensity of a business's spending, the spend-based methodology examines financial transactions that take place within the firm. This methodology offers insights into the procurement processes, operating expenses, and embedded indirect emissions throughout the supply chain. Organizations can prioritize initiatives for emission reduction by identifying high-emission activities and suppliers by analyzing spending data.

Toyota: The automotive manufacturer utilizes activity-based analysis to identify emission hotspots in its manufacturing processes, leading to initiatives such as energy efficiency improvements, waste reduction, and the adoption of renewable energy sources.

### Spend-Based Methodology's Advantages

**Comprehensive Analysis:** Spend-based technique provides a more comprehensive view by taking indirect emissions throughout the supply chain into consideration, in contrast to typical emissions accounting methods that concentrate on direct emissions from internal operations.

**Targeted Interventions:** Organizations can achieve significant emission reductions where they matter most by identifying emission hotspots in supply chain and procurement activities.

**Cost Savings:** By streamlining processes, managing resources, and implementing sustainable practices, identifying and improving high-emission spending areas can result in cost savings.

**Stakeholder Engagement:** By showcasing a dedication to sustainability and environmental responsibility along the whole supply chain, spend-based analysis's transparency promotes stakeholder engagement.

Microsoft: Through its Carbon Fee program, Microsoft incorporates spend-based analysis to internalize the cost of carbon emissions across its operations, incentivizing emission reduction initiatives and investments in renewable energy

## 4. Activity-Based Methodology

The use of activity-based techniques allows for the precise identification of emission sources, including waste management, manufacturing, transportation, and energy consumption, by quantifying greenhouse gas emissions from particular organizational operations.

### Activity-Based Methodology Benefits

- Provides detailed insights into emission sources.
- Enables targeted interventions for efficient emission reductions.
- Enables ongoing performance tracking and evaluation of emission reduction initiatives.
- Facilitates resource allocation based on activity emissions intensity.

### Challenges of Activity-Based Methodology

- **Data Accuracy and Availability:** Robust data gathering procedures and extensive monitoring systems provide technological and logistical obstacles.
- **Complexity and Integration:** It can be challenging to analyze emissions at the activity level, particularly for businesses that engage in a variety of business ventures.
- **Behavioral and Cultural Change:** It is frequently necessary to alter organizational behavior, culture, and procedures to achieve emission reductions. This could lead to employee resistance or apathy.
- **Boundaries and Scope:** It can be difficult to define the limits and scope of activities related to emission assessment.

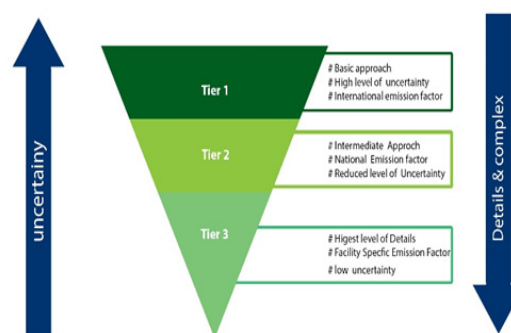
## 5. Hybrid-Based Methodology

A hybrid-based approach to calculating greenhouse gas emissions combines several approaches for a thorough analysis, providing businesses with a customized framework for measuring and controlling their carbon footprint.

The hybrid-based methodology incorporates components from Input-Output Analysis, Spend-Based Methodology, Activity-Based Methodology, and Life Cycle Assessment (LCA) among other GHG emission calculation methodologies. It measures emissions linked to particular activities, analyzes financial transactions and procurement data to pinpoint emission-intensive spending activities within an organization's supply chain, evaluates the environmental effects of products, processes, or activities throughout their whole lifecycle, and assesses interdependencies between sectors.

## 6. IPCC's Tier methodology for GHG Emission Calculation

To estimate greenhouse gas (GHG) emissions, the Intergovernmental Panel on Climate Change (IPCC) uses a tiered approach. This methodology is essential for comprehending and reducing the effects of climate change. The IPCC Guidelines for National Greenhouse Gas Inventories outline this organized system, which provides a thorough framework for countries to compute and report their emissions. The three tiers of the tiered method are each designed to take into account the complexity, accuracy, and availability of data related to the emissions profiles of specific nations.



### 1. IPCC's Tier 1 Methodology

Based on the degree of detail and the availability of data, the Intergovernmental Panel on Climate Change (IPCC) has created tiered approaches for the assessment of greenhouse gas (GHG) emissions. Tier 1 is the most basic estimation level; it provides a rudimentary assessment of emissions using activity data and default emission parameters.

#### Important Elements

**Default Emission Factors:** To estimate GHG emissions related to particular activities or processes, Tier 1 methodology makes use of default emission factors that are provided by the IPCC or other credible sources. These criteria are applied consistently across many areas and industries, and their average values are obtained from empirical data and scientific research.

**Activity Data:** To estimate emissions at a national or sectoral level, Tier 1 methodology uses aggregated activity data, such as energy consumption, production quantities, or population statistics. International databases, national inventories, and statistical sources commonly serve as the sources of these data.

**Simplified Models:** To estimate emissions based on activity data and emission variables, Tier 1 methodology may use simplified models or equations. These models frequently make the assumption that emissions are uniform, which means they could ignore site-specific variables that could have an impact on emissions intensity.

**Limited Scope:** The Tier 1 technique is appropriate for nations or industries with limited data availability or technological competence since it offers a basic estimate of GHG emissions. However, it might not fully represent the complex nature of emission paths and sources, especially in industries with considerable amounts of volatility or uncertainty.

#### Advantages of Tier 1 Methodology

- **Accessibility:** Fits a variety of users, including small-scale businesses and developing nations.
- **Consistency:** Comparability between areas and industries is guaranteed by the use of default emission factors.
- **Cost-Effectiveness:** It is feasible for preliminary emission evaluations because it necessitates little data gathering, processing, or analysis.
- **Policy Support:** Offers a foundation for creating sector-specific or national GHG inventories and aids in the formulation of policies about the adaptation and mitigation of climate change.

#### Limitations on Tier 1 Emission Estimation

- **Uncertainty:** Site-specific factors, emission variability, and technology developments may not be fully reflected in Tier 1 estimates.
- **Scope Restrictions:** Not all emission sources or routes within a sector or region may

- be included by the Tier 1 methodology.
- **Data Quality:** The availability and quality of emission variables and activity data affect reliability.
- **Continuous Improvement:** While Tier 1 is an excellent starting point to start, over time, adjustments and improvements are required.

## 2. Tier 2 Methodology

A systematic method known as the Tier 2 emission calculation methodology is used to calculate greenhouse gas (GHG) emissions from various kinds of sources, including waste management, energy production, transportation, and industrial activities. When compared to Tier 1 approaches, it offers a more thorough and precise assessment because it considers particular metrics and factors related to the emission source.

### Methodology

**Data collection techniques include:**

- i.) Compiling detailed information on activities, procedures, and inputs.
- ii.) Applying emission factors from reliable sources, such as the EPA, IPCC, or industry standards.
  1. Activity Data: Measuring the amount of use or activity related to the source of emissions.
  2. Calculation: To estimate total emissions, multiply activity data by the relevant emission factor.
  3. Adjustments: Adapting for particular situations or variables impacting emissions.
  4. Validation: Verifying computed emissions against real measurements or impartial data sources is the first step in the validation process.
  5. Reporting: For transparent reporting, it is necessary to document the calculated emissions, methodology, assumptions, and uncertainties.

The Tier 2 emission calculation methodology is widely recognized and endorsed by leading organizations in the fields of climate science, environmental management, and sustainability. It builds upon established principles and guidelines developed by reputable institutions such as the IPCC and EPA, ensuring a high level of credibility and reliability in emission estimation practices.

### Tier 3 Methodology

The most sophisticated and comprehensive method for calculating greenhouse gas (GHG) emissions is the Tier 3 emission methodology developed by the Intergovernmental Panel on Climate Change (IPCC). To produce the most precise emission assessments possible, it expands upon Tier 1 and Tier 2 procedures by combining very specific data, sophisticated modeling tools, and site-specific measurements.

The IPCC's Tier 3 emission methodology offers a detailed understanding of GHG emissions, especially from complex sources like industrial facilities, energy production, and agriculture, using advanced techniques and precise data to identify emission hotspots, evaluate mitigation strategies, and inform policy decisions.

### Emission Estimation Methodology

- **Data Collection:** Gathering detailed information on emission source parameters and activities.
- **Advanced Modeling:** Using mathematical models and computer simulations to simulate emission behavior and predict GHG emissions.
- **Site-Specific Measurements:** Conducting direct measurements and sampling at the emission source to validate model predictions and quantify actual emissions.
- **Data Integration:** Combining data from multiple sources to develop a comprehensive

- understanding of emissions.
- **Uncertainty Analysis:** Conducting rigorous analysis to quantify the reliability and accuracy of emission estimates.
- **Continuous Improvement:** Refining and improving the emission estimation process based on new data, scientific advancements, and stakeholder feedback.

Leading scientists, decision-makers, and regulatory agencies from all over the world accept the IPCC's Tier 3 emission methodology, which is generally recognized as the gold standard for estimating greenhouse gas emissions. Drawing from decades of study and development in the domains of atmospheric science, environmental engineering, and data analysis, it represents the state-of-the-art in emission inventory development.

## Exercise



1. What are the key differences between the Input-Output Analysis and Economic Input-Output Life Cycle Assessment (EIO-LCA) methodologies for calculating GHG emissions, and how do each of these methods apply to environmental performance reporting?
2. How does the Spend-Based Methodology help organizations identify and manage emission hotspots, and what are the advantages of using this approach compared to traditional direct emissions accounting methods?
3. What are the benefits and challenges associated with the Activity-Based Methodology for GHG emission calculations, and how can organizations address issues related to data accuracy and scope in this approach?
4. In what ways does the Hybrid-Based Methodology combine different GHG emission calculation approaches, and how does this combination enhance the accuracy and comprehensiveness of emission assessments?
5. How do the IPCC's Tier 1, Tier 2, and Tier 3 methodologies differ in their approach to GHG emission estimation, and what are the key considerations for choosing the appropriate tier based on data availability and the complexity of emission sources?

## Unit 3.4. Appropriate emission factors, global warming potential values, and calculation tools for accurate and reliable GHG emission calculations.

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Appropriate emission factors.
2. Global warming potential values.
3. Accurate, reliable GHG emission calculations.

### 3.4.1 Why need an appropriate emission factor, GWP Values, and tools?

To perform accurate and trustworthy greenhouse gas (GHG) emission calculations, it is essential to choose the right emission factors, global warming potential (GWP) values, and calculation tools. These elements serve as the basis for estimating emissions from different sources and are crucial in establishing how accurate emission inventories are.

#### Importance of Emission Factor

In greenhouse gas (GHG) emission estimates, selecting the right emission components is essential for several reasons.

**Accuracy:** The number of GHGs released per unit of input or activity is represented by the emission factor. Emission estimations are guaranteed to closely resemble actual emissions from the source under assessment if appropriate and accurate emission variables are chosen. Setting goals for reducing emissions, assessing the success of mitigation strategies, and making informed decisions all depend on this precision.

**Precision:** Depending on elements like technology, fuel type, and operating conditions, different activities and processes may have different emission characteristics. GHG emission computations can be made more precisely by using emission factors that are relevant to the features of the emission source, offering more detailed insights into emission profiles and trends.

**Comparability:** Meaningful comparisons between various emission sources, sectors, geographies, and periods are made possible by the consistent application of standardized emission parameters. By making benchmarking, trend analysis, and emission hotspot identification easier, this comparability aids stakeholders in setting priorities for mitigation activities and allocating resources wisely.

**Regulatory Compliance:** As part of voluntary reporting programs or regulatory compliance, many jurisdictions mandate that enterprises report their greenhouse gas emissions. Using the right emission variables improves regulatory reporting procedures' accountability, transparency, and credibility by ensuring that reported emissions follow established methodologies and standards.

**Policy creation:** At the municipal, state, federal, and international levels, policy creation and decision-making are greatly aided by GHG emission estimations. Precise emission variables provide policymakers with dependable information to formulate policies, incentives, and laws that are grounded in science and targeted at curbing emissions and bringing in a low-carbon economy.

#### Importance of Global Warming Potentials Values

The Significance of Correct Global Warming Potential (GWP) Values in the Calculation of GHG Emissions

- **Climate Impact Assessment:** Greenhouse Gas Potential (GWP) values measure how much a certain GHG can warm the planet about CO<sub>2</sub>. Precise evaluation of the climate impact of GHG emissions is made possible by robust and precise GWP values.
- **Comparability:** Standardized GWP values make it easier to compare various GHGs, emission sources, industries, and geographical areas insightfully.
- **Policy Development:** Local, national, and international decision-making is influenced by GWP values. They support the creation of incentive programs, regulatory standards, and emission reduction targets.
- **Emission Reporting:** Reporting emissions that accurately reflect the various greenhouse gas impacts is ensured by the use of suitable GWP values.
- **International Agreements:** GWP values are integral to international agreements like the Kyoto Protocol and the Paris Agreement. Consistent application of GWP values across countries promotes global cooperation in addressing climate change challenges.

### Need for Selecting Appropriate Calculation Tools for GHG Emission Estimates

- **Accuracy and Precision:** By reducing errors and uncertainties, sophisticated algorithms and tools generate accurate emission estimations.
- **Comprehensive Analysis:** With the use of these tools, it is possible to conduct an extensive analysis of GHG emissions from a range of sources, activities, and industries.
- **Cost and Time Efficiency:** The emission estimation process is streamlined by effective calculating tools, which results in time and resource savings.
- **Scenario Modeling:** Stakeholders can evaluate the effects of mitigation techniques, technology advancements, and legislative initiatives on greenhouse gas emissions by utilizing the scenario modeling features that advanced tools frequently offer.
- **Data Integration and Management:** A lot of calculation tools come with features for managing and integrating data, which makes data interchange easy and allows for easy integration of data into emission inventories.
- **Transparency and Documentation:** The proper tools offer elements that promote accountability, transparency, and credibility in the procedures involved in emission reporting and verification.
- **User-Friendly Interface:** For stakeholders with different levels of experience, calculation tools with user-friendly interfaces improve usability and accessibility.
- **Flexibility and customization:** Tools for flexible calculations enable flexibility and customization to meet the demands of certain regulatory frameworks and organizational requirements.

## Exercise



1. Why is it crucial to use appropriate emission factors in GHG emission calculations, and how does their accuracy affect regulatory compliance and policy development?
2. How do Global Warming Potential (GWP) values contribute to climate impact assessment and international agreements, and why is consistency in GWP values important for global cooperation?
3. What are the advantages of using advanced calculation tools for GHG emission estimates, and how do these tools enhance accuracy, efficiency, and transparency in the emission reporting process?
4. How does the choice of emission factors and GWP values influence the comparability of GHG emissions data across different sectors, regions, and time periods?
5. What role do calculation tools play in scenario modeling for GHG emissions, and how can they help stakeholders evaluate the effectiveness of mitigation strategies and technological advancements?

## Unit 3.5. The principles of emissions reporting and information to be incorporated within the GHG inventory report.

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. The core principles of emissions reporting, including Measurement and disclosure.
2. Outline the essential information required in a GHG inventory report

### 3.5.1 Fundamental Principles of Emissions Reporting

Several fundamental ideas underpin emissions reporting, which is necessary to measure and disclose greenhouse gas (GHG) emissions with accuracy. These guidelines guarantee reporting procedures are reliable, transparent, and consistent. The following are the core principles of emissions reporting

**Accuracy:** When it comes to data collecting, measurement, and calculation techniques, emissions reporting must aim for accuracy and precision. Precise reporting guarantees that the declared emissions correspond to the actual environmental effect of an entity's operations.

**Completeness:** Within the specified reporting parameters, reporting must address all noteworthy sources of emissions. This comprises emissions resulting directly from the organization's operations (Scope 1), indirectly from energy purchases (Scope 2), and from other sources (Scope 3) as well.

**Relevance:** Information about emissions that is reported must be relevant to the target audience, decision-makers, and stakeholders. It ought to offer insightful information on the company's environmental performance and its role in climate change.

**Consistency:** Meaningful comparisons over time and between different organizations are made possible by maintaining consistency in reporting techniques and data collection procedures. Conformity to industry-wide reporting systems, like the Greenhouse Gas Protocol, fosters uniformity among various companies and sectors.

**Transparency:** To be transparent, the methods, assumptions, and data sources utilized in emissions reporting must be made publicly available. Transparent reporting builds trust and makes it easier for stakeholders to comprehend the creation and interpretation of emissions data.

**Timeliness:** To reflect current emissions trends and facilitate well-informed decision-making, emissions reporting should be done regularly and promptly. Organizations may monitor their progress toward reducing emissions and efficiently address new environmental concerns when they report on time.

**Verification:** Third-party auditors' independent confirmation or assurance of emissions data raises the dependability and credibility of reporting. Verification guarantees that emissions reported are accurate, comprehensive, and compliant with specified guidelines and procedures.

### 3.5.1 Information to be included in the GHG Inventory Report

A thorough GHG inventory report needs to include several crucial components to precisely evaluate and reveal the greenhouse gas emissions of a company. An overview of the essential details that must be mentioned is provided below

#### I. Introduction:

- Provide a brief explanation of the organization's goals and the parameters of the GHG inventory report.
- Indicate which reporting period the inventory covers.

#### II. Profile of the Organization:

- Give an overview of the organization's goals, functions, and organizational structure.
- Note any modifications to organizational structures or noteworthy operations that took place during the reporting period.

#### III. Sources of Emissions:

- Identify and measure the organization's sources of emissions, such as:
  - i. Emissions falling under scope 1 include direct emissions from sources that the organization owns or controls (e.g., combustion of fossil fuels, onsite waste treatment).
  - ii. Emissions falling under scope 2: Indirect emissions resulting from the organization's purchase of heat, steam, or electricity.
  - iii. Emissions falling under scope 3 include additional indirect emissions from things like employee commutes, business travel, and emissions from the upstream and downstream supplier chains.
- Give the emission factors and methods used to determine the emissions from each source.

#### IV. Quantification methods:

- Explain the data sources and methods used to calculate greenhouse gas emissions.
- Give specifics about activity data, emission factors, formulae used to calculate emissions, and any corrections made for accuracy.

#### V. Information Administration:

- Describe the steps involved in gathering data, validating it, and ensuring quality control to guarantee the dependability and correctness of emissions data.
- Name any software or data management systems that are used to gather and store emissions data.

#### VI. Trends in Emissions and Analysis:

- Display past emissions data to monitor changes over time.
- Examine the variables that affect emissions patterns, such as shifts in production levels, advancements in energy efficiency, or the execution of emission control programs.

#### VII. Performance Indicators and Mitigation Strategies:

- Give an overview of the organization's efforts and steps to cut greenhouse gas emissions.
- Provide a report on the effectiveness of emission-reduction initiatives and how they affect total emission reductions.
- Add appropriate performance metrics, like targets for reducing emissions or emission intensity (emissions per unit of output).

**VIII. Confirmation and Confidence:**

- Keep a record of any independent processes used for assurance or verification to confirm the accuracy and authenticity of the GHG inventory data.
- Add the verification process's results and conclusions, as well as any suggestions for enhancement.

**IX. Conclusion and Prospects for the Future:**

- List the main conclusions and observations from the GHG inventory report.
- Determine areas in need of improvement and potential courses of action to cut greenhouse gas emissions even more.
- Give a declaration outlining your commitment to openness and ongoing development in the management and reporting of emissions.

The information provided is based on globally accepted standards, which are reliable and accepted in the field of greenhouse gas emissions reporting. Examples of these standards are the Greenhouse Gas Protocol and ISO 14064.

## Exercise



1. What key components should be included in the introduction section of a GHG inventory report, and why is it important to define the reporting period and organizational goals?
2. How should an organization identify and measure its sources of emissions under Scope 1, Scope 2, and Scope 3, and what role do emission factors and calculation methods play in this process?
3. What steps should be taken to ensure the reliability and accuracy of data in a GHG inventory report, and how can data management systems and quality control processes enhance information administration?
4. How should trends in emissions and the effectiveness of mitigation strategies be presented in the GHG inventory report, and what role do performance indicators play in evaluating and improving emission reduction efforts?

## Unit 3.6. Interpreting various emissions data and the global trends in GHG reporting.

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Interpreting Emissions Data.
2. Understanding Global GHG Reporting Trends.
3. Identifying Emerging Issues and Challenges.
4. Assessing Impacts and Opportunities.

For sound decision-making and successful climate action, it is essential to understand worldwide trends in greenhouse gas (GHG) reporting and interpret emissions statistics. Here are a few guidelines for interpreting different emissions data and comprehending worldwide patterns in GHG reporting

#### 1. Types of Data on Emissions:

- **Absolute Emissions:** These show how much greenhouse gas a nation or organization has released into the atmosphere during a given period. Absolute emissions offer a quick overview of the total effect on the environment.
- **Emission Intensity:** This statistic, which might be production unit or per capita, quantifies emissions about a particular output or activity. Comparisons between various scales and settings are made possible by emission intensity.
- **Emission Factors:** These are coefficients that are used to estimate emissions from activity data, such as fuel use. Accurately estimating emissions from a variety of sources and sectors requires an understanding of emission variables.

#### 2. Analysis Methods:

- **Trend Analysis:** Patterns and trends, such as rising or falling emission levels, can be seen by analyzing historical emissions data over time. Analyzing trends aids in seeing long-term shifts and evaluating how well emission reduction strategies are working.
- **Benchmarking:** By comparing emissions data to performance standards or industry benchmarks, one can determine areas for improvement and provide insight into relative performance.
- **Correlation analysis:** Correlation analysis is a useful tool for understanding the underlying causes of changes in emissions and informing strategic decision-making. It involves finding correlations between emissions data and external factors, such as energy prices and economic indicators.

#### 3. Global GHG Reporting Trends

- **Historical Trends:** Patterns of increase, stillness, or reduction are revealed by examining past trends in global GHG emissions and reporting efforts. Comprehending past patterns offers a framework for evaluating present emissions information.
- **Principle Motivators:** It is possible to predict future trends and problems by identifying the major factors impacting GHG reporting patterns, such as market dynamics, technical improvements, and regulatory requirements.
- **Regional Differences:** Understanding differences in emission levels, policy priorities, and socioeconomic factors is made possible by acknowledging regional variances in GHG reporting practices and emission reduction initiatives.

#### 4. New Challenges and Difficulties:

- **Data Quality:** In GHG reporting, it can be difficult to guarantee data consistency, accuracy, and completeness, especially for scope 3 emissions and complicated supply chains.
- **Limitations on Scope:** For comprehensive GHG reporting, it is imperative to address scope restrictions and guarantee full coverage of emissions sources.
- **Standards Evolving:** Transparency and disclosure must be continuously prioritized to adjust to changing reporting frameworks and requirements, such as those recommended by the Task Force on Climate-related Financial Disclosures (TCFD).

#### 5. Evaluating Opportunities and Impacts:

- **Impact on Sustainability plans:** Stakeholder views, investment decisions, and business sustainability plans are all impacted by greenhouse gas reporting. Credible and transparent reporting builds confidence and improves reputation.
- **Possibilities for Improvement:** Identifying potential areas for improvement strengthens GHG reporting procedures and advances more general climate goals. Examples of these include improving data transparency, harmonizing with international standards, and utilizing emerging technology.

## Exercise



1. What are the key differences between absolute emissions and emission intensity, and how can each be used to assess environmental impact?
2. Explain how trend analysis can be utilized to evaluate the effectiveness of emission reduction strategies over time.
3. Identify and describe at least two major factors that influence global GHG reporting trends.
4. What are some common challenges in ensuring data quality and completeness in GHG reporting, particularly for Scope 3 emissions?
5. How can organizations leverage global GHG reporting trends to enhance their sustainability plans and identify opportunities for improvement?

## Unit 3.7. The concept of materiality and application in the reporting process

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Importance of materiality
2. Factors Influencing Materiality
3. Materiality in Reporting

### 3.7.1 Introduction

In greenhouse gas (GHG) reporting, materiality is a key concept that guarantees that displayed information appropriately reflects the substantial environmental implications of an organization's operations.

### 3.7.2 What is Materiality in GHG Reporting?

The term “materiality” in greenhouse gas reporting describes how important greenhouse gas emissions and related data are in influencing stakeholders' decisions about climate change mitigation, sustainability, and environmental performance. It involves figuring out if the facts on GHG emissions are significant enough to influence stakeholders' perceptions and decision-making.

### 3.7.3 Importance of Materiality in GHG Reporting

By ensuring that GHG reporting concentrates on relevant emissions sources and activities, materiality guarantees that stakeholders will receive insightful information about an organization's environmental performance. In regions where monitoring and lowering emissions has the biggest impact, it helps prioritize efforts and resources.

### 3.7.4 Application of Materiality in GHG Reporting

- **Finding and Prioritizing Important GHG Emissions Sources:** Materiality helps organizations identify and rank important sources and activities of GHG emissions. They are therefore able to concentrate their reporting initiatives and emission-reduction plans on the regions that have the biggest environmental effects.
- **Determining Reporting Borders:** When it comes to GHG reporting, materiality plays a role in determining reporting borders. Organizations give stakeholders a complete picture of their environmental footprint by determining which emissions sources and activities are relevant to them and include them in the reporting scope.
- **Choosing Emission Calculation methodology:** The choice of emission calculation methodology is guided by materiality concerns. To ensure data dependability and credibility for stakeholders, organizations place a high priority on precise measurement and reporting of emissions from key sources.
- **Disclosure and Transparency:** Organizations use materiality as a guide to decide how much information must be disclosed and in what depth in GHG reporting. Their primary objective is to facilitate informed decision-making and environmental

performance assessments by stakeholders by furnishing them with relevant and transparent information on material emissions sources.

- **Engagement of Stakeholders:** Materiality encourages participation from stakeholders in GHG reporting procedures. To identify important concerns, obtain relevant data, and effectively communicate about initiatives to reduce emissions and their effects on the environment, organizations work with stakeholders.

### Difference between Materiality in GHG reporting and Materiality in finance

|                           | Materiality in GHG Reporting                                                                                                                                                | Materiality in Finance Reporting                                                                                                                                              |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Definition                | Significance of greenhouse gas emissions in influencing stakeholders' understanding and decision-making related to environmental performance and climate change mitigation. | Importance or significance of financial information in influencing economic decisions of users of financial statements, such as investors, creditors, and other stakeholders. |
| Focus                     | Environmental impacts, sustainability, and climate change mitigation.                                                                                                       | Financial performance, financial health, and investment decisions.                                                                                                            |
| Scope                     | GHG emissions sources and activities, emission reduction strategies, and environmental footprint.                                                                           | Financial data, including revenues, expenses, assets, liabilities, and equity.                                                                                                |
| Reporting Boundaries      | Determination of which emissions sources and activities are material to stakeholders and should be included in the reporting scope.                                         | Establishment of materiality thresholds to decide what financial information is significant enough to be included in financial statements.                                    |
| Calculation Methodologies | Selection of emission calculation methodologies to accurately measure and report emissions from significant sources.                                                        | Application of materiality thresholds to assess the impact of financial information on users' economic decisions and determine reporting requirements.                        |
| Disclosure Requirements   | Transparent and relevant reporting on material emissions sources, enabling stakeholders to make informed decisions about environmental performance.                         | Disclosure of material financial information in financial statements and related disclosures to provide stakeholders with relevant information for economic decision-making.  |
| Stakeholder Engagement    | Collaboration with stakeholders to identify material issues, gather relevant data, and communicate effectively about emission reduction efforts and environmental impacts.  | Interact with stakeholders to understand their information needs, address concerns, and provide relevant financial information to support decision-making.                    |

## Exercise



1. What is the concept of materiality in GHG reporting, and why is it important for stakeholders?
2. How does materiality influence the identification and prioritization of significant GHG emission sources and activities?
3. In what ways does materiality affect the determination of reporting boundaries for GHG emissions?
4. Compare and contrast materiality in GHG reporting with materiality in financial reporting. What are the key differences in their focus and application?
5. How can organizations use materiality to enhance transparency and stakeholder engagement in their GHG reporting processes?

## Unit 3.8. Disclosing Greenhouse Gas Emission Inventory Data

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Standards and Frameworks
2. Outline of Reporting Process
3. Disclosure Platforms
4. Support Compliance and Best Practices

Environmental responsibility is becoming more and more important in today's society, especially when it comes to greenhouse gas (GHG) emissions. Organizations must disclose GHG emissions inventory data transparently to show their dedication to sustainability and climate action. To ensure transparency and credibility in reporting methods, this guide will offer a thorough overview of how to disclose GHG emission inventory data by various standards, frameworks, and platforms.

### 3.8.1 GHG Emission Inventory Data

Greenhouse gas (GHG) emissions are chemicals that trap heat in the Earth's atmosphere and cause global warming<sup>2</sup>. Examples of these gases are carbon dioxide (CO<sub>2</sub>), methane (CH<sub>4</sub>), and nitrous oxide (N<sub>2</sub>O).

Emissions generated by an organization's operations, both direct and indirect (from sources like fuel combustion and energy purchases), must be measured, and recorded in GHG emission inventory data.

### 3.8.2. Frameworks and Standards for Reporting GHGs:

1. **GHG Protocol:** The World Resources Institute (WRI) and the World Business Council for Sustainable Development (WBCSD) developed the Greenhouse Gas Protocol (GHG Protocol), which offers a framework for greenhouse gas accounting and reporting that is widely accepted.
2. **ISO 14064:** The international standard ISO 14064 describes the guidelines and specifications for measuring, tracking, and disclosing greenhouse gas emissions.
3. **CDP:** The CDP, or Carbon Disclosure Project: Companies, cities, states, and regions can measure and manage their environmental impacts, including GHG emissions, through the CDP, a global disclosure system.

### 3.8.3 Procedures for Publishing GHG Emission Inventory Information

- **Establish Boundaries:** For GHG accounting, establish the operational (such as scope 1, 2, and 3 emissions) and organizational (such as operational control or financial control) boundaries.
- **Gather Information:** Compile relevant data on fuel usage, energy consumption, transportation, and other activities that affect greenhouse gas emissions.
- **Compute Emissions:** To precisely measure GHG emissions, use emission factors and computation techniques outlined by selected standards.
- **Verify the Data:** To improve the reliability of GHG emission inventory data and guarantee compliance with reporting guidelines, take into account third-party verification.
- **Report Findings:** Write a thorough report with all the information from the GHG emissions inventory, including the methods, sources, and reduction efforts of the emissions.
- **Platforms for Disclosure:** To make GHG emission inventory data publicly available, select the right venues, such as CDP, sustainability reports, or government-mandated registries.

### 3.8.4. Upholding Credibility and Transparency

- record the methods, presumptions, and data sources that were used to calculate GHG emissions.
- Involve stakeholders and ask for input on GHG reporting procedures to promote accountability and transparency.
- Review and update the GHG emission inventory data regularly to account for modifications to operations, emissions variables, or reporting specifications.



Policy and Action Standard (pdf)



CDP High-Quality Mandatory Disclosure.pdf

## Exercise



1. What are the key frameworks and standards for GHG reporting, and what role does each play in the reporting process?
2. Describe the steps involved in the process of disclosing GHG emission inventory data, from establishing boundaries to reporting findings.
3. How can organizations ensure the credibility and transparency of their GHG emission inventory data?
4. What are some of the platforms available for disclosing GHG emission inventory data, and how do they contribute to public transparency?
5. Why is third-party verification important in the context of GHG emission inventory reporting, and how does it enhance the reliability of the reported data?

## Unit 3.9. Designing GHG Emission Reduction Targets

### Unit objectives



After reading this unit, you will be able to understand the following points:

1. Utilizing best practices
2. Considering long-term goals

Businesses must set targets for lowering greenhouse gas (GHG) emissions like how they do for revenue generation and product sales. These objectives assist businesses in monitoring their emissions reduction efforts in the same manner as they monitor their sales goals. Having broad goals for lowering GHG emissions is crucial for businesses developing strategies to reduce emissions from their daily operations and products. Setting company-wide targets is a key component of the plan, even in cases where some components of the organization are subject to regulations regarding their GHG emissions.

### 3.9.1 Importance of GHG reduction targets

- GHG reduction goals are crucial for directing companies' efforts to improve sustainability and lessen climate impact.
- Businesses can better align their goals, investments, and actions towards achieving carbon reductions by setting explicit targets.

Step 1 Obtain senior management commitment<sup>3</sup>

Step 2 Decide on the target type

- Set an absolute or intensity target?

Step 3 Decide on the target boundary

- Which GHGs to include?
- Which direct and indirect emissions?
- Which geographical operations?
- Treat business types separately?

Step 4 Choose the target base year

- Use a fixed or rolling approach?
- Use a single or multi-year approach?

Step 5 Define the target completion date

- Set a long- or short-term target?

Step 6 Define the length of the target commitment period

- Set a one-year or multi-year commitment period?

Step 7 Decide on the use of offsets or credits

<sup>3</sup> [http://pdf.wri.org/ghg\\_protocol\\_2004\\_chp011.pdf](http://pdf.wri.org/ghg_protocol_2004_chp011.pdf)

**Step 8 Establish a target double-counting policy**

- How to deal with double counting of reductions across companies?
- How does GHG trading affect target performance?

**Step 9 Decide on the target level**

- What is business-as-usual? How far to go beyond that?
- How do all the above steps influence the decision?

**Step 10 Track and report progress**

- Make regular performance checks
- Report information about the target

**3.9.2 Considerations for Setting Targets**

- When establishing GHG reduction targets, businesses should take into account their present emission levels, emissions intensity, sectoral benchmarks, and scientific studies on climate change.
- Setting goals that are both challenging and attainable while accounting for the organization's resources, capabilities, and technology improvements is essential.

**3.9.3 Various Target Types**

1. **Absolute targets:** They set a definite decrease in total greenhouse gas emissions over a predetermined period, regardless of growth or modifications to corporate operations.
2. **Intensity targets:** Intensity targets are designed to minimize emissions per unit of production while preserving flexibility to accommodate the expansion of businesses.
3. **Normalized Targets:** These modify the objectives for reducing emissions in response to variations in outside variables such as production levels or weather.

**3.9.4 Timelines for Targets**

**Short-term Targets:** These usually span one to five years and concentrate on upgrades and quick fixes to attain rapid emission reductions.

**Long-term Targets:** Cover broader strategic goals and transitions over five to twenty years or longer to achieve significant and long-lasting reductions in emissions.

**3.9.5 Engagement and Recommendation**

To improve transparency, accountability, and support for emission reduction activities, businesses are encouraged to involve stakeholders in the target-setting process, such as employees, investors, consumers, and communities.

**3.9.6 Monitoring and Reporting Progress**

- Tracking the achievement of GHG reduction targets requires the establishment of strong monitoring, reporting, and verification systems.
- Clear and open reporting of emissions data and updates on progress promotes confidence, makes benchmarking easier, and helps stakeholders evaluate how well emission reduction initiatives are working.

## Exercise



1. What are the key steps in designing effective GHG emission reduction targets, and how does each step contribute to setting a comprehensive target?
2. What are the differences between absolute targets, intensity targets, and normalized targets in GHG reduction, and how might an organization choose between them?
3. How do short-term and long-term GHG reduction targets differ in terms of focus and timeline, and what factors should organizations consider when setting each type of target?
4. What role does senior management commitment play in the process of setting GHG reduction targets, and why is it crucial for the success of these targets?
5. How can organizations ensure transparency and accountability in their GHG reduction efforts, and what are some best practices for monitoring and reporting progress?

## Unit 3.10. Identifying and Implementing Methodologies for GHG Emissions Reduction

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Utilizing international frameworks.
2. Setting targets based on specific methodologies.

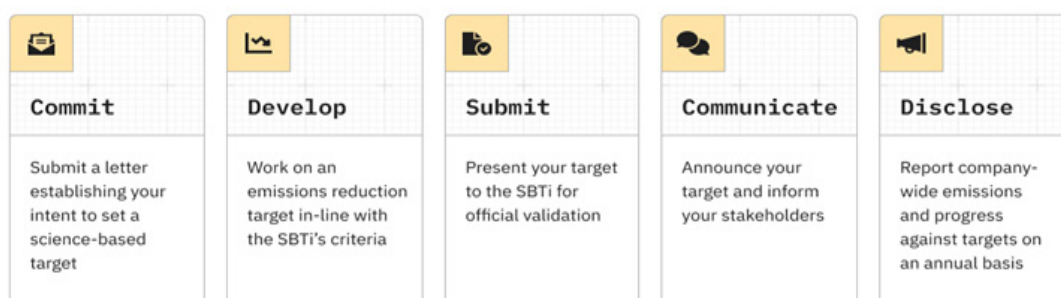
An essential component of addressing climate change is establishing targets for the reduction of greenhouse gas (GHG) emissions. Several international frameworks offer approaches for determining and carrying out these goals. Setting Greenhouse Gas (GHG) emissions reduction targets involves various methodologies and international frameworks.

### 3.10.1 Methodologies for Setting GHG Emissions Reduction Targets

#### 1. Science-Based Targets Initiative (SBTi)

Several organizations, including CDP, the World Wide Fund for Nature (WWF), the United Nations Global Compact (UNGC), and the World Resources Institute (WRI), are working together to lead the Science-Based Targets Initiative (SBTi). In keeping with the most recent climate science, it seeks to support businesses in establishing aggressive goals for reducing greenhouse gas emissions to keep global warming well below 2°C over pre-industrial levels, as stated in the Paris Agreement.

### A Step-by-Step Process



Source: Science Based Targets Initiative Annual Progress Report, 2021

Fig 3.10.1. SBTi

## Components of the SBTi:

### Science-Based Target Criteria:

- Compliance to the degree of decarbonization needed to limit the increase in world temperature to 1.5°C or below.
- Compliance with global and national climate objectives.
- Target setting that is transparent and robust in terms of methodology.

### Target Validation Process

Businesses send the SBTi their suggested targets to be validated. To make sure the targets have the necessary amount of ambition and scientific accuracy, a group of specialists evaluates them in comparison to the SBTi standards.

### Resources and Guidance

The SBTi provides organizations with tools, resources, and guidance materials to aid them in successfully navigating the target-setting process. This includes case studies showing best practices in creating and achieving SBTs as well as sector-specific guidelines.

### Acknowledgment and Support

Businesses are commended for their leadership in climate action when they set and meet SBTs by SBTi standards. To assist businesses in achieving their goals and monitoring their advancement over time, the SBTi offers continuous support and direction.

### Advantages of SBTi

- Enhances the company's reputation as a sustainability leader.
- Aids in identifying and mitigating climate-related risks.
- Encourages innovation in low-carbon technologies and processes.
- Drives investor confidence towards sustainable businesses.

## 2. Top-Down and Bottom-Up Approaches in GHG Emissions Reduction

Top-down and bottom-up are the two primary methodologies that are frequently used. When attempting to address the difficult task of mitigating climate change, each strategy has unique benefits and things to keep in consideration.

### Top-Down Approach

The top-down strategy is defining broad targets for reducing greenhouse gas emissions at a high level, usually at the national or worldwide level, and then dissecting those targets into organizational or sector-specific goals.

**Strategic Focus:** Government regulations, international agreements (like the Paris Agreement), or industry-wide initiatives are usually the sources of top-down approaches. For companies and industries involved in reducing emissions, they offer strategic direction and assistance.

**Standardization:** To ensure uniformity and comparability of emissions reduction efforts, this technique frequently depends on standardized methodologies and protocols for setting objectives and tracking progress.

**National Commitments:** Governments are essential in establishing aggressive goals for reducing greenhouse gas emissions and putting regulations, legislation, and incentives in place to promote compliance and lower emissions.

**Sectoral Allocation:** Based on each sector's relative contribution to overall emissions, targets may be assigned, with particular reduction objectives adapted to suit each sector's capabilities and features.

### Bottom-Up Approach

The bottom-up strategy refers to grassroots initiatives in which certain groups, communities, or industries establish their own goals for reducing their greenhouse gas emissions by their particular needs, interests, and capacities.

**Local Context:** Bottom-up methods are context-specific, accounting for the specific challenges, opportunities, and assets that any community or organization faces. They provide stakeholders the authority to take charge of programs aimed at reducing emissions.

**Flexibility:** Businesses can create customized plans and goals that meet stakeholder expectations, operational limitations, and corporate objectives. This encourages creativity and originality in locating affordable emissions reduction strategies.

**Engagement and Participation:** Bottom-up strategies place a strong emphasis on engaging and collaborating with a variety of stakeholders, including local communities, suppliers, employees, and customers.

**Aggregation:** Although individual targets are established, bottom-up strategies may include combining emissions reduction initiatives from higher levels (such as industry groups and regional coalitions) to attain a larger and more collective impact.

### 3.10.2 Implementing Methodologies

- Compile information on greenhouse gas emissions from different sources inside the organization while adhering to the rules and procedures that the technique has been selected.
- To precisely calculate GHG emissions, use activity data, emission factors, and emission calculation tools.
- To determine baseline emissions and define reduction targets based on reduction scenarios and goals, apply the proper accounting principles and procedures.

### 3.10.3 Integrating with Organizational Strategies

- Incorporate targets for reducing greenhouse gas emissions into your company's overall strategic planning process, taking stakeholder participation, cost-effectiveness, and risk management into account.
- Ascertain conformity with additional sustainability objectives, legal mandates, and industry standards.

### 3.10.4 Monitoring and Reporting Progress

- Provide effective reporting, monitoring, and verification systems to monitor the achievement of GHG emission reduction goals.
- Review and evaluate performance compared to goals regularly, making necessary adjustments to plans and tactics to keep on course.
- Be transparent and forthright in exchanges with stakeholders regarding the obstacles, successes, and state of the GHG emission reduction.

Organizations can effectively develop, implement, and accomplish GHG emissions reduction targets by adhering to these procedures and utilizing approaches from several international frameworks. This will help contribute to global efforts aimed at mitigating climate change.

## Exercise



1. What are the key components and advantages of the Science-Based Targets Initiative (SBTi), and how can organizations benefit from setting targets that align with SBTi criteria?
2. How do top-down and bottom-up approaches differ in setting GHG emissions reduction targets, and what are the unique benefits and considerations associated with each methodology?
3. What steps should organizations take to implement GHG emissions reduction methodologies effectively, and how can they ensure accurate calculation and reporting of emissions?
4. How can organizations integrate GHG emissions reduction targets into their overall strategic planning process, and what factors should be considered to align these targets with broader sustainability goals and regulatory requirements?
5. What best practices should organizations follow for monitoring and reporting progress towards GHG emissions reduction goals, and how can transparency and stakeholder engagement be maintained throughout this process?



## 4. Introduction to Business Responsibility and Sustainability Reporting (BRSR).

- Unit 4.1 The rationale for businesses to report on their sustainability efforts, and recognize how incorporating sustainability practices can contribute to environmental and social sustainability.
- Unit 4.2 Business Responsibility and Sustainability Reporting (BRSR), outline its key stakeholders, and how to assess requirements & interpret the principles of BRSR.
- Unit 4.3 How to comply with the national standards guiding Indian companies in their sustainability efforts.
- Unit 4.4 Differentiate between the earlier BRR and the BRSR framework
- Unit 4.5 How to Assist businesses in integrating sustainability considerations into their operational processes.
- Unit 4.6 Identify the specific disclosure requirements for each section of the BRSR report.
- Unit 4.7 How to develop a comprehensive Business Responsibility and Sustainability Report.
- Unit 4.8 National Guidelines on Responsible Business Conduct principles and how to assess and link a company's ESG performance with that.
- Unit 4.9 Discuss how to conduct a thorough analysis of gaps in BRSR reports to ensure compliance
- Unit 4.10 Discuss how to incorporate the Sustainable Development Goals into BRSR reports
- Unit 4.11 Explain how to recognize the frameworks, guiding principles, and connections of global reporting to BRSR.
- Unit 4.12 Explain the importance of assuring corporate sustainability disclosures to gain an understanding of the requirements and methodology





## Chapter Highlights

**After reading this chapter, you will be able to Understand**

1. Application of Sustainability Reporting Principles.
2. Develop an Understanding of Business Responsibility Reporting (BRR) and Business Responsibility and Sustainability Reporting (BRSR) and its application.
3. Develop a Comprehensive Understanding of BRSR Structure to Navigate and Apply BRSR Guidelines.
4. Understanding the Linkage and Differences between BRSR and International Frameworks and the Importance of Independent Third-Party Assurance.

## Unit 4.1. Business Sustainability Reporting

### Unit Objectives

At the end of this unit, you will be able to:

1. Rationale for sustainability reporting.
2. Recognition of sustainability practices' environmental and social contribution.



In the present economic climate, sustainability has become a critical component for businesses all over the world. Businesses that disclose to stakeholders their environmental, social, and governance (ESG) performance are deemed engaging in sustainability reporting.

### 4.1.1 The Significance of Sustainability Reporting for Businesses

#### Accountability and transparency

Sustainability reporting encourages accountability by enabling stakeholders to assess a company's impact on society and the environment. It increases stakeholder confidence and trust by making companies accountable for their actions.

#### Risk management

Businesses can identify and reduce risks related to social and environmental issues by promoting sustainability efforts and performance. By proactively controlling these risks, long-term business viability can be protected.

#### Competitive Advantage

Businesses can stand out in the market by implementing sustainable practices. Businesses that exhibit a dedication to sustainability tend to draw in investors, workers, and customers who share their values, giving them a competitive advantage.

#### Regulatory Compliance

Sustainability reporting is required by regulation in many places or is financially rewarded. Following regulations guarantees legal compliance and demonstrates a company's commitment to ethical business conduct.

#### Investor Confidence

When making investment decisions, investors are taking environmental, social, and governance (ESG) aspects into account more and more. Evaluating a company's sustainability performance through comprehensive sustainability reporting boosts investor trust and draws sustainable investment.

### 4.1.2 Environmental Sustainability through the Integration of Sustainability Practices

#### **Incorporating Sustainability Resource Efficiency**

Reducing waste, conserving water, and implementing energy-efficient techniques all help to decrease the use of resources and their negative effects on the environment.

#### **Adoption of Renewable Energy**

By moving away from fossil fuels and toward renewable energy sources like solar or wind power, greenhouse gas emissions are reduced, and climate change is mitigated.

#### **Waste Management**

By integrating recycling, reuse, and appropriate disposal into practice, landfill trash is reduced and a circular economy is fostered.

#### **Ecosystem Preservation**

By safeguarding and rehabilitating natural habitats, biodiversity, and ecosystems impacted by their operations, businesses can support environmental sustainability.

### 4.1.3 Integrating Sustainability Methods for Ensuring Social Sustainability

#### **Community Engagement**

Getting involved in neighborhood collaborations, philanthropic endeavors, and community development initiatives promotes social cohesiveness and boosts the local economy.

#### **Labor Practices**

Social sustainability and employee happiness are improved by ensuring fair labor practices, which include equal pay, safe working conditions, and employee welfare programs.

#### **Diversity and Inclusion**

Developing a more diverse and inclusive workforce and supply chain stimulates creativity, improves judgment, and builds a more just society.

#### **Human Rights Protection**

Encouraging social sustainability and maintaining moral standards depend on respecting human rights at every stage of the value chain, including dealing with problems like child labor and forced labor.

By incorporating sustainability practices, businesses can contribute to both environmental and social sustainability, playing a vital role in addressing global challenges and creating a more sustainable future for generations to come.

## Exercise



1. How can businesses effectively leverage sustainability reporting to enhance transparency and build trust with stakeholders?
2. What are some practical steps companies can take to integrate resource efficiency and renewable energy into their operations to improve environmental sustainability?
3. In what ways can sustainability reporting help businesses manage risks related to social and environmental issues, and how can this contribute to long-term viability?
4. How can businesses ensure that their labor practices and diversity initiatives align with social sustainability goals while fostering a positive and inclusive work environment?
5. What are the key benefits of adopting sustainability practices in terms of competitive advantage, regulatory compliance, and attracting investors?

## Unit 4.2. Business Responsibility and Sustainability Reporting (BRSR)

### Unit Objectives

At the end of this unit, you will be able to:

1. Key stakeholders.
2. Assesses requirements and interprets principles.



The “Voluntary Guidelines on Corporate Social Responsibility” are the source of Business Responsibility and Sustainability Reporting (BRSR) in India. The Ministry of Corporate Affairs in India developed these rules in 2009, and they are regarded as the initial move in bringing the idea of corporate responsibility to the Indian market.

### 4.2.1 What is BRSR?

The Securities and Exchange Board of India (SEBI) standardized ESG disclosures for listed firms, focusing on Business Responsibility and Sustainability.

#### SEBI’s Guidelines for Business Responsibility and Sustainability (BRSR)

- By emphasizing the environmental, social, and governance components, BRSR seeks to optimize business impact.
- Businesses can compare sustainability reports with globally recognized disclosure guidelines such as TCFD, SASB, GRI, etc.
- Places a strong emphasis on education and awareness, mandating businesses to provide information about worker health and safety protocols, anti-corruption initiatives, and labor upskilling.
- Disclosures on any Environmental or Social Impact Assessments that listed companies have completed are requested under the new reporting structure.
- The updated framework pushes businesses to report on “essential” and “leadership” metrics and supports the use of the Key Performance metrics (KPIs) model.

#### Business Sustainability’s (BRSR) benefits <sup>1</sup>

- **Greater Value Creation:** Organizations that include sustainability in their core competencies can build more resilient, competitively stronger enterprises over time. The growing use of ESG performance data in investment choices has led to an increase in investor interest in global sustainability frameworks such as SASB and GRI.
- **Stronger Brand Positioning:** Sustainability raises a company’s brand value and attracts investors, which helps with funding.
- **Talent Attraction and Retention:** Organizations that employ industry best practices and responsibly oversee operations are drawing in and keeping top talent.

<sup>1</sup> Business Responsibility and Sustainability Reporting (BRSR) Report by EY

### Key Stakeholders

|              |                                                                                                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------|
| Investors    | Benefit from strong financial returns, efficient operations employing best practices, and corporate governance. |
| Shareholders | Integrity and transparency, creating and adhering to a code of conduct for business                             |
| Government   | Enough benefits and pay for employees, as well as occupational health and safety                                |
| Communities  | Value for money, high-quality of goods and services, safety and content of products                             |
| Employees    | Abiding by rules and regulations and collaborating openly and honestly with government agencies                 |
| Clients      | Local employment, little environmental impact, adherence to regulations                                         |

### Structure of BRSR

- 1) **Section A: General Disclosures**
  - a. Offers fundamental details about the listed company, such as its assets, joint ventures, subsidiary firms, workforce, operations, and standards for transparency.
- 2) **Section B: Disclosures Regarding Management and Process**
  - a. Demands that businesses publish their NGRBC-related policies and procedures, including those about stakeholder participation, leadership, and governance.
  - b. Information mostly relates to management procedures, leadership concerns, governance, and supervision.
- 3) **Section C: Performance Disclosures according to Principles**
  - a. Demands that businesses show their dedication to ethical business practices by their deeds and results.
  - b. Businesses have to provide KPIs that support the nine responsible business conduct principles set forth by the NGRBC.
- 4) Leadership indicators are optional but expected for increased accountability and transparency.
- 5) Essential indicators are required and include environmental statistics, training, community activities, and social effects.

C



Image Source :BSRS-EY

Fig 4.2.1

## Exercise



1. How does the Business Responsibility and Sustainability Reporting (BRSR) framework enhance transparency and accountability for listed companies in India?
2. What are the key differences between the “essential” and “leadership” metrics in the BRSR framework, and how do they impact reporting practices?
3. In what ways does BRSR benefit various stakeholders, including investors, shareholders, and employees?
4. How can businesses effectively integrate the BRSR guidelines with globally recognized sustainability frameworks such as TCFD, SASB, and GRI?
5. What are the key components of the BRSR structure, and how do the General Disclosures, Management and Process Disclosures, and Performance Disclosures contribute to overall business sustainability?

## Unit 4.3. Compliance with National Standards (Indian Standards)

### Unit Objectives

At the end of this unit, you will be able to:

1. Understanding National Standards.
2. Integrating Sustainability into Business Operations
3. Environmental and Social Compliance



Indian corporations have placed an increasing emphasis on ESG reporting and disclosure in recent times. Regulators like the Securities and Exchange Board of India (SEBI) have pushed for the voluntary adoption of ESG reporting criteria by listed corporations. Furthermore, entities such as the National Stock Exchange (NSE) have introduced ESG indexes to monitor the success of businesses according to their ESG ratings.

### 1. Environmental, Social, and Governance (ESG) Criteria

ESG (environment, social, and governance) factors are becoming more and more significant for both businesses and investors. Indian businesses are under close examination for their effects on the environment, society, and governance standards in addition to their financial performance.

#### Environmental Criteria

Deforestation, pollution, water scarcity, and climate change are just a few of the major environmental issues India faces. The evaluation of Indian enterprises is dependent upon their endeavors to address these concerns and foster environmental sustainability. When assessing a company's environmental performance, it is important to consider factors including energy efficiency, waste management techniques, adoption of clean technology, compliance with environmental standards, and measures to reduce carbon footprint.

#### Social Criteria

In a nation as diverse as India, social concerns are crucial. Indian businesses are evaluated on how they treat employees, interact with the community, and address larger societal issues. Social criteria encompass several aspects such as labor practices, diversity and welfare of employees, health and safety regulations, community development programs, assistance for education and skill enhancement, and endeavors to tackle social disparities.

#### Governance Criteria

To guarantee accountability, ethics, and transparency in Indian businesses, governance standards are essential. Over time, India's corporate governance practices have significantly improved, but there are still issues, especially about board independence, CEO compensation, shareholder rights, and financial reporting transparency. Companies with robust risk management programs, solid governance

frameworks, and a dedication to following ethical guidelines are sought after by investors.

## 2. Business Responsibility and Sustainability Report (BRSR)

With significant importance, the Business Responsibility and Sustainability Report (BRSR) is a crucial component of corporate reporting in India.

### Why it matters?

**Regulatory Influence:** To promote ESG (Environmental, Social, and Governance) reporting among listed businesses, SEBI is a vital player. Top-ranked corporations are required to submit the Business Responsibility and Sustainability Report (BRSR)

**Objective and Content:** A company's financial results and its environmental, social, and governance (ESG) performance are linked by the BRSR. In keeping with the Ministry of Corporate Affairs' National Guidelines on Responsible Business Conduct, it provides a thorough summary of a company's business performance and its impacts.

**Challenges and Structure:** Compiling and disclosing ESG information presents difficulties, necessitating accuracy and accountability. About 300 data elements make up the BRSR framework, which emphasizes the significance of data availability, accessibility, correctness, and auditability.

## 3. National Guidelines on Responsible Business Conduct

In India, a framework for advancing moral and sustainable business practices in a range of industries is provided by the National Guidelines on Responsible Business Conduct. The Ministry of Corporate Affairs has released these principles to motivate companies to conduct their operations ethically while taking into account their effects on government, the environment, and society.

### Indian Business Guidelines

- Encourage ethical business practices among Indian enterprises.
- Address topics including corporate governance, consumer protection, environmental protection, labor policies, and human rights.
- Relevant to all kinds of businesses, including multinationals, SMEs, and large corporations.
- Founded on values such as human rights dignity, accountability, transparency, and equity.
- Promote voluntary compliance while offering corporate governance guidelines.
- Promote reporting on guidelines compliance to show a dedication to moral and environmentally friendly behavior.

## 4. India and Sustainability Standards (ISS)

India and Sustainability Standards (ISS) describes the steps and activities that India has made in several economic sectors to uphold international sustainability standards and encourage sustainability. To ensure ethical and sustainable corporate practices, these standards take environmental, social, and governance (ESG) factors into account.

### India's Sustainability Standards and Policies

#### a. Adoption of International Standards

- India has adopted international frameworks and guidelines from organizations like ISO, GRI, and UNGC.

#### b. Development of National Standards:

- The Bureau of Indian Standards (BIS) has developed standards for environmental management, energy efficiency, and CSR reporting.

#### c. Integration into Policies and Regulations

- Sustainability standards are integrated into national and state policies and regulations.
- SEBI mandates companies to disclose ESG performance, and MoEFCC promotes sustainable practices in industries.

**d. Capacity Building and Awareness:**

- India actively supports sustainability standards implementation through training programs, workshops, seminars, and awareness campaigns.

**e. Partnerships and Collaborations:**

- India collaborates with international organizations, governments, civil society groups, and businesses to advance sustainability standards and practices.

**5. National Voluntary standards (NVGs)**

The Government of India introduced the National Voluntary Standards (NVGs) as a collection of standards to encourage companies to voluntarily embrace sustainable and ethical practices. These recommendations give companies a framework for incorporating ethical, social, and environmental factors into their daily operations and decision-making.

**The Development and Goals of NVGs**

- Created by the Ministry of Corporate Affairs in association with relevant parties to encourage corporate citizenship and ethical business practices.
- Addresses corporate governance, stakeholder participation, labor practices, environmental sustainability, and human rights.
- Relevant to all kinds of enterprises, including multinationals, SMEs, and big organizations.

**Key Principles**

- Predicated on human rights respect, responsibility, openness, and equity.
- Places a strong emphasis on how business strategies include social and environmental factors.
- Promotes appropriate and transparent stakeholder involvement.

**Execution and Documentation**

- Although NVG adoption as part of a company's corporate governance framework is elective, it is encouraged.
- Adoption boosts a company's competitiveness and reputation.

Indian businesses may make sure they are meeting national sustainability requirements by following these rules and principles. Furthermore, it is critical to remember that these standards are always changing, so being up to date on the most recent advancements is essential.

## Exercise



1. How do the Environmental, Social, and Governance (ESG) criteria impact the assessment of Indian businesses, and what specific factors are considered under each criterion?
2. What are the main objectives and content of the Business Responsibility and Sustainability Report (BRSR), and what challenges do companies face in compiling and disclosing ESG information under this framework?
3. How do the National Guidelines on Responsible Business Conduct (NGRBC) influence corporate practices in India, and what are the key areas addressed by these guidelines?
4. In what ways has India integrated international sustainability standards into its national policies, and how does this integration support ethical and sustainable corporate practices?
5. What are the key principles and goals of the National Voluntary Standards (NVGs), and how do they encourage businesses to adopt sustainable and ethical practices?

## Unit 4.4. BRSR Framework vs BRR

### Unit Objectives

At the end of this unit, you will be able to:

1. Differentiate between earlier frameworks.
2. BRSR framework for better decision-making.



### Key Difference between BRSR and BRR <sup>2</sup>

| Aspect                       | BRR (Business Responsibility Reporting)                                          | BRSR (Business Responsibility and Sustainability Reporting)                                                        |
|------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Focus                        | Emphasis on ethical practices, business integrity, and social responsibilities.  | Expands beyond responsibility to encompass sustainability and ESG factors.                                         |
| Scope                        | Primarily covers ethical behavior, transparency, social welfare, and governance. | Encompasses environmental conservation, social impact, sustainable development, and governance for sustainability. |
| Environmental Considerations | May touch on limited environmental aspects, but not a primary focus.             | Places a strong emphasis on a company's environmental impact and sustainable practices.                            |
| Social Impact                | Addresses social welfare initiatives and ethical conduct.                        | Goes deeper into human rights, labor practices, supply chain ethics, and social equity.                            |
| Sustainable Development      | Not a core focus, although some initiatives may align with sustainability.       | Highlights strategies and actions toward long-term sustainable development.                                        |
| ESG Integration              | Not inherently integrated; focuses on business responsibility.                   | Inherently integrates ESG (Environmental, Social, Governance) principles.                                          |
| Strategic Implications       | Demonstrates commitment to ethics and social responsibility.                     | Showcases commitment to sustainable development, environmental consciousness, and holistic stakeholder value.      |

<sup>2</sup> BRSR Report –4C Consulting Private Limited

| Aspect                    | BRR (Business Responsibility Reporting)                                                 | BRSR (Business Responsibility and Sustainability Reporting)                                           |
|---------------------------|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Alignment with ESG Trends | May not fully align with evolving ESG (Environmental, Social, Governance) expectations. | Aligns well with increasing emphasis on ESG integration in reporting.                                 |
| Future Readiness          | May require additional reporting to address sustainability and ESG demands.             | Addresses both current and anticipated demands for sustainability and ESG disclosures.                |
| Stakeholder Signaling     | Signals commitment to ethical conduct and social well-being.                            | Signals proactive engagement with sustainability, environmental issues, and long-term value creation. |

## Exercise



1. How does the focus of BRSR on sustainability and ESG factors differ from BRR's emphasis on ethical practices and social responsibilities?
2. What are the key environmental and social considerations that BRSR covers that BRR may not, and how does this reflect a shift towards comprehensive sustainability reporting?
3. In what ways does BRSR align with current ESG trends compared to BRR, and how does this impact the strategic implications for companies adopting BRSR?
4. How does the adoption of BRSR address future demands for sustainability and ESG disclosures compared to BRR, and what additional reporting requirements might companies face with BRR?

## Unit 4.5. Business Sustainability Integration

### Unit Objectives

At the end of this unit, you will be able to:

1. Assisting in operational processes.
2. Ensuring sustainability considerations.



Promoting ethical and sustainable business practices requires helping companies incorporate sustainability concerns into their daily operations.

### Evaluation and Awareness

To begin, ascertain the present condition of sustainability in the company. Determine the sustainability-related possibilities, risks, vulnerabilities, and strengths. Inform important stakeholders on the value of sustainability and its possible advantages for the company and society at large, such as management, staff, and suppliers.

### Establish Specific Goals and Objectives

Establish specific sustainability goals and objectives that are in line with the overarching purpose, vision, and values of the company. They ought to be SMART objectives—specific, measurable, realistic, relevant, and time-bound. Reducing carbon emissions, cutting waste production, improving employee well-being, and encouraging community involvement are a few examples of sustainability objectives.

### Integrating Operations with Strategy

Incorporate sustainability concerns into the strategic planning procedure of the company. Make sure that the creation of new products, purchasing, production, distribution, marketing, sales, and customer service are all connected with sustainability. Promote interdepartmental cooperation and communication to guarantee that sustainability is taken into account along the whole value chain.

### Employee Education and Engagement

Involve staff members at all organizational levels in the sustainability initiative. Offer staff capacity-building and training programs to improve their knowledge and abilities about sustainability. Employees should be given the freedom to see areas for improvement and carry out long-term fixes within their spheres of influence.

### Engagement and Collaboration with Stakeholders

Work together to address sustainability concerns with external stakeholders, including suppliers, customers, regulators, non-governmental organizations, and local communities. Create open lines of communication to get input, resolve issues, and form alliances for sustainable

Growth. Participate in peer networks and industry groups to exchange best practices and lessons discovered.

### Monitoring, Measuring, and Reporting:

Create measurements and key performance indicators (KPIs) to monitor the advancement of sustainability goals and objectives. Put in place procedures and systems for tracking and evaluating economic, social, and environmental performance. Provide internal and external stakeholders with regular updates on the company's sustainability performance through websites, annual reports, sustainability reports, and other channels of communication.

## Innovation and Continuous Improvement

To promote sustainable innovation inside the company, cultivate an environment that values innovation and continuous improvement. To find and use innovative sustainable practices, business models, and technology, promote experimentation and risk-taking. Recognize accomplishments and draw lessons from mistakes to keep improving the company's sustainability performance.

## Compliance and Risk Management

Make sure that all applicable laws, rules, and international sustainability standards are followed. Put in place effective risk management systems to recognize, evaluate, reduce, and keep an eye on risks associated with sustainability. To stay ahead of the curve, proactively address new sustainability challenges and forecast developing trends.

Some examples of successful sustainability integration

- Rothy's: This Company uses recycled plastic bottles as a production material for shoes, bags, and face coverings.
- Bank of America: They have made significant strides in sustainability, including committing to carbon neutrality and 100% renewable electricity by 2020
- Etsy: Etsy achieved carbon-neutral shipping, meaning they offset the carbon dioxide produced by shipping items<sup>3</sup>.
- AstraZeneca: This pharmaceutical company has an 'Ambition Zero Carbon' strategy that aims to eliminate emissions by 2025 and be carbon-negative across their entire value chain by 2030<sup>4</sup>.
- Ben & Jerry's: This ice cream company is known for its social activism and commitment to fair trade and non-GMO ingredients<sup>5</sup>.
- The High Line, New York City: This renowned urban park was built on a historic elevated railway track. It involved transforming a disused industrial structure into a green public space<sup>6</sup>.

3 <https://www.etsy.com/news/on-etsy-every-purchase-makes-a-positive-impact>

4 <https://www.astrazeneca.com/media-centre/press-releases/2020/astrazenecas-ambition-zero-carbon-strategy-to-eliminate-emissions-by-2025-and-be-carbon-negative-across-the-entire-value-chain-by-2030-22012020.html>

5 <https://www.linkedin.com/pulse/socially-responsible-practices-ben-jerrys-divya-yadav>

6 <https://instituteprojectmanagement.com/blog/sustainability-integration-in-project-management>

## Exercise



1. What initial steps should a company take to evaluate its current sustainability practices and raise awareness among stakeholders?
2. How can companies establish SMART sustainability goals and objectives that align with their overall mission and values?
3. In what ways can sustainability concerns be integrated into a company's strategic planning and daily operations?
4. What strategies can be employed to effectively engage and educate employees about sustainability, and why is their involvement crucial?
5. How should companies approach collaboration with external stakeholders to address sustainability issues and foster sustainable growth?
6. What are some key performance indicators (KPIs) and measurement systems that companies can use to track and report their sustainability progress?
7. How can companies cultivate a culture of innovation and continuous improvement to advance their sustainability efforts?
8. What steps should companies take to ensure compliance with sustainability regulations and effectively manage associated risks?

## Unit 4.6. BRSR Report Disclosure Requirements

### Unit Objectives

At the end of this unit, you will be able to:

1. Identify specific sections.
2. Ensure clarity.



The new BRSR has been designed to be compatible with other globally recognized reporting frameworks, including the Global Reporting Initiative (GRI), the Sustainability Accounting Standards Board (SASB), and the Task Force on Climate-Related Financial Disclosures (TCFD).

The BRSR starting with April 2022 to March 2023, compels Indian businesses to publish quantitative measurements on sustainability-related aspects.\*

**What are the main disclosure areas under the BRSR?**

#### *Section A: The general disclosures*

This includes the company's office and plant addresses, product and service details, stock exchanges on which the company is listed, and reporting boundaries, which indicate whether the disclosures pertain to a single company or a consolidated portfolio of companies. In addition, information on employee turnover and the total number of complaints and grievances lodged against the business are necessary. An employee counts and representation by gender, diversity, and inclusion, and differently-abled status are also necessary.

#### *Section B: Management and Process Disclosures*

The purpose of this part is to assist businesses in showcasing the systems, guidelines, and procedures they have implemented to embrace the NGRBC Core Elements and Principles.

#### *Section c: principle performance disclosure*

The purpose of this part is to assist entities in demonstrating how well they integrate the Principles and Core Elements with important choices and procedures. The requested material falls within the categories of "Leadership" and "Essential." While all organizations required to file this report are obliged to reveal the essential indicators, entities that aim to advance to a higher degree of socially, ecologically, and morally responsible behavior may choose to voluntarily publish the leadership indicators.

**Annexure I****BUSINESS RESPONSIBILITY & SUSTAINABILITY REPORTING FORMAT****SECTION A: GENERAL DISCLOSURES****I. Details of the listed entity**

1. Corporate Identity Number (CIN) of the Listed Entity
2. Name of the Listed Entity
3. Year of incorporation
4. Registered office address
5. Corporate address
6. E-mail
7. Telephone
8. Website
9. Financial year for which reporting is being done
10. Name of the Stock Exchange(s) where shares are listed
11. Paid-up Capital
12. Name and contact details (telephone, email address) of the person who may be contacted in case of any queries on the BRSR report
13. Reporting boundary – Are the disclosures under this report made on a standalone basis (i.e. only for the entity) or on a consolidated basis (i.e. for the entity and all the entities which form a part of its consolidated financial statements, taken together).

**II. Products/services**

14. Details of business activities (accounting for 90% of the turnover):

| S. No. | Description of Main Activity | Description of Business Activity | % of Turnover of the entity |
|--------|------------------------------|----------------------------------|-----------------------------|
|        |                              |                                  |                             |
|        |                              |                                  |                             |

15. Products/Services sold by the entity (accounting for 90% of the entity's Turnover):

| S. No. | Product/Service | NIC Code | % of total Turnover contributed |
|--------|-----------------|----------|---------------------------------|
|        |                 |          |                                 |
|        |                 |          |                                 |
|        |                 |          |                                 |

**III. Operations**

16. Number of locations where plants and/or operations/offices of the entity are situated:

| Location      | Number of plants | Number of offices | Total |
|---------------|------------------|-------------------|-------|
| National      |                  |                   |       |
| International |                  |                   |       |

17. Markets served by the entity:

a. Number of locations

| Locations                        | Number |
|----------------------------------|--------|
| National (No. of States)         |        |
| International (No. of Countries) |        |

b. What is the contribution of exports as a percentage of the total turnover of the entity?

c. A brief on types of customers

**IV. Employees**

18. Details as at the end of Financial Year:

a. Employees and workers (including differently abled):

| S. No.           | Particulars              | Total (A) | Male    |           | Female  |           |
|------------------|--------------------------|-----------|---------|-----------|---------|-----------|
|                  |                          |           | No. (B) | % (B / A) | No. (C) | % (C / A) |
| <b>EMPLOYEES</b> |                          |           |         |           |         |           |
| 1.               | Permanent (D)            |           |         |           |         |           |
| 2.               | Other than Permanent (E) |           |         |           |         |           |
| 3.               | Total employees (D + E)  |           |         |           |         |           |
| <b>WORKERS</b>   |                          |           |         |           |         |           |
| 4.               | Permanent (F)            |           |         |           |         |           |
| 5.               | Other than Permanent (G) |           |         |           |         |           |
| 6.               | Total workers (F + G)    |           |         |           |         |           |

**b. Differently abled Employees and workers:**

| S. No                                     | Particulars                               | Total<br>{A} | Male    |           | Female  |           |
|-------------------------------------------|-------------------------------------------|--------------|---------|-----------|---------|-----------|
|                                           |                                           |              | No. {B} | % {B / A} | No. {C} | % {C / A} |
| <b><u>DIFFERENTLY ABLED EMPLOYEES</u></b> |                                           |              |         |           |         |           |
| 1.                                        | Permanent (D)                             |              |         |           |         |           |
| 2.                                        | Other than Permanent (E)                  |              |         |           |         |           |
| 3.                                        | Total differently abled employees (D + E) |              |         |           |         |           |
| <b><u>DIFFERENTLY ABLED WORKERS</u></b>   |                                           |              |         |           |         |           |
| 4.                                        | Permanent (F)                             |              |         |           |         |           |
| 5.                                        | Other than permanent (G)                  |              |         |           |         |           |
| 6.                                        | Total differently abled workers (F + G)   |              |         |           |         |           |

**19. Participation/Inclusion/Representation of women**

|                          | Total<br>(A) | No. and percentage of Females |           |
|--------------------------|--------------|-------------------------------|-----------|
|                          |              | No. (B)                       | % (B / A) |
| Board of Directors       |              |                               |           |
| Key Management Personnel |              |                               |           |

## 20. Turnover rate for permanent employees and workers

*(Disclose trends for the past 3 years)*

[illegible]

V. Holding, Subsidiary and Associate Companies (including joint ventures)

21. (a) Names of holding / subsidiary / associate companies / joint ventures

| S. No. | Name of the holding / subsidiary / associate companies / joint ventures (A) | Indicate whether holding/ Subsidiary/ Associate/ Joint Venture | % of shares held by listed entity | Does the entity indicated at column A, participate in the Business Responsibility initiatives of the listed entity? (Yes/No) |
|--------|-----------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|        |                                                                             |                                                                |                                   |                                                                                                                              |

VI. CSR Details

22. (i) Whether CSR is applicable as per section 135 of Companies Act, 2013: (Yes/No)  
 (ii) Turnover (in Rs.)  
 (iii) Net worth (in Rs.)

VII. Transparency and Disclosures Compliances

23. Complaints/Grievances on any of the principles (Principles 1 to 9) under the National Guidelines on Responsible Business Conduct:

| Stakeholder group from whom complaint is received | Grievance Redressal Mechanism in Place (Yes/No)<br><br>(If Yes, then provide web-link for grievance redress policy) | FY ____<br>Current Financial Year          |                                                              |         | FY ____<br>Previous Financial Year         |                                                              |         |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------------------------------|---------|--------------------------------------------|--------------------------------------------------------------|---------|
|                                                   |                                                                                                                     | Number of complaints filed during the year | Number of complaints pending resolution at close of the year | Remarks | Number of complaints filed during the year | Number of complaints pending resolution at close of the year | Remarks |
| Communities                                       |                                                                                                                     |                                            |                                                              |         |                                            |                                                              |         |
| Investors (other than shareholders)               |                                                                                                                     |                                            |                                                              |         |                                            |                                                              |         |

| Stakeholder group from whom complaint is received | Grievance Redressal Mechanism in Place (Yes/No)<br><br>(If Yes, then provide web-link for grievance redress policy) | FY ____<br>Current Financial Year          |                                                              |         | FY ____<br>Previous Financial Year         |                                                              |         |
|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------------------------------------------------------|---------|--------------------------------------------|--------------------------------------------------------------|---------|
|                                                   |                                                                                                                     | Number of complaints filed during the year | Number of complaints pending resolution at close of the year | Remarks | Number of complaints filed during the year | Number of complaints pending resolution at close of the year | Remarks |
| Shareholders                                      |                                                                                                                     |                                            |                                                              |         |                                            |                                                              |         |
| Employees and workers                             |                                                                                                                     |                                            |                                                              |         |                                            |                                                              |         |
| Customers                                         |                                                                                                                     |                                            |                                                              |         |                                            |                                                              |         |
| Value Chain Partners                              |                                                                                                                     |                                            |                                                              |         |                                            |                                                              |         |
| Other (please specify)                            |                                                                                                                     |                                            |                                                              |         |                                            |                                                              |         |

#### 24. Overview of the entity's material responsible business conduct issues

Please indicate material responsible business conduct and sustainability issues pertaining to environmental and social matters that present a risk or an opportunity to your business, rationale for identifying the same, approach to adapt or mitigate the risk along-with its financial implications, as per the following format

| S. No. | Material identified issue | Indicate whether risk or opportunity (R/O) | Rationale for identifying the risk / opportunity | In case of risk, approach to adapt or mitigate | Financial implications of the risk or opportunity (Indicate positive or negative implications) |
|--------|---------------------------|--------------------------------------------|--------------------------------------------------|------------------------------------------------|------------------------------------------------------------------------------------------------|
|        |                           |                                            |                                                  |                                                |                                                                                                |
|        |                           |                                            |                                                  |                                                |                                                                                                |

## SECTION R: MANAGEMENT AND PROCESS DISCLOSURES

This section is aimed at helping businesses demonstrate the structures, policies and processes put in place towards adopting the NGRBC Principles and Core Elements.

| Disclosure Questions                                                                                                                                                                                                                                       | P<br>1 | P<br>2 | P<br>3 | P<br>4 | P<br>5 | P<br>6 | P<br>7 | P<br>8 | P<br>9 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|--------|--------|
| <b>Policy and management processes</b>                                                                                                                                                                                                                     |        |        |        |        |        |        |        |        |        |
| 1. a. Whether your entity's policy/policies cover each principle and its core elements of the NGRBCs. (Yes/No)                                                                                                                                             |        |        |        |        |        |        |        |        |        |
| b. Has the policy been approved by the Board? (Yes/No)                                                                                                                                                                                                     |        |        |        |        |        |        |        |        |        |
| c. Web Link of the Policies, if available                                                                                                                                                                                                                  |        |        |        |        |        |        |        |        |        |
| 2. Whether the entity has translated the policy into procedures. (Yes / No)                                                                                                                                                                                |        |        |        |        |        |        |        |        |        |
| 3. Do the enlisted policies extend to your value chain partners? (Yes/No)                                                                                                                                                                                  |        |        |        |        |        |        |        |        |        |
| 4. Name of the national and international codes/certifications/labels/ standards (e.g. Forest Stewardship Council, Fairtrade, Rainforest Alliance, Trustea) standards (e.g. SA 8000, OHSAS, ISO, BIS) adopted by your entity and mapped to each principle. |        |        |        |        |        |        |        |        |        |
| 5. Specific commitments, goals and targets set by the entity with defined timelines, if any.                                                                                                                                                               |        |        |        |        |        |        |        |        |        |
| 6. Performance of the entity against the specific commitments, goals and targets along-with reasons in case the same are not met.                                                                                                                          |        |        |        |        |        |        |        |        |        |
| <b>Governance, leadership and oversight</b>                                                                                                                                                                                                                |        |        |        |        |        |        |        |        |        |
| 7. Statement by director responsible for the business responsibility report, highlighting ESG related challenges, targets and achievements <i>(listed entity has flexibility regarding the placement of this disclosure)</i>                               |        |        |        |        |        |        |        |        |        |
| 8. Details of the highest authority responsible for implementation and oversight of the Business Responsibility policy (ies).                                                                                                                              |        |        |        |        |        |        |        |        |        |
| 9. Does the entity have a specified Committee of the Board/ Director responsible for decision making on sustainability related issues? (Yes / No). If yes, provide details.                                                                                |        |        |        |        |        |        |        |        |        |

#### 10. Details of Review of NGRBCs by the Company:

| Subject for Review                                                                                                                                                    | Indicate whether review was undertaken by Director / Committee of the Board/ Any other Committee |     |     |     |     |     |     |     |     | Frequency (Annually/ Half yearly/ Quarterly/ Any other – please specify) |     |     |     |     |     |     |     |     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|--------------------------------------------------------------------------|-----|-----|-----|-----|-----|-----|-----|-----|
|                                                                                                                                                                       | P 1                                                                                              | P 2 | P 3 | P 4 | P 5 | P 6 | P 7 | P 8 | P 9 | P 1                                                                      | P 2 | P 3 | P 4 | P 5 | P 6 | P 7 | P 8 | P 9 |
| Performance against above policies and follow up action                                                                                                               |                                                                                                  |     |     |     |     |     |     |     |     |                                                                          |     |     |     |     |     |     |     |     |
| Compliance with statutory requirements of relevance to the principles, and, rectification of any non-compliances                                                      |                                                                                                  |     |     |     |     |     |     |     |     |                                                                          |     |     |     |     |     |     |     |     |
| 11. Has the entity carried out independent assessment/ evaluation of the working of its policies by an external agency? (Yes/No). If yes, provide name of the agency. |                                                                                                  |     |     |     |     |     |     |     |     | P 1                                                                      | P 2 | P 3 | P 4 | P 5 | P 6 | P 7 | P 8 | P 9 |
|                                                                                                                                                                       |                                                                                                  |     |     |     |     |     |     |     |     |                                                                          |     |     |     |     |     |     |     |     |

12. If answer to question (1) above is "No" i.e. not all Principles are covered by a policy, reasons to be stated:

[illegible]

**SECTION C: PRINCIPLE WISE PERFORMANCE DISCLOSURE**

This section is aimed at helping entities demonstrate their performance in integrating the Principles and Core Elements with key processes and decisions. The information sought is categorized as "Essential" and "Leadership". While the essential indicators are expected to be disclosed by every entity that is mandated to file this report, the leadership indicators may be voluntarily disclosed by entities which aspire to progress to a higher level in their quest to be socially, environmentally and ethically responsible.

**PRINCIPLE 1 Businesses should conduct and govern themselves with integrity, and in a manner that is Ethical, Transparent and Accountable.**

**Essential Indicators**

1. Percentage coverage by training and awareness programmes on any of the Principles during the financial year:

| Segment                           | Total number of training and awareness programmes held | Topics / principles covered under the training and its impact | %age of persons in respective category covered by the awareness programmes |
|-----------------------------------|--------------------------------------------------------|---------------------------------------------------------------|----------------------------------------------------------------------------|
| Board of Directors                |                                                        |                                                               |                                                                            |
| Key Managerial Personnel          |                                                        |                                                               |                                                                            |
| Employees other than BoD and KMPs |                                                        |                                                               |                                                                            |
| Workers                           |                                                        |                                                               |                                                                            |

2. Details of fines / penalties /punishment/ award/ compounding fees/ settlement amount paid in proceedings (by the entity or by directors / KMPs) with regulators/ law enforcement agencies/ judicial institutions, in the financial year, in the following format (Note: the entity shall make disclosures on the basis of materiality as specified in Regulation 30 of SEBI (Listing Obligations and Disclosure Obligations) Regulations, 2015 and as disclosed on the entity's website):

| <b>Monetary</b> |                 |                                     |                 |                   |                    |
|-----------------|-----------------|-------------------------------------|-----------------|-------------------|--------------------|
|                 | NGRBC Principle | Name of the regulatory/ enforcement | Amount (In INR) | Brief of the Case | Has an appeal been |

|                            |                            |                                                                                                |                          |                                                   |                                |
|----------------------------|----------------------------|------------------------------------------------------------------------------------------------|--------------------------|---------------------------------------------------|--------------------------------|
|                            |                            | <b>agencies/<br/>judicial<br/>institutions</b>                                                 |                          |                                                   | <b>preferred?<br/>(Yes/No)</b> |
| <b>Penalty/ Fine</b>       |                            |                                                                                                |                          |                                                   |                                |
| <b>Settlement</b>          |                            |                                                                                                |                          |                                                   |                                |
| <b>Compounding<br/>fee</b> |                            |                                                                                                |                          |                                                   |                                |
|                            | <b>Non-Monetary</b>        |                                                                                                |                          |                                                   |                                |
|                            | <b>NGRBC<br/>Principle</b> | <b>Name of the<br/>regulatory/<br/>enforcement<br/>agencies/<br/>judicial<br/>institutions</b> | <b>Brief of the Case</b> | <b>Has an appeal been<br/>preferred? (Yes/No)</b> |                                |
| <b>Imprisonment</b>        |                            |                                                                                                |                          |                                                   |                                |
| <b>Punishment</b>          |                            |                                                                                                |                          |                                                   |                                |

3. Of the instances disclosed in Question 2 above, details of the Appeal/ Revision preferred in cases where monetary or non-monetary action has been appealed.

|                     |                                                                                |
|---------------------|--------------------------------------------------------------------------------|
| <b>Case Details</b> | <b>Name of the regulatory/ enforcement<br/>agencies/ judicial institutions</b> |
|                     |                                                                                |

4. Does the entity have an anti-corruption or anti-bribery policy? If yes, provide details in brief and if available, provide a web-link to the policy.
5. Number of Directors/KMPs/employees/workers against whom disciplinary action was taken by any law enforcement agency for the charges of bribery/ corruption:

|                  | <b>FY ____<br/>(Current Financial<br/>Year)</b> | <b>FY ____<br/>(Previous Financial Year)</b> |
|------------------|-------------------------------------------------|----------------------------------------------|
| <b>Directors</b> |                                                 |                                              |
| <b>KMPs</b>      |                                                 |                                              |
| <b>Employees</b> |                                                 |                                              |
| <b>Workers</b>   |                                                 |                                              |

6. Details of complaints with regard to conflict of interest:

|  | <b>FY ____<br/>(Current Financial Year)</b> | <b>FY ____<br/>(Previous Financial Year)</b> |
|--|---------------------------------------------|----------------------------------------------|
|--|---------------------------------------------|----------------------------------------------|

|                                                                                              | Number | Remarks | Number | Remarks |
|----------------------------------------------------------------------------------------------|--------|---------|--------|---------|
| Number of complaints received in relation to issues of Conflict of Interest of the Directors |        |         |        |         |
| Number of complaints received in relation to issues of Conflict of Interest of the KMPs      |        |         |        |         |

7. Provide details of any corrective action taken or underway on issues related to fines / penalties / action taken by regulators/ law enforcement agencies/ judicial institutions, on cases of corruption and conflicts of interest.

#### Leadership Indicators

1. Awareness programmes conducted for value chain partners on any of the Principles during the financial year:

| Total number of awareness programmes held | Topics / principles covered under the training | %age of value chain partners covered (by value of business done with such partners) under the awareness programmes |
|-------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
|                                           |                                                |                                                                                                                    |

2. Does the entity have processes in place to avoid/ manage conflict of interests involving members of the Board? (Yes/No) If Yes, provide details of the same.

## PRINCIPLE 2 Businesses should provide goods and services in a manner that is sustainable and safe

### Essential Indicators

1. Percentage of R&D and capital expenditure (capex) investments in specific technologies to improve the environmental and social impacts of product and processes to total R&D and capex investments made by the entity, respectively.

|       | Current Financial Year | Previous Financial Year | Details of improvements in environmental and social impacts |
|-------|------------------------|-------------------------|-------------------------------------------------------------|
| R&D   |                        |                         |                                                             |
| Capex |                        |                         |                                                             |

2. a. Does the entity have procedures in place for sustainable sourcing? (Yes/No)  
b. If yes, what percentage of inputs were sourced sustainably?
3. Describe the processes in place to safely reclaim your products for reusing, recycling and disposing at the end of life, for (a) Plastics (including packaging) (b) E-waste (c) Hazardous waste and (d) other waste.
4. Whether Extended Producer Responsibility (EPR) is applicable to the entity's activities (Yes / No). If yes, whether the waste collection plan is in line with the Extended Producer Responsibility (EPR) plan submitted to Pollution Control Boards? If not, provide steps taken to address the same.

### Leadership Indicators

1. Has the entity conducted Life Cycle Perspective / Assessments (LCA) for any of its products (for manufacturing industry) or for its services (for service industry)? If yes, provide details in the following format?

| NIC Code | Name of Product /Service | % of total Turnover contributed | Boundary for which the Life Cycle Perspective / Assessment was conducted | Whether conducted by independent external agency (Yes/No) | Results communicated in public domain (Yes/No)<br><br>If yes, provide the web-link. |
|----------|--------------------------|---------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------|
|          |                          |                                 |                                                                          |                                                           |                                                                                     |

2. If there are any significant social or environmental concerns and/or risks arising from production or disposal of your products / services, as identified in the Life Cycle Perspective / Assessments (LCA) or through any other means, briefly describe the same along-with action taken to mitigate the same.

| Name of Product / Service | Description of the risk / concern | Action Taken |
|---------------------------|-----------------------------------|--------------|
|                           |                                   |              |
|                           |                                   |              |
|                           |                                   |              |

3. Percentage of recycled or reused input material to total material (by value) used in production (for manufacturing industry) or providing services (for service industry).

| Indicate input material | Recycled or re-used input material to total material |                                    |
|-------------------------|------------------------------------------------------|------------------------------------|
|                         | FY ____<br>Current Financial Year                    | FY ____<br>Previous Financial Year |
|                         |                                                      |                                    |

4. Of the products and packaging reclaimed at end of life of products, amount (in metric tonnes) reused, recycled, and safely disposed, as per the following format:

|                                | FY ____<br>Current Financial Year |          |                 | FY ____<br>Previous Financial Year |          |                 |
|--------------------------------|-----------------------------------|----------|-----------------|------------------------------------|----------|-----------------|
|                                | Re-Used                           | Recycled | Safely Disposed | Re-Used                            | Recycled | Safely Disposed |
| Plastics (including packaging) |                                   |          |                 |                                    |          |                 |
| E-waste                        |                                   |          |                 |                                    |          |                 |
| Hazardous waste                |                                   |          |                 |                                    |          |                 |
| Other waste                    |                                   |          |                 |                                    |          |                 |

5. Reclaimed products and their packaging materials (as percentage of products sold) for each product category.

| Indicate product category | Reclaimed products and their packaging materials as % of total products sold in respective category |
|---------------------------|-----------------------------------------------------------------------------------------------------|
|                           |                                                                                                     |

### PRINCIPLE 3 Businesses should respect and promote the well-being of all employees, including those in their value chains

| Essential Indicators                                                           |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
|--------------------------------------------------------------------------------|----------------------------------------------------|------------------------------------------------|------------------------------------------------------|----------------------------------------------------|------------------------------------------------|------------------------------------------------------|-----------|--------------------|-----------|---------------------|-----------|
| 1. a. Details of measures for the well-being of employees:                     |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Category                                                                       | Total<br>(A)                                       | % of employees covered by                      |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
|                                                                                |                                                    | Health insurance                               |                                                      | Accident insurance                                 |                                                | Maternity benefits                                   |           | Paternity Benefits |           | Day Care facilities |           |
|                                                                                |                                                    | Number<br>(B)                                  | % (B / A)                                            | Number<br>(C)                                      | % (C / A)                                      | Number<br>(D)                                        | % (D / A) | Number<br>(E)      | % (E / A) | Number<br>(F)       | % (F / A) |
| Permanent employees                                                            |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Male                                                                           |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Female                                                                         |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Total                                                                          |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Other than Permanent employees                                                 |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Male                                                                           |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Female                                                                         |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Total                                                                          |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| b. Details of measures for the well-being of workers:                          |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Category                                                                       | Total<br>(A)                                       | % of workers covered by                        |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
|                                                                                |                                                    | Health insurance                               |                                                      | Accident insurance                                 |                                                | Maternity benefits                                   |           | Paternity Benefits |           | Day Care facilities |           |
|                                                                                |                                                    | Number<br>(B)                                  | % (B / A)                                            | Number<br>(C)                                      | % (C / A)                                      | Number<br>(D)                                        | % (D / A) | Number<br>(E)      | % (E / A) | Number<br>(F)       | % (F / A) |
| Permanent workers                                                              |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Male                                                                           |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Female                                                                         |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Total                                                                          |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Other than Permanent workers                                                   |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Male                                                                           |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Female                                                                         |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Total                                                                          |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| 2. Details of retirement benefits, for Current FY and Previous Financial Year. |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| Benefits                                                                       | FY ____<br>Current Financial Year                  |                                                |                                                      | FY ____<br>Previous Financial Year                 |                                                |                                                      |           |                    |           |                     |           |
|                                                                                | No. of employees covered as a % of total employees | No. of workers covered as a % of total workers | Deducted and deposited with the authority (Y/N/N/A.) | No. of employees covered as a % of total employees | No. of workers covered as a % of total workers | Deducted and deposited with the authority (Y/N/N/A.) |           |                    |           |                     |           |
|                                                                                |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |
| PF                                                                             |                                                    |                                                |                                                      |                                                    |                                                |                                                      |           |                    |           |                     |           |

|                         |  |  |  |  |  |  |
|-------------------------|--|--|--|--|--|--|
| Gratuity                |  |  |  |  |  |  |
| ESI                     |  |  |  |  |  |  |
| Others – please specify |  |  |  |  |  |  |

### 3. Accessibility of workplaces

Are the premises / offices of the entity accessible to differently abled employees and workers, as per the requirements of the Rights of Persons with Disabilities Act, 2016? If not, whether any steps are being taken by the entity in this regard.

4. Does the entity have an equal opportunity policy as per the Rights of Persons with Disabilities Act, 2016? If so, provide a web-link to the policy.

5. Return to work and Retention rates of permanent employees and workers that took parental leave.

|        | Permanent employees |                | Permanent workers   |                |
|--------|---------------------|----------------|---------------------|----------------|
| Gender | Return to work rate | Retention rate | Return to work rate | Retention rate |
| Male   |                     |                |                     |                |
| Female |                     |                |                     |                |
| Total  |                     |                |                     |                |

6. Is there a mechanism available to receive and redress grievances for the following categories of employees and worker? If yes, give details of the mechanism in brief.

|                                | Yes/No<br>(If Yes, then give details of the mechanism in brief) |
|--------------------------------|-----------------------------------------------------------------|
| Permanent Workers              |                                                                 |
| Other than Permanent Workers   |                                                                 |
| Permanent Employees            |                                                                 |
| Other than Permanent Employees |                                                                 |

7. Membership of employees and worker in association(s) or Unions recognised by the listed entity:

| Category | FY ____<br>(Current Financial Year)                  |                                                                                                |           | FY ____<br>(Previous Financial Year)                 |                                                                                                |           |
|----------|------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------|------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------|
|          | Total employees / workers in respective category (A) | No. of employees / workers in respective category, who are part of association(s) or Union (B) | % (B / A) | Total employees / workers in respective category (C) | No. of employees / workers in respective category, who are part of association(s) or Union (D) | % (D / C) |

|                                  |  |  |  |  |  |  |
|----------------------------------|--|--|--|--|--|--|
| <b>Total Permanent Employees</b> |  |  |  |  |  |  |
| - Male                           |  |  |  |  |  |  |
| - Female                         |  |  |  |  |  |  |
| <b>Total Permanent Workers</b>   |  |  |  |  |  |  |
| - Male                           |  |  |  |  |  |  |
| - Female                         |  |  |  |  |  |  |

#### 8. Details of training given to employees and workers:

2. Details of training given to employees and workers.

| Category  | FY ____                |                               |           |                      |           | FY ____                 |                               |           |                      |           |
|-----------|------------------------|-------------------------------|-----------|----------------------|-----------|-------------------------|-------------------------------|-----------|----------------------|-----------|
|           | Current Financial Year |                               |           |                      |           | Previous Financial Year |                               |           |                      |           |
|           | Total<br>(A)           | On Health and safety measures |           | On Skill upgradation |           | Total<br>(D)            | On Health and safety measures |           | On Skill upgradation |           |
|           |                        | No.<br>(B)                    | % (B / A) | No.<br>(C)           | % (C / A) |                         | No.<br>(E)                    | % (E / D) | No.<br>(F)           | % (F / D) |
| Employees |                        |                               |           |                      |           |                         |                               |           |                      |           |
| Male      |                        |                               |           |                      |           |                         |                               |           |                      |           |
| Female    |                        |                               |           |                      |           |                         |                               |           |                      |           |
| Total     |                        |                               |           |                      |           |                         |                               |           |                      |           |
| Workers   |                        |                               |           |                      |           |                         |                               |           |                      |           |
| Male      |                        |                               |           |                      |           |                         |                               |           |                      |           |
| Female    |                        |                               |           |                      |           |                         |                               |           |                      |           |
| Total     |                        |                               |           |                      |           |                         |                               |           |                      |           |

#### 9. Details of performance and career development reviews of employees and workers:

| Category  | FY ____                |         |           | FY ____                 |         |           |
|-----------|------------------------|---------|-----------|-------------------------|---------|-----------|
|           | Current Financial Year |         |           | Previous Financial Year |         |           |
|           | Total<br>(A)           | No. (B) | % (B / A) | Total<br>(C)            | No. (D) | % (D / C) |
| Employees |                        |         |           |                         |         |           |
| Male      |                        |         |           |                         |         |           |
| Female    |                        |         |           |                         |         |           |
| Total     |                        |         |           |                         |         |           |
| Workers   |                        |         |           |                         |         |           |
| Male      |                        |         |           |                         |         |           |
| Female    |                        |         |           |                         |         |           |
| Total     |                        |         |           |                         |         |           |

#### 10. Health and safety management systems:

- Whether an occupational health and safety management system has been implemented by the entity? (Yes/ No). If yes, the coverage such system?
- What are the processes used to identify work-related hazards and assess risks on a routine and non-routine basis by the entity?
- Whether you have processes for workers to report the work related hazards and to remove themselves from such risks. (Y/N)

- d. Do the employees/ worker of the entity have access to non-occupational medical and healthcare services? (Yes/ No)

11. Details of safety related incidents, in the following format:

| Safety Incident/Number                                                        | Category  | FY ____<br>Current Financial Year | FY ____<br>Previous Financial Year |
|-------------------------------------------------------------------------------|-----------|-----------------------------------|------------------------------------|
| Lost Time Injury Frequency Rate (LTIFR) (per one million-person hours worked) | Employees |                                   |                                    |
|                                                                               | Workers   |                                   |                                    |
| Total recordable work-related injuries                                        | Employees |                                   |                                    |
|                                                                               | Workers   |                                   |                                    |
| No. of fatalities                                                             | Employees |                                   |                                    |
|                                                                               | Workers   |                                   |                                    |
| High consequence work-related injury or ill-health (excluding fatalities)     | Employees |                                   |                                    |
|                                                                               | Workers   |                                   |                                    |

12. Describe the measures taken by the entity to ensure a safe and healthy work place.

13. Number of Complaints on the following made by employees and workers:

|                    | FY ____<br>(Current Financial Year) |                                       |         | FY ____<br>(Previous Financial Year) |                                       |         |
|--------------------|-------------------------------------|---------------------------------------|---------|--------------------------------------|---------------------------------------|---------|
|                    | Filed during the year               | Pending resolution at the end of year | Remarks | Filed during the year                | Pending resolution at the end of year | Remarks |
| Working Conditions |                                     |                                       |         |                                      |                                       |         |
| Health & Safety    |                                     |                                       |         |                                      |                                       |         |

14. Assessments for the year:

|                             | % of your plants and offices that were assessed (by entity or statutory authorities or third parties) |
|-----------------------------|-------------------------------------------------------------------------------------------------------|
| Health and safety practices |                                                                                                       |
| Working Conditions          |                                                                                                       |

15. Provide details of any corrective action taken or underway to address safety-related incidents (if any) and on significant risks / concerns arising from assessments of health & safety practices and working conditions.

### Leadership Indicators

1. Does the entity extend any life insurance or any compensatory package in the event of death of (A) Employees (Y/N) (B) Workers (Y/N).
2. Provide the measures undertaken by the entity to ensure that statutory dues have been deducted and deposited by the value chain partners.
3. Provide the number of employees / workers having suffered high consequence work-related injury / ill-health / fatalities (as reported in Q11 of Essential Indicators above), who have been are rehabilitated and placed in suitable employment or whose family members have been placed in suitable employment:

|                  | Total no. of affected employees/ workers |                                      | No. of employees/workers that are rehabilitated and placed in suitable employment or whose family members have been placed in suitable employment |                                      |
|------------------|------------------------------------------|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
|                  | FY ____<br>(Current Financial Year)      | FY ____<br>(Previous Financial Year) | FY ____<br>(Current Financial Year)                                                                                                               | FY ____<br>(Previous Financial Year) |
| <i>Employees</i> |                                          |                                      |                                                                                                                                                   |                                      |
| <i>Workers</i>   |                                          |                                      |                                                                                                                                                   |                                      |

4. Does the entity provide transition assistance programs to facilitate continued employability and the management of career endings resulting from retirement or termination of employment? (Yes/ No)

5. Details on assessment of value chain partners:

|                             | % of value chain partners (by value of business done with such partners) that were assessed |
|-----------------------------|---------------------------------------------------------------------------------------------|
| Health and safety practices |                                                                                             |
| Working Conditions          |                                                                                             |

6. Provide details of any corrective actions taken or underway to address significant risks / concerns arising from assessments of health and safety practices and working conditions of value chain partners.

## **PRINCIPLE 4: Businesses should respect the interests of and be responsive to all its stakeholders**

### **Essential Indicators**

1. Describe the processes for identifying key stakeholder groups of the entity.
2. List stakeholder groups identified as key for your entity and the frequency of engagement with each stakeholder group.

| <b>Stakeholder Group</b> | <b>Whether identified as Vulnerable &amp; Marginalized Group (Yes/No)</b> | <b>Channels of communication (Email, SMS, Newspaper, Pamphlets, Advertisement, Community Meetings, Notice Board, Website), Other</b> | <b>Frequency of engagement (Annually/ Half yearly/ Quarterly / others – please specify)</b> | <b>Purpose and scope of engagement including key topics and concerns raised during such engagement</b> |
|--------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
|                          |                                                                           |                                                                                                                                      |                                                                                             |                                                                                                        |
|                          |                                                                           |                                                                                                                                      |                                                                                             |                                                                                                        |

### **Leadership Indicators**

1. Provide the processes for consultation between stakeholders and the Board on economic, environmental, and social topics or if consultation is delegated, how is feedback from such consultations provided to the Board.
2. Whether stakeholder consultation is used to support the identification and management of environmental, and social topics (Yes / No). If so, provide details of instances as to how the inputs received from stakeholders on these topics were incorporated into policies and activities of the entity.
3. Provide details of instances of engagement with, and actions taken to, address the concerns of vulnerable/ marginalized stakeholder groups.

## PRINCIPLE 5 Businesses should respect and promote human rights

### Essential Indicators

1. Employees and workers who have been provided training on human rights issues and policy(ies) of the entity, in the following format:

| Category             | FY ____                |                                        |           | FY ____                 |                                        |           |
|----------------------|------------------------|----------------------------------------|-----------|-------------------------|----------------------------------------|-----------|
|                      | Current Financial Year |                                        |           | Previous Financial Year |                                        |           |
|                      | Total (A)              | No. of employees / workers covered (B) | % (B / A) | Total (C)               | No. of employees / workers covered (D) | % (D / C) |
| <b>Employees</b>     |                        |                                        |           |                         |                                        |           |
| Permanent            |                        |                                        |           |                         |                                        |           |
| Other than permanent |                        |                                        |           |                         |                                        |           |
| Total Employees      |                        |                                        |           |                         |                                        |           |
| <b>Workers</b>       |                        |                                        |           |                         |                                        |           |
| Permanent            |                        |                                        |           |                         |                                        |           |
| Other than permanent |                        |                                        |           |                         |                                        |           |
| Total Workers        |                        |                                        |           |                         |                                        |           |

2. Details of minimum wages paid to employees and workers, in the following format:

[illegible]

## 3. Details of remuneration/salary/wages, in the following format:

|                                  | Male   |                                                           | Female |                                                           |
|----------------------------------|--------|-----------------------------------------------------------|--------|-----------------------------------------------------------|
|                                  | Number | Median remuneration/ salary/ wages of respective category | Number | Median remuneration/ salary/ wages of respective category |
| Board of Directors (BoD)         |        |                                                           |        |                                                           |
| Key Managerial Personnel         |        |                                                           |        |                                                           |
| Employees other than BoD and KMP |        |                                                           |        |                                                           |
| Workers                          |        |                                                           |        |                                                           |

4. Do you have a focal point (Individual/ Committee) responsible for addressing human rights impacts or issues caused or contributed to by the business? (Yes/No)

5. Describe the internal mechanisms in place to redress grievances related to human rights issues.

## 6. Number of Complaints on the following made by employees and workers:

|                                   | FY ____<br>Current Financial Year |                                       |         | FY ____<br>Previous Financial Year |                                       |         |
|-----------------------------------|-----------------------------------|---------------------------------------|---------|------------------------------------|---------------------------------------|---------|
|                                   | Filed during the year             | Pending resolution at the end of year | Remarks | Filed during the year              | Pending resolution at the end of year | Remarks |
| Sexual Harassment                 |                                   |                                       |         |                                    |                                       |         |
| Discrimination at workplace       |                                   |                                       |         |                                    |                                       |         |
| Child Labour                      |                                   |                                       |         |                                    |                                       |         |
| Forced Labour/Involuntary Labour  |                                   |                                       |         |                                    |                                       |         |
| Wages                             |                                   |                                       |         |                                    |                                       |         |
| Other human rights related issues |                                   |                                       |         |                                    |                                       |         |

7. Mechanisms to prevent adverse consequences to the complainant in discrimination and harassment cases.

8. Do human rights requirements form part of your business agreements and contracts? (Yes/No)

**9. Assessments for the year:**

|                             | <b>% of your plants and offices that were assessed<br/>(by entity or statutory authorities or third parties)</b> |
|-----------------------------|------------------------------------------------------------------------------------------------------------------|
| Child labour                |                                                                                                                  |
| Forced/involuntary labour   |                                                                                                                  |
| Sexual harassment           |                                                                                                                  |
| Discrimination at workplace |                                                                                                                  |
| Wages                       |                                                                                                                  |
| Others – please specify     |                                                                                                                  |

10. Provide details of any corrective actions taken or underway to address significant risks / concerns arising from the assessments at Question 9 above.

**Leadership Indicators**

1. Details of a business process being modified / introduced as a result of addressing human rights grievances/complaints.
2. Details of the scope and coverage of any Human rights due-diligence conducted.
3. Is the premise/office of the entity accessible to differently abled visitors, as per the requirements of the Rights of Persons with Disabilities Act, 2016?

**4. Details on assessment of value chain partners:**

|                                  | <b>% of value chain partners (by value of business done with such partners) that were assessed</b> |
|----------------------------------|----------------------------------------------------------------------------------------------------|
| Sexual Harassment                |                                                                                                    |
| Discrimination at workplace      |                                                                                                    |
| Child Labour                     |                                                                                                    |
| Forced Labour/Involuntary Labour |                                                                                                    |
| Wages                            |                                                                                                    |
| Others – please specify          |                                                                                                    |

5. Provide details of any corrective actions taken or underway to address significant risks / concerns arising from the assessments at Question 4 above.

## PRINCIPLE 6: Businesses should respect and make efforts to protect and restore the environment

### Essential Indicators

1. Details of total energy consumption (in Joules or multiples) and energy intensity, in the following format:

| Parameter                                                                                   | FY ____<br>(Current Financial Year) | FY ____<br>(Previous Financial Year) |
|---------------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------|
| Total electricity consumption (A)                                                           |                                     |                                      |
| Total fuel consumption (B)                                                                  |                                     |                                      |
| Energy consumption through other sources (C)                                                |                                     |                                      |
| Total energy consumption (A+B+C)                                                            |                                     |                                      |
| Energy intensity per rupee of turnover<br>(Total energy consumption/<br>turnover in rupees) |                                     |                                      |
| Energy intensity (optional) – the relevant metric may be selected by the entity             |                                     |                                      |

Note: Indicate if any independent assessment/ evaluation/assurance has been carried out by an external agency? (Y/N) If yes, name of the external agency.

2. Does the entity have any sites / facilities identified as designated consumers (DCs) under the Performance, Achieve and Trade (PAT) Scheme of the Government of India? (Y/N) If yes, disclose whether targets set under the PAT scheme have been achieved. In case targets have not been achieved, provide the remedial action taken, if any.
3. Provide details of the following disclosures related to water, in the following format:

| Parameter                                  | FY ____<br>(Current Financial Year) | FY ____<br>(Previous Financial Year) |
|--------------------------------------------|-------------------------------------|--------------------------------------|
| Water withdrawal by source (in kilolitres) |                                     |                                      |

|                                                                                       |  |  |
|---------------------------------------------------------------------------------------|--|--|
| (i) Surface water                                                                     |  |  |
| (ii) Groundwater                                                                      |  |  |
| (iii) Third party water                                                               |  |  |
| (iv) Seawater / desalinated water                                                     |  |  |
| (v) Others                                                                            |  |  |
| <b>Total volume of water withdrawn (in kilolitres) (i + ii + iii + iv + v)</b>        |  |  |
| <b>Total volume of water consumption (in kilolitres)</b>                              |  |  |
| <b>Water intensity per rupee of turnover (Water consumed / turnover)</b>              |  |  |
| <b>Water intensity (optional) – the relevant metric may be selected by the entity</b> |  |  |

Note: Indicate if any independent assessment/ evaluation/assurance has been carried out by an external agency? (Y/N) If yes, name of the external agency.

- Has the entity implemented a mechanism for Zero Liquid Discharge? If yes, provide details of its coverage and implementation.
- Please provide details of air emissions (other than GHG emissions) by the entity, in the following format:

| <b>Parameter</b>                    | <b>Please specify unit</b> | <b>FY ____<br/>(Current Financial Year)</b> | <b>FY ____<br/>(Previous Financial Year)</b> |
|-------------------------------------|----------------------------|---------------------------------------------|----------------------------------------------|
| NO <sub>x</sub>                     |                            |                                             |                                              |
| SO <sub>x</sub>                     |                            |                                             |                                              |
| Particulate matter (PM)             |                            |                                             |                                              |
| Persistent organic pollutants (POP) |                            |                                             |                                              |
| Volatile organic compounds (VOC)    |                            |                                             |                                              |
| Hazardous air pollutants (HAP)      |                            |                                             |                                              |
| Others – please specify             |                            |                                             |                                              |

Note: Indicate if any independent assessment/ evaluation/assurance has been carried out by an external agency? (Y/N) If yes, name of the external agency.

6. Provide details of greenhouse gas emissions (Scope 1 and Scope 2 emissions) & its intensity, in the following format:

| <i>Parameter</i>                                                                                                                                                         | <i>Unit</i>                                       | <i>FY ____<br/>(Current Financial Year)</i> | <i>FY ____<br/>(Previous Financial Year)</i> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|---------------------------------------------|----------------------------------------------|
| Total Scope 1 emissions<br>(Break-up of the GHG into CO <sub>2</sub> , CH <sub>4</sub> , N <sub>2</sub> O, HFCs, PFCs, SF <sub>6</sub> , NF <sub>3</sub> , if available) | <i>Metric tonnes of CO<sub>2</sub> equivalent</i> |                                             |                                              |
| Total Scope 2 emissions<br>(Break-up of the GHG into CO <sub>2</sub> , CH <sub>4</sub> , N <sub>2</sub> O, HFCs, PFCs, SF <sub>6</sub> , NF <sub>3</sub> , if available) | <i>Metric tonnes of CO<sub>2</sub> equivalent</i> |                                             |                                              |
| Total Scope 1 and Scope 2 emissions per rupee of turnover                                                                                                                |                                                   |                                             |                                              |
| Total Scope 1 and Scope 2 emission intensity (optional)<br>– the relevant metric may be selected by the entity                                                           |                                                   |                                             |                                              |

Note: Indicate if any independent assessment/ evaluation/assurance has been carried out by an external agency? (Y/N) If yes, name of the external agency.

7. Does the entity have any project related to reducing Green House Gas emission? If Yes, then provide details.

8. Provide details related to waste management by the entity, in the following format:

| <i>Parameter</i>                                | <i>FY ____<br/>(Current Financial Year)</i> | <i>FY ____<br/>(Previous Financial Year)</i> |
|-------------------------------------------------|---------------------------------------------|----------------------------------------------|
| <b>Total Waste generated (in metric tonnes)</b> |                                             |                                              |
| Plastic waste (A)                               |                                             |                                              |
| E-waste (B)                                     |                                             |                                              |
| Bio-medical waste (C)                           |                                             |                                              |
| Construction and demolition waste (D)           |                                             |                                              |
| Battery waste (E)                               |                                             |                                              |
| Radioactive waste (F)                           |                                             |                                              |

|                                                                                                                                                |  |  |
|------------------------------------------------------------------------------------------------------------------------------------------------|--|--|
| Other Hazardous waste. Please specify, if any. (G)                                                                                             |  |  |
| Other Non-hazardous waste generated (H). Please specify, if any.<br>(Break-up by composition i.e. by materials relevant to the sector)         |  |  |
| <b>Total (A+B + C + D + E + F + G + H)</b>                                                                                                     |  |  |
| <b>For each category of waste generated, total waste recovered through recycling, re-using or other recovery operations (in metric tonnes)</b> |  |  |
| <b>Category of waste</b>                                                                                                                       |  |  |
| (i) Recycled                                                                                                                                   |  |  |
| (ii) Re-used                                                                                                                                   |  |  |
| (iii) Other recovery operations                                                                                                                |  |  |
| <b>Total</b>                                                                                                                                   |  |  |
| <b>For each category of waste generated, total waste disposed by nature of disposal method (in metric tonnes)</b>                              |  |  |
| <b>Category of waste</b>                                                                                                                       |  |  |
| (i) Incineration                                                                                                                               |  |  |
| (ii) Landfilling                                                                                                                               |  |  |
| (iii) Other disposal operations                                                                                                                |  |  |
| <b>Total</b>                                                                                                                                   |  |  |

Note: Indicate if any independent assessment/ evaluation/assurance has been carried out by an external agency? (Y/N) If yes, name of the external agency.

- Briefly describe the waste management practices adopted in your establishments. Describe the strategy adopted by your company to reduce usage of hazardous and toxic chemicals in your products and processes and the practices adopted to manage such wastes.
- If the entity has operations/offices in/around ecologically sensitive areas (such as national parks, wildlife sanctuaries, biosphere reserves, wetlands, biodiversity hotspots, forests, coastal regulation zones etc.) where environmental approvals / clearances are required, please specify details in the following format:

| S. No. | Location of operations/offices | Type of operations | Whether the conditions of environmental approval / clearance are being complied with? (Y/N)<br>If no, the reasons thereof and corrective action taken, if any. |
|--------|--------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|        |                                |                    |                                                                                                                                                                |
|        |                                |                    |                                                                                                                                                                |

11. Details of environmental impact assessments of projects undertaken by the entity based on applicable laws, in the current financial year:

| Name and brief details of project | EIA Notification No. | Date | Whether conducted by independent external agency (Yes / No) | Results communicated in public domain (Yes / No) | Relevant Web link |
|-----------------------------------|----------------------|------|-------------------------------------------------------------|--------------------------------------------------|-------------------|
|                                   |                      |      |                                                             |                                                  |                   |
|                                   |                      |      |                                                             |                                                  |                   |

12. Is the entity compliant with the applicable environmental law/ regulations/ guidelines in India; such as the Water (Prevention and Control of Pollution) Act, Air (Prevention and Control of Pollution) Act, Environment protection act and rules thereunder (Y/N). If not, provide details of all such non-compliances, in the following format:

| S. No. | Specify the law / regulation / guidelines which was not complied with | Provide details of the non-compliance | Any fines / penalties / action taken by regulatory agencies such as pollution control boards or by courts | Corrective action taken, if any |
|--------|-----------------------------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------|---------------------------------|
|        |                                                                       |                                       |                                                                                                           |                                 |
|        |                                                                       |                                       |                                                                                                           |                                 |

### Leadership Indicators

1. Provide break-up of the total energy consumed (in Joules or multiples) from renewable and non-renewable sources, in the following format:

| Parameter                                            | FY ____<br>(Current Financial Year) | FY ____<br>(Previous Financial Year) |
|------------------------------------------------------|-------------------------------------|--------------------------------------|
| From renewable sources                               |                                     |                                      |
| Total electricity consumption (A)                    |                                     |                                      |
| Total fuel consumption (B)                           |                                     |                                      |
| Energy consumption through other sources (C)         |                                     |                                      |
| Total energy consumed from renewable sources (A+B+C) |                                     |                                      |
| From non-renewable sources                           |                                     |                                      |

|                                                          |  |  |
|----------------------------------------------------------|--|--|
| Total electricity consumption (D)                        |  |  |
| Total fuel consumption (E)                               |  |  |
| Energy consumption through other sources (F)             |  |  |
| Total energy consumed from non-renewable sources (D+E+F) |  |  |

Note: Indicate if any independent assessment/ evaluation/assurance has been carried out by an external agency? (Y/N) If yes, name of the external agency.

2. Provide the following details related to water discharged:

| Parameter                                                                    | FY ____<br>(Current Financial Year) | FY ____<br>(Previous Financial Year) |
|------------------------------------------------------------------------------|-------------------------------------|--------------------------------------|
| <b>Water discharge by destination and level of treatment (in kilolitres)</b> |                                     |                                      |
| (i) To Surface water                                                         |                                     |                                      |
| - No treatment                                                               |                                     |                                      |
| - With treatment – please specify level of treatment                         |                                     |                                      |
| (ii) To Groundwater                                                          |                                     |                                      |
| - No treatment                                                               |                                     |                                      |
| - With treatment – please specify level of treatment                         |                                     |                                      |
| (iii) To Seawater                                                            |                                     |                                      |
| - No treatment                                                               |                                     |                                      |
| - With treatment – please specify level of treatment                         |                                     |                                      |
| (iv) Sent to third-parties                                                   |                                     |                                      |
| - No treatment                                                               |                                     |                                      |
| - With treatment – please specify level of treatment                         |                                     |                                      |
| (v) Others                                                                   |                                     |                                      |
| - No treatment                                                               |                                     |                                      |
| - With treatment – please specify level of treatment                         |                                     |                                      |
| Total water discharged (in kilolitres)                                       |                                     |                                      |

Note: Indicate if any independent assessment/ evaluation/assurance has been carried out by an external agency? (Y/N) If yes, name of the external agency.

### 3. Water withdrawal, consumption and discharge in areas of water stress (in kilolitres):

For each facility / plant located in areas of water stress, provide the following information:

- (i) Name of the area
- (ii) Nature of operations
- (iii) Water withdrawal, consumption and discharge in the following format:

| Parameter                                                                             | FY ____<br>(Current Financial Year) | FY ____<br>(Previous Financial Year) |
|---------------------------------------------------------------------------------------|-------------------------------------|--------------------------------------|
| <b>Water withdrawal by source (in kilolitres)</b>                                     |                                     |                                      |
| (i) Surface water                                                                     |                                     |                                      |
| (ii) Groundwater                                                                      |                                     |                                      |
| (iii) Third party water                                                               |                                     |                                      |
| (iv) Seawater / desalinated water                                                     |                                     |                                      |
| (v) Others                                                                            |                                     |                                      |
| <b>Total volume of water withdrawn (in kilolitres)</b>                                |                                     |                                      |
| <b>Total volume of water consumption (in kilolitres)</b>                              |                                     |                                      |
| <b>Water intensity per rupee of turnover (Water consumed / turnover)</b>              |                                     |                                      |
| <b>Water intensity (optional) – the relevant metric may be selected by the entity</b> |                                     |                                      |
| <b>Water discharge by destination and level of treatment (in kilolitres)</b>          |                                     |                                      |
| (i) Into Surface water                                                                |                                     |                                      |
| - No treatment                                                                        |                                     |                                      |
| - With treatment – please specify level of treatment                                  |                                     |                                      |
| (ii) Into Groundwater                                                                 |                                     |                                      |
| - No treatment                                                                        |                                     |                                      |
| - With treatment – please specify level of treatment                                  |                                     |                                      |
| (iii) Into Seawater                                                                   |                                     |                                      |
| - No treatment                                                                        |                                     |                                      |
| - With treatment – please specify level of treatment                                  |                                     |                                      |
| (iv) Sent to third-parties                                                            |                                     |                                      |

|                                                      |  |  |
|------------------------------------------------------|--|--|
| - No treatment                                       |  |  |
| - With treatment – please specify level of treatment |  |  |
| (v) Others                                           |  |  |
| - No treatment                                       |  |  |
| - With treatment – please specify level of treatment |  |  |
| <b>Total water discharged (in kilolitres)</b>        |  |  |

Note: Indicate if any independent assessment/ evaluation/assurance has been carried out by an external agency? (Y/N) If yes, name of the external agency.

4. Please provide details of total Scope 3 emissions & its intensity, in the following format:

| <i>Parameter</i>                                                                                                                                                                | <i>Unit</i>                                       | <i>FY____</i><br><i>(Current Financial Year)</i> | <i>FY____</i><br><i>(Previous Financial Year)</i> |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|--------------------------------------------------|---------------------------------------------------|
| <b>Total Scope 3 emissions</b><br>(Break-up of the GHG into CO <sub>2</sub> , CH <sub>4</sub> , N <sub>2</sub> O, HFCs, PFCs, SF <sub>6</sub> , NF <sub>3</sub> , if available) | <i>Metric tonnes of CO<sub>2</sub> equivalent</i> |                                                  |                                                   |
| <b>Total Scope 3 emissions per rupee of turnover</b>                                                                                                                            |                                                   |                                                  |                                                   |
| <b>Total Scope 3 emission intensity (optional) – the relevant metric may be selected by the entity</b>                                                                          |                                                   |                                                  |                                                   |

Note: Indicate if any independent assessment/ evaluation/assurance has been carried out by an external agency? (Y/N) If yes, name of the external agency.

5. With respect to the ecologically sensitive areas reported at Question 10 of Essential Indicators above, provide details of significant direct & indirect impact of the entity on biodiversity in such areas along-with prevention and remediation activities.
6. If the entity has undertaken any specific initiatives or used innovative technology or solutions to improve resource efficiency, or reduce impact due to emissions / effluent discharge / waste generated, please provide details of the same as well as outcome of such initiatives, as per the following format:

| <b>Sr. No</b> | <b>Initiative undertaken</b> | <b>Details of the initiative (Web-link, if any, may be provided along-with summary)</b> | <b>Outcome of the initiative</b> |
|---------------|------------------------------|-----------------------------------------------------------------------------------------|----------------------------------|
|               |                              |                                                                                         |                                  |

|  |  |  |  |
|--|--|--|--|
|  |  |  |  |
|--|--|--|--|

7. Does the entity have a business continuity and disaster management plan? Give details in 100 words/ web link.
8. Disclose any significant adverse impact to the environment, arising from the value chain of the entity. What mitigation or adaptation measures have been taken by the entity in this regard.
9. Percentage of value chain partners (by value of business done with such partners) that were assessed for environmental impacts.

**PRINCIPLE 7 Businesses, when engaging in influencing public and regulatory policy, should do so in a manner that is responsible and transparent**

**Essential Indicators**

1. a. Number of affiliations with trade and industry chambers/ associations.
- b. List the top 10 trade and industry chambers/ associations (determined based on the total members of such body) the entity is a member of/ affiliated to.

| S. No. | Name of the trade and industry chambers/ associations | Reach of trade and industry chambers/ associations (State/National) |
|--------|-------------------------------------------------------|---------------------------------------------------------------------|
| 1      |                                                       |                                                                     |
| 2      |                                                       |                                                                     |
| 3      |                                                       |                                                                     |
| 4      |                                                       |                                                                     |
| 5      |                                                       |                                                                     |
| 6      |                                                       |                                                                     |
| 7      |                                                       |                                                                     |
| 8      |                                                       |                                                                     |
| 9      |                                                       |                                                                     |
| 10     |                                                       |                                                                     |

2. Provide details of corrective action taken or underway on any issues related to anti-competitive conduct by the entity, based on adverse orders from regulatory authorities.

| Name of authority | Brief of the case | Corrective action taken |
|-------------------|-------------------|-------------------------|
|                   |                   |                         |
|                   |                   |                         |

**Leadership Indicators**

1. Details of public policy positions advocated by the entity:

| S. No. | Public policy advocated | Method resorted for such advocacy | Whether information available in public domain? (Yes/No) | Frequency of Review by Board (Annually/ Half yearly/ Quarterly / Others – please specify) | Web Link, if available |
|--------|-------------------------|-----------------------------------|----------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------|
|        |                         |                                   |                                                          |                                                                                           |                        |
|        |                         |                                   |                                                          |                                                                                           |                        |

## PRINCIPLE 8 Businesses should promote inclusive growth and equitable development

### Essential Indicators

1. Details of Social Impact Assessments (SIA) of projects undertaken by the entity based on applicable laws, in the current financial year.

| Name and brief details of project | SIA Notification No. | Date of notification | Whether conducted by independent external agency (Yes / No) | Results communicated in public domain (Yes / No) | Relevant Web link |
|-----------------------------------|----------------------|----------------------|-------------------------------------------------------------|--------------------------------------------------|-------------------|
|                                   |                      |                      |                                                             |                                                  |                   |
|                                   |                      |                      |                                                             |                                                  |                   |

2. Provide information on project(s) for which ongoing Rehabilitation and Resettlement (R&R) is being undertaken by your entity, in the following format:

| S. No. | Name of Project for which R&R is ongoing | State | District | No. of Project Affected Families (PAFs) | % of PAFs covered by R&R | Amounts paid to PAFs in the FY (In INR) |
|--------|------------------------------------------|-------|----------|-----------------------------------------|--------------------------|-----------------------------------------|
|        |                                          |       |          |                                         |                          |                                         |
|        |                                          |       |          |                                         |                          |                                         |

3. Describe the mechanisms to receive and redress grievances of the community.
4. Percentage of input material (inputs to total inputs by value) sourced from suppliers:

|                                                                      | FY ____<br>Current Financial Year | FY ____<br>Previous Financial Year |
|----------------------------------------------------------------------|-----------------------------------|------------------------------------|
| Directly sourced from MSMEs/ small producers                         |                                   |                                    |
| Sourced directly from within the district and neighbouring districts |                                   |                                    |

### Leadership Indicators

1. Provide details of actions taken to mitigate any negative social impacts identified in the Social Impact Assessments (Reference: Question 1 of Essential Indicators above):

| Details of negative social impact identified | Corrective action taken |
|----------------------------------------------|-------------------------|
|                                              |                         |
|                                              |                         |

2. Provide the following information on CSR projects undertaken by your entity in designated aspirational districts as identified by government bodies:

| S. No. | State | Aspirational District | Amount spent (In INR) |
|--------|-------|-----------------------|-----------------------|
|        |       |                       |                       |
|        |       |                       |                       |

3. (a) Do you have a preferential procurement policy where you give preference to purchase from suppliers comprising marginalized /vulnerable groups? (Yes/No)  
 (b) From which marginalized /vulnerable groups do you procure?  
 (c) What percentage of total procurement (by value) does it constitute?
4. Details of the benefits derived and shared from the intellectual properties owned or acquired by your entity (in the current financial year), based on traditional knowledge:

| S. No. | Intellectual Property based on traditional knowledge | Owned/ Acquired (Yes/No) | Benefit shared (Yes / No) | Basis of calculating benefit share |
|--------|------------------------------------------------------|--------------------------|---------------------------|------------------------------------|
|        |                                                      |                          |                           |                                    |
|        |                                                      |                          |                           |                                    |

5. Details of corrective actions taken or underway, based on any adverse order in intellectual property related disputes wherein usage of traditional knowledge is involved.

| Name of authority | Brief of the Case | Corrective action taken |
|-------------------|-------------------|-------------------------|
|                   |                   |                         |
|                   |                   |                         |

6. Details of beneficiaries of CSR Projects:

| S. No. | CSR Project | No. of persons benefitted from CSR Projects | % of beneficiaries from vulnerable and marginalized groups |
|--------|-------------|---------------------------------------------|------------------------------------------------------------|
|        |             |                                             |                                                            |
|        |             |                                             |                                                            |

## PRINCIPLE 9 Businesses should engage with and provide value to their consumers in a responsible manner

| Essential Indicators                                                                                                                                                                                                                                                                                                  |                                     |                                   |                    |                                      |                                   |         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------|--------------------|--------------------------------------|-----------------------------------|---------|
| 1. Describe the mechanisms in place to receive and respond to consumer complaints and feedback.                                                                                                                                                                                                                       |                                     |                                   |                    |                                      |                                   |         |
| 2. Turnover of products and/ services as a percentage of turnover from all products/service that carry information about:                                                                                                                                                                                             |                                     |                                   |                    |                                      |                                   |         |
|                                                                                                                                                                                                                                                                                                                       |                                     |                                   |                    |                                      | As a percentage to total turnover |         |
| Environmental and social parameters relevant to the product                                                                                                                                                                                                                                                           |                                     |                                   |                    |                                      |                                   |         |
| Safe and responsible usage                                                                                                                                                                                                                                                                                            |                                     |                                   |                    |                                      |                                   |         |
| Recycling and/or safe disposal                                                                                                                                                                                                                                                                                        |                                     |                                   |                    |                                      |                                   |         |
| 3. Number of consumer complaints in respect of the following:                                                                                                                                                                                                                                                         |                                     |                                   |                    |                                      |                                   |         |
|                                                                                                                                                                                                                                                                                                                       | FY ____<br>(Current Financial Year) |                                   | Remarks            | FY ____<br>(Previous Financial Year) |                                   | Remarks |
|                                                                                                                                                                                                                                                                                                                       | Received during the year            | Pending resolution at end of year |                    | Received during the year             | Pending resolution at end of year |         |
| Data privacy                                                                                                                                                                                                                                                                                                          |                                     |                                   |                    |                                      |                                   |         |
| Advertising                                                                                                                                                                                                                                                                                                           |                                     |                                   |                    |                                      |                                   |         |
| Cyber-security                                                                                                                                                                                                                                                                                                        |                                     |                                   |                    |                                      |                                   |         |
| Delivery of essential services                                                                                                                                                                                                                                                                                        |                                     |                                   |                    |                                      |                                   |         |
| Restrictive Trade Practices                                                                                                                                                                                                                                                                                           |                                     |                                   |                    |                                      |                                   |         |
| Unfair Trade Practices                                                                                                                                                                                                                                                                                                |                                     |                                   |                    |                                      |                                   |         |
| Other                                                                                                                                                                                                                                                                                                                 |                                     |                                   |                    |                                      |                                   |         |
| 4. Details of instances of product recalls on account of safety issues:                                                                                                                                                                                                                                               |                                     |                                   |                    |                                      |                                   |         |
|                                                                                                                                                                                                                                                                                                                       | Number                              |                                   | Reasons for recall |                                      |                                   |         |
| Voluntary recalls                                                                                                                                                                                                                                                                                                     |                                     |                                   |                    |                                      |                                   |         |
| Forced recalls                                                                                                                                                                                                                                                                                                        |                                     |                                   |                    |                                      |                                   |         |
| 5. Does the entity have a framework/ policy on cyber security and risks related to data privacy? (Yes/No) If available, provide a web-link of the policy.                                                                                                                                                             |                                     |                                   |                    |                                      |                                   |         |
| 6. Provide details of any corrective actions taken or underway on issues relating to advertising, and delivery of essential services; cyber security and data privacy of customers; re-occurrence of instances of product recalls; penalty / action taken by regulatory authorities on safety of products / services. |                                     |                                   |                    |                                      |                                   |         |

**Leadership Indicators**

1. Channels / platforms where information on products and services of the entity can be accessed (provide web link, if available).
2. Steps taken to inform and educate consumers about safe and responsible usage of products and/or services.
3. Mechanisms in place to inform consumers of any risk of disruption/discontinuation of essential services.
4. Does the entity display product information on the product over and above what is mandated as per local laws? (Yes/No/Not Applicable) If yes, provide details in brief. Did your entity carry out any survey with regard to consumer satisfaction relating to the major products / services of the entity, significant locations of operation of the entity or the entity as a whole? (Yes/No)
5. Provide the following information relating to data breaches:
  - a. Number of instances of data breaches along-with impact
  - b. Percentage of data breaches involving personally identifiable information of customers

## Unit 4.7. Developing Comprehensive Business Responsibility and Sustainability Report

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Conducting Materiality Assessment
2. Collecting and Verifying Data
3. Setting Performance Indicators
4. Structuring Report Content
5. Engaging with Stakeholders
6. Reviewing and Assuring Accuracy

### Publishing and Communicating Findings

To guarantee that the Business Responsibility and Sustainability Report (BRSR) correctly represents the organization's commitment to sustainable business practices, several actions must be taken in its development.

#### Identify the requirements and standards for reporting:

Know about the applicable reporting requirements, such as those established by international frameworks like the Task Force on Climate-related Financial Disclosures (TCFD), the Sustainability Accounting Requirements Board (SASB), and the Global Reporting Initiative (GRI). In India, these standards are determined by regulatory agencies like SEBI.

- **Determine Relevant Subjects**
- Find the most important social, environmental, and economic issues that influence the stakeholders and your company by conducting a materiality evaluation.

Communicate with important stakeholders, including staff members, clients, investors, vendors, and communities, to learn about their expectations and points of view on sustainability-related issues.

- **Gather Information and Data**
- Collect relevant information and data about the material aspects that have been identified. Data from different organizational departments, both quantitative and qualitative, may be included in this.

Establish rigorous procedures for data collection and verification to guarantee data consistency, correctness, and dependability.

- **Determine the Performance Criteria**
- To monitor and assess how well the company is doing in tackling the important sustainability concerns, and set up key performance indicators (KPIs).

Establish goals and standards so that advancement may be evaluated over time.

- **Report Format and Content**
- Include sections on governance, environmental performance, social effect, stakeholder involvement, and future commitments in the clearly defined framework of the BRSR report.
- To aid stakeholders in understanding the organization's approach to sustainability and

ethical business practices, provide background information and context.

To highlight the organization's efforts and accomplishments, provide case studies, best practices, and examples.

- **Stakeholder Engagement**
- Include key stakeholders in the reporting process from the beginning to the end to get their input, confirm the results, and make sure the report is credible and relevant.

Be open and honest in your communication regarding challenges, failures, and opportunities for development to show that you are dedicated to ongoing development.

- **Evaluate and Approve**
- Examine the draft BRSR report carefully to make sure that the material is correct, comprehensive, and understandable.

To increase the report's credibility and dependability, think about getting external assurance or verification.

- **Publish and communicate**
- Communicate the BRSR report via the proper channels, including the corporate website, legal documents, or specialized sustainability portals.

Share the report's conclusions and insights with internal and external stakeholders using a variety of platforms, including social media, press releases, and stakeholder meetings.

- **Monitoring and updating**
- Develop systems for tracking results about the goals and KPIs listed in the BRSR report.

Review and update the report often to reflect modifications to the organization's aims, strategy, and performance in terms of sustainability.

Organizations can develop a comprehensive BRSR report that provides stakeholders with meaningful insights into the organization's sustainability efforts and contributes to building trust and transparency.

## Exercise



1. What are the key international frameworks and regulatory standards that organizations must be aware of when developing their Business Responsibility and Sustainability Report (BRSR)?
2. How can companies effectively conduct a materiality evaluation to determine the most significant social, environmental, and economic issues for their BRSR?
3. What procedures should be implemented to ensure the consistency, accuracy, and reliability of the data collected for the BRSR?
4. What performance criteria and key performance indicators (KPIs) should organizations establish to monitor and assess their sustainability efforts in the BRSR?
5. What should be included in the BRSR report's format and content to ensure it provides a clear and comprehensive overview of the company's sustainability practices?
6. How can organizations engage stakeholders effectively throughout the BRSR reporting process to ensure the report is credible and relevant?
7. What steps should be taken to review and approve the draft BRSR report, and why might external assurance or verification be beneficial?
8. How should organizations communicate and publish the BRSR report, and what strategies can be used to share its insights with both internal and external stakeholders?
9. What systems should be put in place to monitor results against the goals and KPIs outlined in the BRSR report, and how often should the report be updated?

## Unit 4.8. National Guidelines on Responsible Business Conduct

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Principles of Responsible Business Conduct
2. Assessing a Company's ESG Performance
3. Linking ESG Performance with National Guidelines

### 4.8.1 Guidance on Adoption of NGRBC

#### 1. Understanding Adoption

- Adoption refers to the integration of the Principles and Core Elements into the core business strategy and operations.
- The Governance Structure of a business can determine successful integration by having policies and guidance that cover all Principles.
- Key performance indicators should incorporate all Principles of the NGRBC and those Core Elements that significantly impact or are impacted by the business.
- Operating procedures should also cover those Core Elements that significantly impact or are impacted by the business.

#### 2. Key Enablers for Successful Integration

- Leadership commitment: The Governance Structure of the business needs to convince the Governance Structure that the success of the business depends upon adopting the NGRBC.
- Employee engagement: The top management of the company must map out its stakeholders and build strong relationships with them on a consistent and continuous basis.
- Disclosure and reporting: Proactive public disclosure by companies of their impact on society and the environment is essential.

#### 3. The Business Case for Adoption

- Businesses are expected to conduct themselves responsibly as an essential component of the social contract between businesses and society.
- The Business Case Matrix can help businesses determine the benefits of responsible business conduct.

#### 4. The Adoption Process

- Prioritizing the Core Elements: The business must identify the Core Elements that align with or contribute to the Principles.
- Legal and voluntary compliances: Businesses must know the specific laws and regulations, current and emerging, which pertain to each of the Core Elements.

- Integrating Prioritized Core Elements into Core Business: The business must review its current policies and guidelines to ensure they cover all the prioritized Core Elements.

## 5. Business Sustainability Strategy

- Mapping the Value Chain: Businesses should create a map of their direct and indirect value chain partners, covering physical locations, resource dependencies, contractual/compliance status, financial status, and storage/logistics arrangements.
- Prioritizing Core Elements for Value Chain Partners: Businesses should prioritize Core Elements important for their value chain partners to align with, best done in consultation with their major Tier 1 partners.
- Setting Policy Frameworks: Businesses should have policies and guidelines that facilitate the adoption of the NGRBC across their value chain partners. These should be endorsed by the business's Governance Structure.
- Establishing Engagement Processes: Prioritized Core Elements should be embedded in all points of engagement with value chain partners, including diligence, selection, monitoring, periodic engagement, feedback, and recognition of good practice.
- Cascading Responsibility: Businesses should guide and support their value chain partners in adopting the Principles and Core Elements, including collaborating with other businesses and supporting Tier 1 partners to cascade this process to subsequent tiers.

### 4.8.1 National Guidelines on Responsible Business Conduct for MSMEs

#### Significance of MSMEs for NGRBC

- MSMEs significantly contribute to India's GDP, employment, and social equity.
- They are the second largest employer, employing an estimated 100 million people.
- MSMEs are present in almost all economic activities, from crafts to services and high-end industrial activities.
- They are connected with rural markets to global value chains, contributing to 37% of industrial output, 40% of manufactured exports, and 37.54% of the GDP.

#### Classification of MSMEs<sup>7</sup>

- MSMEs can be categorized into two types: MSMEs organized around local and regional value chains, MSMEs part of national & global value chains, and emerging high-growth start-ups.
- The business case for MSMEs to adopt the NGRBC includes increased access to markets and customers, better preparedness for compliance, and increased market and customer engagement.

#### Benefits of Adopting the Principles and Core Elements

- Increased access to markets and customers: MSMEs that are part of national and global value chains have to conform to sustainability codes and standards.
- Better preparedness for compliance: India was a signatory to two global agreements in 2015 – the Sustainable Development Goals and the Paris Agreement on Climate Change.

### MSMEs Adoption Steps

#### 1. Prioritizing Core Elements:

- Map Core Elements against the MSME's vision, mission, values, and business success factors.
- Prioritize Core Elements governed by laws or regulations.
- Study and understand buyer/customer codes.

#### 2. Embedding Prioritized Core Elements:

- Integrate all prioritized Core Elements into the core business.
- Set ambitions at the Essential level.
- Higher ambitions can yield benefits in the medium to long term.

### 4.8.3. Business Responsibility Reporting Framework

The reporting framework is associated with the National Guidelines for Responsible Business Conduct. It consists of three sections:

- **Section A** – *General Disclosures, covering operational, financial, and ownership-related information,*
- **Section B** – *Management and Process Disclosures covering the structures, policies, and processes to integrate the Guidelines, and*
- **Section C** – *Principle-wise Performance Indicators covering how well businesses are performing in pursuit of these Guidelines.*

### 4.8.4 Guidance for Businesses on Using the BRRF as a Self-Assessment Tool

- BRRF is a tool for businesses to assess their alignment with the National Guidelines for Responsible Business Conduct (NGRBC).
- Two dimensions of alignment are “completeness” and “ambition,” which measure the extent to which the Principles and Core Elements are tracked by the business.
- Businesses must complete all three sections of the BRRF, with Sections A and B having all businesses expected to complete. Section C includes Essential and Leadership Indicators, which companies more experienced in business responsibility and sustainability should be able to complete.
- Businesses can assess their completeness by rating themselves against Section B on each of the Principles.
- To achieve an overall score, businesses can assign scores to the level of completeness against each section and sub-section.
- Equal weights are advised for each Principle as they are interdependent, interrelated, and non-divisible.

## Exercise



1. What is the significance of integrating the Principles and Core Elements of the NGRBC into a company's core business strategy and operations?
2. How can a business's governance structure impact the successful adoption of NGRBC principles?
3. What role does leadership commitment play in the successful integration of NGRBC principles within a company?
4. How can employee engagement contribute to the effective adoption of the NGRBC principles?
5. What is the purpose of the Business Case Matrix in helping businesses understand the benefits of responsible business conduct?
6. What are the steps involved in the adoption process of the NGRBC, and how should businesses prioritize the Core Elements?
7. How should businesses map and prioritize Core Elements for their value chain partners, and why is this important?
8. What are the benefits for MSMEs adopting the NGRBC principles, and how can it impact their market access and compliance preparedness?
9. What are the three sections of the Business Responsibility Reporting Framework (BRRF), and what does each section cover?
10. How can businesses use the BRRF as a self-assessment tool to evaluate their alignment with NGRBC principles, and what dimensions are important for this assessment?

## Unit 4.9. Analyzing BRSR Report Gaps for Compliance

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Reviewing Reporting Requirements
2. Comparing with Best Practices
3. Conducting Gap Analysis

To guarantee adherence to reporting guidelines and requirements, a comprehensive examination of the gaps in Business Responsibility and Sustainability Reports (BRSR) is essential.

#### Get familiar with Reporting Standards

To start, identify the reporting standards and frameworks that apply to BRSR. These may include regulatory organizations' (like SEBI's) standards or international standards like TCFD, SASB, GRI, or SASB. These standards offer instructions on what details to include and how to style the report.

#### Examine the Reporting Requirements

Pay close attention to the particular reporting requirements that are specified in the selected framework. Determine whether important topics and metrics, such as governance, environmental performance, social impact, stakeholder involvement, and future commitments, should be covered in the report.

#### Compare with Best Practices

Evaluate the BRSR report by comparing it to industry and similar organization best practices. This comparison might assist in identifying any gaps in the report or potential areas for development to increase accountability and transparency.

#### Gap Analysis

Perform a Gap Analysis by methodically contrasting the BRSR report's content with the specifications and guidelines provided in the best practices and reporting standards. Look for any places where the report could be lacking, such as gaps, contradictions, or non-compliance.

#### Analyse Data Accuracy and Completeness

Determine if the data in the BRSR report is accurate and comprehensive. Make sure that the data sources are trustworthy and verifiable and that all pertinent information is provided. Resolve any discrepancies or errors found throughout the examination.

#### Analyse Disclosure Quality

Rate the BRSR report's disclosures' level of quality. Seek information that is transparent, detailed, and easy to understand. Make sure the report gives stakeholders enough background information and justification to comprehend the performance and effects of the business.

#### Examine Stakeholder Feedback

Pay attention to the opinions of important parties, including communities, suppliers, workers, investors,

and consumers. In light of their expectations and concerns about sustainable performance and ethical business practices, evaluate whether the BRSR report sufficiently addresses them.

**Identify Feasible Suggestions**

Using the gap analysis results as a guide, create practical suggestions for enhancing the BRSR report. Sort suggestions into priority lists according to their significance and viability, taking the organization's resources and capabilities into account.

**Implement Recommendations**

Collaborate with relevant organization stakeholders to put the suggested BRSR report enhancements into practice. To guarantee compliance and alignment with best practices, this may require upgrading data gathering procedures, improving disclosure methods, or modifying reporting structures.

**Monitor and evaluate**

To guarantee continued adherence to reporting guidelines and standards, monitor and evaluate the BRSR report regularly. Update the report regularly to account for shifts in the organization's objectives, performance, and strategy.

Organizations should take proactive steps to guarantee compliance, improve transparency, and show a commitment to sustainable business practices and responsible practices by thoroughly analyzing any gaps in BRSR reports.

## Exercise



1. What steps should be taken to ensure that a Business Responsibility and Sustainability Report (BRSR) adheres to relevant reporting standards and frameworks?
2. How can a gap analysis help in improving the quality and compliance of a BRSR report, and what specific areas should be examined during this analysis?
3. What are the key considerations for evaluating data accuracy, disclosure quality, and stakeholder feedback in the BRSR report, and how can organizations address identified gaps and implement improvements?

## Unit 4.10. Incorporating Sustainable Development Goals into BRSR Reports

### Unit objectives

After reading this unit, you will be able to understand the following points:

1. Sustainable Development Goals (SDGs)
2. Identifying Relevance to Business Operations



To demonstrate a company's commitment to sustainable development, Business Responsibility and Sustainability Reports (BRSR) must include the Sustainable Development Goals (SDGs). The following are some tips for including SDGs in BRSR reports:

#### Understand SDGs

Gain an in-depth knowledge of the 17 Sustainable Development Goals (SDGs) and the goals that correspond with them. The SDGs address a wide variety of social, economic, and environmental challenges to achieve sustainable development on a global scale.

#### Select Relevant SDGs

Choose the SDGs that have the most bearing on the operations, goods, services, and value chain of your business. Think about how the business will affect the SDGs and set priorities.

#### Map SDGs to BRSR structure

Ensure that every pertinent SDG is in line with the BRSR's reporting structure. To do this, each SDG's indicators, metrics, and performance goals must be identified. These must then be integrated into the BRSR report's pertinent parts, which include governance, environmental performance, social impact, and stakeholder engagement.

#### Evaluate current Performance

Evaluate the organization's present performance about the chosen SDGs in great detail. Examine current sustainability policies, programs, practices, and projects to determine their advantages, disadvantages, opportunities, and gaps concerning SDG objectives and indicators.

#### Establish Goals and Performance Measures

To direct the organization's efforts toward making significant progress toward each SDG, clearly define objectives and performance metrics. Make sure that these goals are in line with the company's overarching sustainability plan and that they are SMART (specific, measurable, attainable, relevant, and time-bound).

#### Integrate SDGs into Business Strategy

Incorporate the Sustainable Development Goals (SDGs) into the company's operational planning, decision-making procedures, and overall business strategy. Incorporate sustainability and SDG goals into company governance, supply chain management, product creation, and risk management, among other key business operations.

#### Involve Stakeholders

Communicate the company's commitment to the SDGs to important stakeholders, such as workers, clients, investors, suppliers, communities, and governmental organizations. You should also ask for input and comments on your efforts to integrate the SDGs.

### Monitor and Report Progress

To monitor the advancement of SDG integration projects, and implement strong reporting and monitoring systems. Utilizing both qualitative and quantitative methods, gather, evaluate, and report data on SDG-related performance indicators regularly. In the BRSR report, provide a clear and thorough explanation of the company's progress toward the SDGs, taking into account achievements, obstacles, and lessons discovered.

### Challenges

Identify and resolve potential barriers to the integration of the SDGs, including corporate culture, conflicting agendas, resource limitations, and regulatory complexity. Create plans to deal with these obstacles and take advantage of chances to move the SDG implementation forward.

By following these steps, companies can effectively incorporate the Sustainable Development Goals into their BRSR reports

## SDGs Mapped Against NGRBC

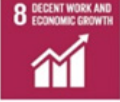
|                                                                                     |                    |                                                                                     |                |                                                                                       |                    |
|-------------------------------------------------------------------------------------|--------------------|-------------------------------------------------------------------------------------|----------------|---------------------------------------------------------------------------------------|--------------------|
|    | P3, P4, P8         |    | P2, P6, P7     |    | P2, P6, P7, P8     |
|   | P2, P6, P7, P8, P9 |   | P2, P3, P5, P8 |   | P2, P6, P7, P8, P9 |
|  | P3, P6, P8         |  | P3, P4, P8     |  | P2, P6, P7, P8, P9 |
|  | P3, P8, P9         |  | P2, P6, P7     |  | P1, P3, P4, P5, P8 |
|  | P3, P4, P5, P8     |  | P3, P4, P7, P8 |  | P1, P7, P8         |
|  | P2, P6, P8         |  | P2, P6, P9     |                                                                                       | ★                  |

Fig 4.10.1 1

## Exercise



1. How can companies select and prioritize the Sustainable Development Goals (SDGs) that are most relevant to their operations and value chain for inclusion in their Business Responsibility and Sustainability Reports (BRSR)?
2. What are the key steps involved in mapping SDGs to the BRSR structure, and how should indicators and metrics for each SDG be integrated into the report?
3. What methods can be used to evaluate an organization's current performance in relation to the chosen SDGs, and how can gaps and opportunities be identified?
4. How should companies establish and align SMART objectives and performance measures for the SDGs within their overall business strategy and operations?
5. What are the common challenges faced when integrating SDGs into BRSR reports, and how can companies address these challenges to ensure effective implementation and reporting?

## Unit 4.11. Importance of Corporate Sustainability Disclosures

### Unit objectives

After reading this unit, you will be able to understand the following points:



1. Understanding requirements and methodology.
2. Ensuring sustainability disclosures.

Enhancing the transparency, credibility, and reliability of the data presented in sustainability reports requires ensuring corporate sustainability disclosures.

**Improving Credibility and Trust:** Assurance offers a third-party verification of the completeness and correctness of sustainability disclosures, boosting stakeholders' trust in the quality of the data provided. Consequently, this raises the organization's reputation and cultivates trust among stakeholders such as workers, consumers, regulators, and investors.

**Ensuring Reporting Standards Compliance:** A significant number of sustainability reporting frameworks and standards advise against or need external assurance to guarantee reporting standards compliance. To improve the credibility of their sustainability reports, for instance, organizations are urged by the Global Reporting Initiative (GRI) to seek external confirmation.

**Risk Mitigation:** Assurance assists in identifying and reducing the risks brought on by incomplete or inaccurate sustainability disclosures. Assurance assists companies in detecting and mitigating possible vulnerabilities in their sustainability reporting through the verification of data accuracy, evaluation of the strength of reporting procedures, and identification of areas that require improvement.

**Displaying Accountability and Transparency:** By putting its sustainability disclosures under unbiased examination, Assurance shows the organization's dedication to accountability and transparency. It makes it abundantly evident to stakeholders that

The company takes its responsibility for sustainability reporting seriously and is prepared to be held accountable for its actions.

**Encouraging Decision-Making:** Stakeholders may make well-informed decisions when they have access to trustworthy and guaranteed sustainability information. Investors are in a better position to appraise the company's sustainability performance and long-term potential for wealth development. Consumers will be able to make more environmentally friendly selections, and staff members will have faith in the company's dedication to moral and responsible business conduct.

### Assurance Provider Requirements and Methodology

#### Requirements

- **Independence:** Ensures objectivity and impartiality.
- **Competence:** Possesses necessary expertise in sustainability reporting, auditing, and assurance practices.
- **Relevance:** Tailored to the organization's specific requirements and nature of sustainability disclosures.
- **Transparency:** Clear communication of the scope, methodology, findings, and conclusions of the assurance engagement.

#### Methodology

- **Planning:** Understand the organization's sustainability reporting objectives and material issues.

- **Data Collection and Analysis:** Collect relevant data to assess the accuracy and reliability of sustainability disclosures.
- **Testing:** Tests the effectiveness of internal controls and processes related to sustainability reporting.
- **Evaluation:** Evaluate the organization's adherence to reporting standards and best practices.
- **Reporting:** Communicates findings, conclusions, and recommendations through an assurance report.

Organizations may improve the credibility, transparency, and accountability of their reporting methods by successfully ensuring their sustainability disclosures by adhering to these principles and a methodical methodology.

## Exercise



1. How does external assurance contribute to enhancing the credibility and trustworthiness of sustainability reports, and what impact does it have on stakeholder perception?
2. What are the key requirements and methodologies for assurance providers to ensure effective verification of sustainability disclosures, and how do these requirements affect the quality of the assurance process?
3. In what ways does external assurance help mitigate risks associated with incomplete or inaccurate sustainability disclosures, and how does it support organizations in improving their reporting practices?



## Annexure



SCGJ/N5001

## Abbreviations & Acronyms

### A – C

BRSR – Business Responsibility and Sustainability Reporting

BRR – Business Responsibility Report

CBDR-RC – Common but Differentiated Responsibilities and Respective Capabilities

CDM – Clean Development Mechanism

CDP – Carbon Disclosure Project

CIF – Climate Investment Funds

CO<sub>2</sub> – Carbon Dioxide

COP – Conference of the Parties (UNFCCC)

CSR – Corporate Social Responsibility

### D – G

ETS – Emissions Trading System

ESG – Environmental, Social, and Governance

EU-ETS – European Union Emissions Trading System

GHG – Greenhouse Gas

GCF – Green Climate Fund

GCP – Green Climate Partnership

GEMS – Global Environmental Monitoring System

GEO – Group on Earth Observations

GHG Protocol – Greenhouse Gas Protocol

GRI – Global Reporting Initiative

GWP – Global Warming Potential

**H – N**

HFCs – Hydrofluorocarbons

IPCC – Intergovernmental Panel on Climate Change

ISO – International Organization for Standardization

IoT – Internet of Things

LCA – Life Cycle Assessment

MDB – Multilateral Development Bank

MoEFCC – Ministry of Environment, Forest, and Climate Change (India)

MRV – Measurement, Reporting, and Verification

NAPCC – National Action Plan on Climate Change

NDCs – Nationally Determined Contributions

NGOs – Non-Governmental Organizations

NOAA – National Oceanic and Atmospheric Administration

N<sub>2</sub>O – Nitrous Oxide

**O – S**

OECD – Organisation for Economic Co-operation and Development

PFCs – Perfluorocarbons

REDD+ – Reducing Emissions from Deforestation and Forest Degradation

SBTs – Science-Based Targets

SBTi – Science-Based Targets Initiative

SDGs – Sustainable Development Goals

SEBI – Securities and Exchange Board of India

SF<sub>6</sub> – Sulfur Hexafluoride

SMEs – Small and Medium Enterprises

SAPCC – State Action Plan on Climate Change

**T – Z**

TCFD – Task Force on Climate-related Financial Disclosures

UNEP – United Nations Environment Programme

UNDP – United Nations Development Programme

UNFCCC – United Nations Framework Convention on Climate Change

WBCSD – World Business Council for Sustainable Development

WRI – World Resources Institute

WWF – World Wide Fund for Nature

## Glossary

1. **Anthropogenic Climate Change** - Climate change resulting from human activities, such as fossil fuel combustion, deforestation, and industrial processes.
2. **Baseline Emissions** - The level of emissions that would occur without any intervention or mitigation efforts.
3. **Business Responsibility and Sustainability Reporting (BRSR)** - A framework for corporate sustainability disclosures in India, replacing the previous Business Responsibility Reporting (BRR) framework.
4. **Carbon Footprint** - The total amount of greenhouse gases emitted directly or indirectly by an individual, organization, event, or product.
5. **Carbon Market** - A trading system where emission allowances or credits can be bought and sold to incentivize emission reductions.
6. **Clean Development Mechanism (CDM)** - A mechanism under the Kyoto Protocol allowing industrialized countries to invest in emission reduction projects in developing countries.
7. **Climate Finance** - Financial resources allocated to support climate mitigation and adaptation actions, sourced from public, private, or multilateral institutions.
8. **Emission Factor** - A coefficient that quantifies emissions per unit of activity, used to estimate GHG emissions from different sources.
9. **Green Climate Fund (GCF)** - A financial mechanism under the UNFCCC that assists developing nations in climate adaptation and mitigation.
10. **Greenhouse Gas (GHG)** - Gases that trap heat in the Earth's atmosphere, including CO<sub>2</sub>, CH<sub>4</sub>, N<sub>2</sub>O, and fluorinated gases.
11. **Global Warming Potential (GWP)** - A metric comparing the heat-trapping ability of different greenhouse gases relative to CO<sub>2</sub> over a specific period.
12. **Low-Carbon Development** - A strategy that promotes economic growth while minimizing carbon emissions through renewable energy, efficiency measures, and sustainable practices.
13. **Nationally Determined Contributions (NDCs)** - Climate action plans submitted by countries under the Paris Agreement outlining their mitigation and adaptation goals.
14. **Net-Zero Emissions** - Achieving a balance between emitted and removed greenhouse gases through mitigation and carbon offsetting measures.
15. **Paris Agreement** - A legally binding international treaty adopted in 2015 under the UNFCCC to limit global temperature rise to well below 2°C.

## Recommended Links

| Module | Unit | Topic Name                                         | URL                                                                                                                 | QR Code                                                                               |
|--------|------|----------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1      | 1.2  | Greenhouse Gases And Anthropogenic Climate Change. | <a href="https://youtu.be/Tcv4gawvglc?si=Stu-cralypzqwtcqm">https://youtu.be/Tcv4gawvglc?si=Stu-cralypzqwtcqm</a>   |    |
|        |      |                                                    | <a href="https://www.youtube.com/watch?v=Cht-Ixve-hya">https://www.youtube.com/watch?v=Cht-Ixve-hya</a>             |    |
|        | 1.6  | Paris Agreement                                    | <a href="https://www.youtube.com/watch?v=Wigd0ogk2ug">https://www.youtube.com/watch?v=Wigd0ogk2ug</a>               |    |
|        | 1.8  | Climate Risk Assessment                            | <a href="https://www.youtube.com/watch?v=Hjdrdcp-bqz8">https://www.youtube.com/watch?v=Hjdrdcp-bqz8</a>             |   |
|        |      |                                                    | <a href="https://youtu.be/Pmlbh-2scn0?si=Snpl2z95-Cdm-d09p">https://youtu.be/Pmlbh-2scn0?si=Snpl2z95-Cdm-d09p</a>   |  |
|        |      |                                                    | <a href="https://youtu.be/Hnc-qawzz5f0?si=Dd2cbkw-viklgyypd">https://youtu.be/Hnc-qawzz5f0?si=Dd2cbkw-viklgyypd</a> |  |
|        | 1.10 | Green Climate Fund                                 | <a href="https://youtu.be/K-K6ut6ilhg?si=Iyf-39dvzwcjpjppkm">https://youtu.be/K-K6ut6ilhg?si=Iyf-39dvzwcjpjppkm</a> |  |
|        |      |                                                    | <a href="https://youtu.be/Cgsxy0s-olea?si=Tilnqo3k_Ui7tftd">https://youtu.be/Cgsxy0s-olea?si=Tilnqo3k_Ui7tftd</a>   |  |
|        | 1.12 | Sustainable Agriculture                            | <a href="https://youtu.be/X4d-zltdsecm?si=GYJ71iisOf-W3l8XX">https://youtu.be/X4d-zltdsecm?si=GYJ71iisOf-W3l8XX</a> |  |

| Module | Unit | Topic Name                                            | URL                                                                                                                 | QR Code                                                                               |
|--------|------|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 3      | 3.2  | Carbon Foot Printing                                  | <a href="https://youtu.be/A9yo-K8mwI0?Si=M-vm1srg5svtzsbo6">https://youtu.be/A9yo-K8mwI0?Si=M-vm1srg5svtzsbo6</a>   |    |
|        |      |                                                       | <a href="https://youtu.be/By-b7ylsxvzg?Si=-Wiorhg-dr8kznnys">https://youtu.be/By-b7ylsxvzg?Si=-Wiorhg-dr8kznnys</a> |    |
|        |      |                                                       | <a href="https://youtu.be/8q7Av8elue?Si=Esn2_Qldl6vbmjfl">https://youtu.be/8q7Av8elue?Si=Esn2_Qldl6vbmjfl</a>       |    |
|        |      |                                                       | <a href="https://www.youtube.com/watch?v=J_Idck-dawba">https://www.youtube.com/watch?v=J_Idck-dawba</a>             |   |
| 4      | 4.3  | Compliance With National Standards (Indian Standards) | <a href="https://youtu.be/Akbgz-3cyvqe?Si=Lalxcpe9yxe-g0rgf">https://youtu.be/Akbgz-3cyvqe?Si=Lalxcpe9yxe-g0rgf</a> |  |

## Reference for Further Reading

- 1. United Nations Framework Convention on Climate Change (UNFCCC)**
- 2. Official site: <https://unfccc.int>**  
**Intergovernmental Panel on Climate Change (IPCC) Reports**
- 3. IPCC Assessments and Guidelines: <https://www.ipcc.ch>**  
**Greenhouse Gas Protocol (GHG Protocol)**
- 4. Corporate GHG Accounting Standards: <https://ghgprotocol.org>**  
**International Energy Agency (IEA) Reports**
- 5. Renewable energy and emissions trends: <https://www.iea.org>**  
**Global Reporting Initiative (GRI) Standards**
- 6. Sustainability Reporting Standards: <https://www.globalreporting.org>**  
**Carbon Disclosure Project (CDP)**
- 7. Environmental disclosure reports: <https://www.cdp.net>**  
**World Bank Climate Finance**
- 8. Reports and funding mechanisms: <https://www.worldbank.org/en/topic/climatechange>**  
**ISO 14064 - International Standard for GHG Emissions Reporting**
- 9. Guidelines for GHG quantification and reporting.**  
**Business Responsibility and Sustainability Reporting (BRSR) Framework**
- 10. Securities and Exchange Board of India (SEBI): <https://www.sebi.gov.in>**





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